



STANDARD

DNV-ST-B203

Edition October 2022

Additive manufacturing of metallic parts

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FOREWORD

DNV standards contain requirements, principles and acceptance criteria for objects, personnel, organisations and/or operations.

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CHANGES – CURRENT

This document supersedes the May 2020 edition of DNVGL-ST-B203.
The numbering and/or title of items containing changes is highlighted in red.

Changes October 2022

<i>Topic</i>	<i>Reference</i>	<i>Description</i>
General scope and definitions	Sec.1	Changed scope to include more AM technologies, made adjustments to definitions and updated references.
Restructure	Sec.2 , Sec.3 , Sec.4 , Sec.5 , Sec.6 , Sec.7 , Sec.8 , App.A , App.B , App.C , App.D	Restructured the document.
Additive manufacturing categories (AMC)	[2.2]	Adjusted definition of AMC 1-3, and introduced AMC 0 as a category for non-critical parts.
	Sec.5 , Sec.6 , Sec.7 , Sec.8	Adjusted qualification, production and inspection requirements.
Documentation and workflow	Sec.3	Added section to address the relationship between the production documents, work process, and disclosure of information.
Part family	[4.3]	Introduced the concept of part families for the purpose of cost efficiency.
Production testing	[6.6]	Included production testing to a varying extent based on additive manufacturing category (AMC), for the purpose of making AMC 1 a viable option for more parts and alignment with other standards.
Qualification testing and validity	Sec.7	Adjusted and aligned the test scope for qualification and the validity of qualification.
Directed energy deposition (DED) with integrated substrate	[7.2] , [7.3]	Updated the document with requirements for when a DED process is used to add material to an existing part, i.e. repair or feature addition.
Laser-based directed energy deposition	[7.3]	Included requirements and guidance related to DED-LB.
Electron beam-based powder bed fusion	[7.5]	Included requirements and guidance related to PBF-EB.
Binder jetting technology	[7.6]	Included requirements and guidance related to BJT.
Non-destructive testing (NDT) acceptance criteria	[8.3]	Introduced the NDT acceptance criteria.
Selection of AMC	App.B	Introduced guidance on selecting the AMC.
Part families	App.C	Introduced guidance on the definition of part families.
NDT and monitoring methods	App.D	Introduced guidance on how to qualify inspection and monitoring techniques.

Editorial corrections

In addition to the above stated changes, editorial corrections may have been made.

Acknowledgements

This document has been developed based on the results of two (2) joint industry projects (JIPs). The companies listed in the first table are acknowledged for their contributions to the development of the initial edition of this document. The companies listed in the second one are acknowledged for their contribution to the further improvements and extensions made in Phase II.

<i>ProGRAM JIP Phase I participants</i>			
Additive Industries	Aidro	BP	Equinor
Guaranteed / OCAS	HIPtec	Immensa Technology Labs	IMI Critical Engineering
Ivaldi	Kongsberg Maritime	Sandvik	Shell
Siemens Energy	SLM Solutions	TecnipFMC	Total
National Manufacturing Institute Scotland	Vallourec	voestalpine	Quintus Technologies

<i>ProGRAM JIP Phase II participants</i>			
Addilan	BMT Aerospace	Equinor	ExOne
FIT Additive Manufacturing Group	Guaranteed	Howco Group	IMI Critical Engineering
Imphytek Powders	Intertek	Kongsberg Maritime	Kongsberg Ferrotech
National Manufacturing Institute Scotland	Sandvik	Saudi Aramco	Siemens Energy
voestalpine	XDM 3D		

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SECTION 1 GENERAL

1.1 Introduction

Additive manufacturing (AM) is a computer controlled process that creates three-dimensional objects by depositing materials, usually layer by layer. Due to its unique capabilities compared to more conventional manufacturing methods, AM may enable the energy and maritime industries to take advantage of new part designs, improved lead times, reduction of waste from manufacturing, and new types of supply chains. This standard provides guidance and requirements for the quality management of additive manufacturing and additively manufactured metal parts, while complying with and complementing international AM standards to enable AM in the energy and maritime industry sectors.

DNV has led a joint industry project (JIP) over two (2) phases, with 30 participating companies representing the whole value chain, from feedstock supplier to end user. The goal of the JIP has been to establish a framework for qualifying additively manufactured metal parts. This standard is based on the results of that work.

This standard is part of DNV's portfolio of additive manufacturing documents, in which [DNV-CG-0197](#), [DNV-CP-0267](#) and [DNV-CP-0291](#) are related to maritime classification services.

1.2 Objective

The objective of this standard is to provide an internationally acceptable framework for additive manufacturing with the aim of obtaining a consistent quality of additively manufactured metal parts and of providing predictability in the supply chain to reduce lead time and costs.

1.3 Scope

This standard specifies requirements and guidance for:

- the qualification of parts made by AM for the energy and maritime industries and related industries
- the purchasing of AM parts
- quality management by manufacturers of AM parts
- the manufacturing of AM parts.

The standard specifies requirements for the following specific metallic AM technologies:

- directed energy deposition, using an electric arc as the heat source (DED-arc)
- directed energy deposition, using a laser beam as the heat source (DED-LB)
- laser beam powder bed fusion (PBF-LB)
- electron beam powder bed fusion (PBF-EB)
- binder jetting technology (BJT).

1.4 Application

This document is intended to be used:

- as a contractual reference document
- for creating specifications when purchasing additively manufactured metal parts
- for documenting and proving the quality management of manufacturers and integrity of manufactured parts
- to provide guidance for purchasers and manufacturers of additively manufactured parts.

The requirements in this standard are in addition to the requirements in other relevant recognized standards, see [\[1.5\]](#).

This standard is not intended to inhibit a vendor from offering, or a purchaser from accepting, alternative materials or manufacturing processes. The principles stated in this document may be applied with additional agreed requirements on a case-by-case basis.

1.5 References

The standards in [Table 1-1](#) and [Table 1-2](#) include requirements and guidance which, through reference, constitute requirements and guidance of this standard. The latest edition applies, unless dated references are given.

Other recognized standards may be used provided it can be demonstrated that they meet or exceed the requirements of the referenced standards.

Table 1-1 DNV references

<i>Document code</i>	<i>Title</i>
DNV-CG-0197	Additive manufacturing - qualification and certification process for materials and components
DNV-CP-0267	Additive manufacturing
DNV-CP-0291	Additive manufacturing feedstock
DNV-RP-A203	Technology qualification
DNV-SE-0568	Qualification of additive manufacturing service providers, manufacturers and parts

Table 1-2 Other references

<i>Document code</i>	<i>Title</i>
API Spec 6A	Specification for Wellhead and Tree Equipment
API Std 20S	Additively Manufactured Metallic Components for Use in the Petroleum and Natural Gas Industries
ASME BPVC.IX	ASME Boiler and Pressure Vessel Code, Section IX: Welding and Brazing Qualifications
ASNT SNT-TC-1A	Personnel Qualification and Certification in Nondestructive Testing
ASTM A751	Standard Test Methods and Practices for Chemical Analysis of Steel Products
ASTM A991/A991M	Standard Test Method for Conducting Temperature Uniformity Surveys of Furnaces Used to Heat Treat Steel Products
ASTM D3951	Standard Practice for Commercial Packaging
ASTM E11	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E23	Standard Test Methods for Notched Bar Impact Testing of Metallic Materials
ASTM E29	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specification
ASTM E290	Standard Test Methods for Bend Testing of Material for Ductility
ASTM E3	Standard Guide for Preparation of Metallographic Specimens
ASTM E3166	Standard Guide for Nondestructive Examination of Metal Additively Manufactured Aerospace Parts After Build
ASTM E384	Standard Test Method for Microindentation Hardness of Materials

<i>Document code</i>	<i>Title</i>
ASTM E407	Standard Practice for Microetching Metals and Alloys
ASTM E466	Standard Practice for Conducting Force Controlled Constant Amplitude Axial Fatigue Tests of Metallic Materials
ASTM E8/E8M	Standard Test Methods for Tension Testing of Metallic Materials
ASTM F2924	Standard Specification for Additive Manufacturing Titanium-6 Aluminum-4 Vanadium with Powder Bed Fusion
ASTM F3001	Standard Specification for Additive Manufacturing Titanium-6 Aluminum-4 Vanadium ELI (Extra Low Interstitial) with Powder Bed Fusion
ASTM F3055	Standard Specification for Additive Manufacturing Nickel Alloy (UNS N07718) with Powder Bed Fusion
ASTM F3056	Standard Specification for Additive Manufacturing Nickel Alloy (UNS N06625) with Powder Bed Fusion
ASTM F3122	Standard Guide for Evaluating Mechanical Properties of Metal Materials Made via Additive Manufacturing Processes
ASTM F3184	Standard Specification for Additive Manufacturing Stainless Steel Alloy (UNS31603) with Powder Bed Fusion
ASTM F3187	Standard Guide for Directed Energy Deposition of Metals
ASTM F3301	Standard for Additive Manufacturing - Post Processing Methods - Standard Specification for Thermal Post-Processing Metal Parts Made Via Powder Bed Fusion
ASTM F3302	Standard for Additive Manufacturing - Finished Part Properties - Standard Specification for Titanium Alloys via Powder Bed Fusion
AWS A5.01M/A5.01	Welding and Brazing Consumables - Procurement of Filler Metals and Fluxes
AWS A5.32M/A5.32	Welding Consumables - Gases and Gas Mixtures for Fusion Welding and Allied Processes
EN 10204	Metallic products - Types of inspection documents
EN 13018	Non-destructive testing - Visual testing - General principles
EN 4179	Aerospace series - Qualification and approval of personnel for non-destructive testing
ISO 10423	Petroleum and natural gas industries - Drilling and production equipment - Wellhead and tree equipment
ISO 13916	Welding - Measurement of preheating temperature, interpass temperature and preheat maintenance temperature
ISO 14175	Welding consumables - Gases and gas mixtures for fusion welding and allied processes
ISO 14344	Welding consumables - Procurement of filler materials and fluxes
ISO 148-1	Metallic materials - Charpy pendulum impact test - Part 1: Test method
ISO 15549	Non-destructive testing - Eddy current testing - General principles
ISO 15614 series	Specification and qualification of welding procedures for metallic materials - Welding procedure test
ISO 15708 series	Non-destructive testing - Radiation methods for computed tomography
ISO 16810	Non-destructive testing - Ultrasonic testing - General principles

<i>Document code</i>	<i>Title</i>
ISO 17296-3	Additive manufacturing - General principles - Part 3: Main characteristics and corresponding test methods
ISO 17635	Non-destructive testing of welds - General rules for metallic materials
ISO 17663	Welding - Quality requirements for heat treatment in connection with welding and allied processes
ISO 3452 series	Non-destructive testing - Penetrant testing
ISO 4136	Destructive tests on welds in metallic materials - Transverse tensile test
ISO 5173	Destructive tests on welds in metallic materials - Bend tests
ISO 5579	Non-destructive testing - Radiographic testing of metallic materials using film and X- or gamma rays - Basic rules
ISO 5817	Welding - Fusion-welded joints in steel, nickel, titanium and their alloys (beam welding excluded) - Quality levels for imperfections
ISO 6507-1	Metallic Materials - Vickers hardness test - Part 1: Test method
ISO 6892-1	Metallic materials - Tensile testing - Part 1: Method of test at room temperature
ISO 6947	Welding and allied processes - Welding positions
ISO 9000	Quality management systems - Fundamentals and vocabulary
ISO 9001	Quality management systems - Requirements
ISO 9044	Industrial woven wire cloth - Technical requirements and tests
ISO 9712	Non-destructive testing - Qualification and certification of NDT personnel
ISO 9934 series	Non-destructive testing - Magnetic particle testing
ISO/ASTM 52900	Additive manufacturing - General principles - Fundamentals and vocabulary
ISO/ASTM 52901	Additive manufacturing - General principles - Requirements for purchased AM parts
ISO/ASTM 52907	Additive manufacturing - Feedstock materials - Methods to characterize metal powders
ISO/ASTM 52910	Additive Manufacturing - Design - Requirements, guidelines and recommendations
ISO/ASTM 52911 series	Additive manufacturing - Design
ISO/ASTM 52921	Standard terminology for additive manufacturing - Coordinate systems and test methodologies
ISO/IEC 17000	Conformity assessment - Vocabulary and general principles
ISO/IEC 17020	Conformity assessment - Requirements for the operation of various types of bodies performing inspection
ISO/IEC 17025	General requirements for the competence of testing and calibration laboratories
ISO/IEC 27001	Information technology - Security techniques - Information security management systems - Requirements
ISO/IEC/IEEE 12207	Systems and software engineering - Software life cycle processes
ISO/TR 18491	Welding and allied processes - Guidelines for measurement of welding energies
SAE AMS2750	Pyrometry

<i>Document code</i>	<i>Title</i>
SAE AMS-H-6875	Heat Treatment of Steel Raw Materials

1.6 Definitions and abbreviations

1.6.1 Definition of verbal forms

The verbal forms defined in [Table 1-3](#) are used in this document.

Table 1-3 Definition of verbal forms

<i>Term</i>	<i>Description</i>
shall	verbal form used to indicate requirements strictly to be followed in order to conform to the document
should	verbal form used to indicate that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others
may	verbal form used to indicate a course of action permissible within the limits of the document

1.6.2 Definition of terms

In addition to [Table 1-4](#), the terminology defined in ISO/ASTM 52900 and ISO/ASTM 52921 applies.

Table 1-4 Definition of terms

<i>Term</i>	<i>Definition</i>
additive manufacturing category	term used to describe the qualification approach based on the importance of a part to meet requirements
batch (powder)	given amount of powder traceable to a unique number, either virgin, used or a blend, to be applied as feedstock
batch (wire)	individual production unit of wire rods or electrodes, as well as a combination of a single production unit each of fluxes and wire electrodes when relevant
building	layer-by-layer process resulting in the near-net-shape of a part, often referred to as 'printing', not including post processing steps
build file	data file, or collection of data files, containing the information required to produce a specific build This is similar to the digital part product definition in API Std 20S.
build job	building of a part or set of parts, from deposition of initial layer to the final layer
build process	layer-by-layer deposition which occurs during a build job, defined by means of essential parameters
build process qualification	work process leading to a qualified build process, including building, post processing, testing and documentation
build process qualification record	document containing information produced during the build process qualification, including all essential parameters, log files and test results

<i>Term</i>	<i>Definition</i>
build process qualification testing	testing of a defined set of samples as part of the build process qualification
build report	report containing information about a build job and a summary of selected logged parameters
case-by-case basis	assessment made by relevant stakeholders taking into consideration the specific part and its application, as opposed to generic considerations
certificate	certification deliverable
certification	third-party issuance of a statement, based on a decision, following a review, that fulfilment of specified requirements has been demonstrated related to products, processes or systems (ISO/IEC 17000)
deposition path	direction which the depositing head travels as a layer is built in DED manufacturing
designer	company responsible for the design of a part
end user	company intending to use the part, which may be the same company as the purchaser and/or manufacturer
essential parameters	process or equipment parameters that when altered are likely to result in a change to the resulting material properties
logging	recording of parameters or values
log file	file containing information about a build job after completion, including sensor data
lot (powder)	powder made in the same production run, which may be divided into several powder batches
manufacturer	organization that manufactures the part or carries out part production that determines the quality of a part, or does the final assembly of a product
manufacturing	complete process from receipt of digital model to a finished part, including post processing steps
machine/equipment manufacturer	company manufacturing the hardware and associated firmware/software used for AM, which may be a complete machine or manufacturing equipment integrated into a system
monitoring	recording of parameter(s) or values through a sensor and comparing against pre-defined values
near-net-shape	condition where the parts require little post-processing to meet dimensional tolerance (adapted from ISO/ASTM 52900)
part	final product from an AM manufacturing process
part family	defined collection of parts in a defined group of similar application, design and manufacturing processes
part production specification	document containing all the relevant information in order to build a specific part with repeatable properties
part qualification testing	testing performed to validate that a defined part has the properties intended
part qualification record	document containing information produced during part qualification, that is sufficient to replicate the part and its properties
post-processing	process steps taken after the completion of a build job in order to achieve the desired properties in the final part (adapted from ISO/ASTM 52900)

<i>Term</i>	<i>Definition</i>
printability	part's ease of building, dependent on e.g. AM technology, geometry and material
production testing	testing performed to verify that results obtained during qualification are also obtained during production
prolongation	material added to the end of a part and used for testing
purchaser	company ordering a part, which may or may not be the designer
qualification	process of demonstrating whether an entity is capable of fulfilling specified requirements (ISO/IEC/IEEE 12207)
qualified facility	manufacturing facility where the right equipment, procedures and competence are in place to ensure the production of parts according to this standard
qualified manufacturer	manufacturer with a qualified facility and a qualified build process according to this standard
quality management system	set of policies, processes and procedures required for the planning and execution of manufacturing to ensure a consistent and repeatable quality that meets relevant requirements
supplier	company supplying consumables (wire, powder or gas) to the manufacturer
support structure	material produced along with a part to enable or improve its manufacturing
standard qualification build	defined geometry or geometries from which a required set of test specimens shall be extracted or machined and tested as a part of the build process qualification testing (BPQT)
test coupon	test piece that represents a selected section of the part
thermal history	heating and cooling cycles that a part or material has undergone during building and heat treatment, which may differ at different sections of a part based on part geometry and the manufacturing process
verification	confirmation, through the provision of objective evidence, that specified requirements have been fulfilled (ISO 9000)

1.6.3 Definition of symbols

$H/4$ = one quarter of the relevant part's full height down from full height

$T/2$ = mid-thickness

$T/4$ = one quarter of the relevant part's full thickness in from the surface.

1.6.4 Abbreviations

The abbreviations described in [Table 1-5](#) are used in this document.

Table 1-5 Abbreviations

<i>Abbreviation</i>	<i>Description</i>
ACCP	ASTN Central Certification Program
AM	additive manufacturing
AMC	additive manufacturing category

<i>Abbreviation</i>	<i>Description</i>
API	American Petroleum Institute
ASTM	American Society for Testing and Materials
BJT	binder jetting technology
BPQ	build process qualification
BPQT	build process qualification testing
BPQR	build process qualification record
CAD	computer aided design
CT	computed tomography
DED	directed energy deposition
DED-arc	directed energy deposition with electric arc
DED-LB	directed energy deposition with laser beam
ET	eddy current testing
FE	finite element
HIP	hot isostatic pressing
HSE	health, safety and environment
HT	heat treatment
ITP	inspection and test plan
MPS	manufacturing procedure specification
MT	magnetic particle testing
N-DP	normal to deposition path
NDT	non-destructive testing
P-BD	parallel to build direction
P-DP	parallel to deposition path
PBF	powder bed fusion
PBF-EB	electron beam powder bed fusion
PBF-LB	laser beam powder bed fusion
PPE	personal protective equipment
PPS	part production specification
PQR	part qualification record
PT	penetrant testing
QMS	quality management system
RT	radiographic testing
SI	International System of Units, i.e. the metric system

<i>Abbreviation</i>	<i>Description</i>
UT	ultrasonic testing
UTS	ultimate tensile strength
VT	visual testing
YS	yield strength

SECTION 2 PURCHASING

2.1 General

This section provides guidance and requirements to the purchaser for the design and procurement of AM parts.

2.2 Additive manufacturing category

This document presents three (3) different qualification schemes for a part, based on the selected additive manufacturing category (AMC), designated AMC 1, AMC 2, and AMC 3. The three (3) main categories reflect increasing criticality or specification levels, hence there are increasing requirements from AMC 1 to AMC 3: AMC 1 is intended for low criticality parts, AMC 2 for medium criticality parts and AMC 3 for high criticality parts.

In addition, a simple category is defined without a required qualification scheme and production testing scheme, designated AMC 0. This category is intended for non-critical parts, and will only ensure traceability and general quality management by the manufacturer. Therefore the qualification requirements and quality control requirements that are defined for AMC 1, AMC 2 and AMC 3 are not applicable to AMC 0.

An overview of the requirements for each additive manufacturing category is given in [Table 2-1](#). The AMC shall be defined by the purchaser or design owner, see [App.B](#) for guidance.

Table 2-1 Requirements for the different AM categories

<i>Additive manufacturing category</i>	<i>Requirements</i>	<i>Typical deliveries</i>
AMC 0	Part built by a qualified facility	Part(s) with verifiable traceability information
AMC 1	Part built using a qualified build process by qualified manufacturer Simplified quality control of delivered parts Other requirements as agreed with purchaser	Part(s) with verifiable traceability information Details of PPS and BPQ or BPQR Testing and inspection reports
AMC 2 and AMC 3	Part built using a qualified build process and qualified part production specification by a qualified manufacturer Medium and higher degree of quality control of delivered parts Other requirements as agreed with purchaser	Part(s) with verifiable traceability information Details of BPQ or BPQR Details of PPS and PQ or PQR Testing and inspection reports

Guidance note:

When deciding on the relevant AMC, the following should be considered:

- consequences of failure for health and safety
- economic consequences of failure
- environmental consequences of failure
- loss of reputation
- regulatory requirements.

A part's designated criticality level may be affected by other measures, e.g. tests or control steps added outside of the scope of this document. Such measures could be full scale testing or a design safety factor. This should be assessed by the purchaser.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

2.3 Purchase order information

2.3.1 General

2.3.1.1 Upon entering into a purchase agreement, the purchaser shall submit a written purchase order to the manufacturer. The manufacturer shall be provided with all relevant information that is required for the part to serve its intended function and comply with relevant standards.

2.3.1.2 ISO/ASTM 52901 should be consulted for AM specific topics related to the purchasing of AM products.

2.3.1.3 As a minimum, the following information shall be specified in the purchase order:

- 1) Ordering information:
 - a) Customer organization, contact person and shipping address.
 - b) Traceability or marking requirements.
 - c) Required delivery date for single orders, or required delivery frequency and quantity for multiple orders.
 - d) Disposal of rejected material.
 - e) Reference to this standard.
 - f) Preservation and shipment requirements.
- 2) Part details:
 - a) Clear definition of the part(s) through digital model(s) in a format accepted by the manufacturer. The part definition shall contain information on the unit of measurement used. The unit system used shall be either SI or the US customary system of units.
 - b) Requested AMC.
 - c) Quantity of part(s).
 - d) Material selection.
 - e) For AMC 3 only: definition of critical section(s) of the requested part(s), see [\[2.3.2\]](#).
 - f) Post processing, e.g. thermal processing, surface treatment or machining.
 - g) Surfaces of the part(s) where supports are not to be added, if applicable.
- 3) Acceptance criteria:
 - a) Dimensional tolerances, preferably through product manufacturing information or drawing.
 - b) Mechanical, chemical, and metallurgical acceptance criteria.
 - c) NDT acceptance criteria.
 - d) Inspection and test plan, indicating the frequency of production testing and NDT.
 - e) Surface condition, e.g. as-printed, machined, processed by blasting, tribo-finishing, electrochemical polishing, or passivation treatment.
 - f) Additional relevant requirements from ISO/ASTM 52901.
 - g) Any additional purchaser requirements or testing, e.g. other applicable standards or regulations, additional functional testing or quality control testing, 3.2 certificate in accordance with EN 10204 or independent third-party involvement.
 - h) Additional test specimens, including type, amount, test frequency, location, and orientation, to be included for qualification or production testing, if relevant.

2.3.1.4 If the digital model (CAD file) of the requested part is prepared by the manufacturer or a third party on behalf of the purchaser, the purchaser shall review and accept the digital model.

2.3.1.5 A purchase order template is given in [App.A](#).

Guidance note:

- Care should be taken when referring to standards or requirements that are usually relevant for parts produced by conventional methods. Assumptions made when producing conventionally may not be applicable when using AM technologies, e.g. certain complex shapes have not been possible to manufacture using conventional methods, and thus not considered in conventional standards or requirements.
- When specifying additional testing that is not described in this standard, other relevant standards or criteria should be specified, for example as specified in ISO 17296-3.
- If accepted by the manufacturer, CAD models may be transferred via interface formats such as STEP or IGES, where the manufacturer converts the relevant file to the appropriate file format for build preparation.
- If an STL file is provided by the purchaser, the purchaser should clearly specify what unit of measurement the STL is intended to be in, as the STL metadata does not contain this information.
- As support structures are removed from a part, the remaining surface may have a different topography than the as-printed surface.
- It may be relevant to contact the designer in order to define the critical section(s) of a part.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

2.3.2 Defining the critical section

2.3.2.1 The critical section(s) of an AMC 3 part shall be defined based on information about how the part is intended to be used, and how the part is built. The following two (2) topics shall be assessed:

- The critical section with regard to its application. It is recommended that the definition of critical sections is based on an appropriate analysis, e.g. finite element (FE) analysis.
- The part's geometry and intended manufacturing procedure. Areas where it is most likely the specified acceptance criteria will not be achieved should be defined as critical sections.

2.3.2.2 Critical section(s) defined by the designer shall be used when relevant. A part may contain several critical sections.

Guidance note:

To ensure a proper consideration of critical sections, a structured approach should be followed, e.g. through a failure mode and effect analysis (FMEA). The following are suggested topics to be included when assessing areas at risk of not achieving the specified acceptance criteria, in addition to the manufacturer's own experience with the relevant setup and feedstock:

- thermal history
- stress concentrations
- sections to be welded
- start and stop locations and deposition strategy, for DED
- support locations.

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2.4 Ordering parts designed for conventional manufacturing

Although additive manufacturing is capable of producing a wide range of geometries, there are design rules specific to AM that introduce new limitations compared to conventional manufacturing methods. The non-exhaustive list below is intended to highlight some of the aspects that are different in cases where the purchaser orders a part that was originally intended for another means of production.

The purchaser and manufacturer should assess the topics below and agree on appropriate measures:

- Design optimizations for more efficient production may or may not be possible or relevant.
- Caution should be taken when using additive manufacturing to produce parts designed for conventional manufacturing due to the implicit assumptions made. One such assumption may be isotropic material properties.

- ISO/ASTM 52910, and especially section 7 *Warnings to designers*, should be consulted when designing or assessing the build strategy or printability of a part.
- Some AM processes are capable of producing thin-walled features. With thin-wall thicknesses defects may be expected to have more pronounced effect.
- The fatigue life of cyclically loaded parts, may be different to that of conventionally produced parts.

SECTION 3 DOCUMENTATION AND WORKFLOW

3.1 General

AM parts shall be manufactured according to documented processes that ensure traceability regarding the purchaser's requirements, the manufacturer's quality system, work procedures, and the qualification of processes and personnel. This section gives a general overview of the type of information that shall be prepared and made available. It also provides a high-level description of the workflow when producing, documenting, and delivering an AM part.

3.2 Documents

In addition to testing and quality control requirements, this standard stipulates a requirement of documents to describe the part and the qualification of the part. These documents are defined below based on their intent and purpose. A document in the context of this standard can be a package of information given in any format.

- Manufacturing procedure specification (MPS): an overall plan and description of the manufacturing and post-processing of the ordered part, described further in [6.2.1].
- Part production specification (PPS): a specification of technical details and parameters for building the ordered part and defining the minimum post-processing steps to be carried out based on the qualification, described further in [6.2.2].
- Build process qualification record (BPQR): a record of the qualification of the build process, described further in [4.5.2].
- Part qualification record (PQR): a record of the qualification of a specific part, described further in [4.5.3].
- Compliance record: a document confirming the compliance of a delivered part, described further in [6.6.3].

Manufacturing shall be performed according to the manufacturing procedure specification (MPS) and a part production specification (PPS) that are supported by a build process qualification (BPQ), and, if applicable, a part qualification.

Figure 3-1 gives an overview of the different documents and how they refer to each other. The purchaser is responsible for the purchase order and for clarifying any specific requirements that will apply. The manufacturer is responsible for the production documents and qualification documents.

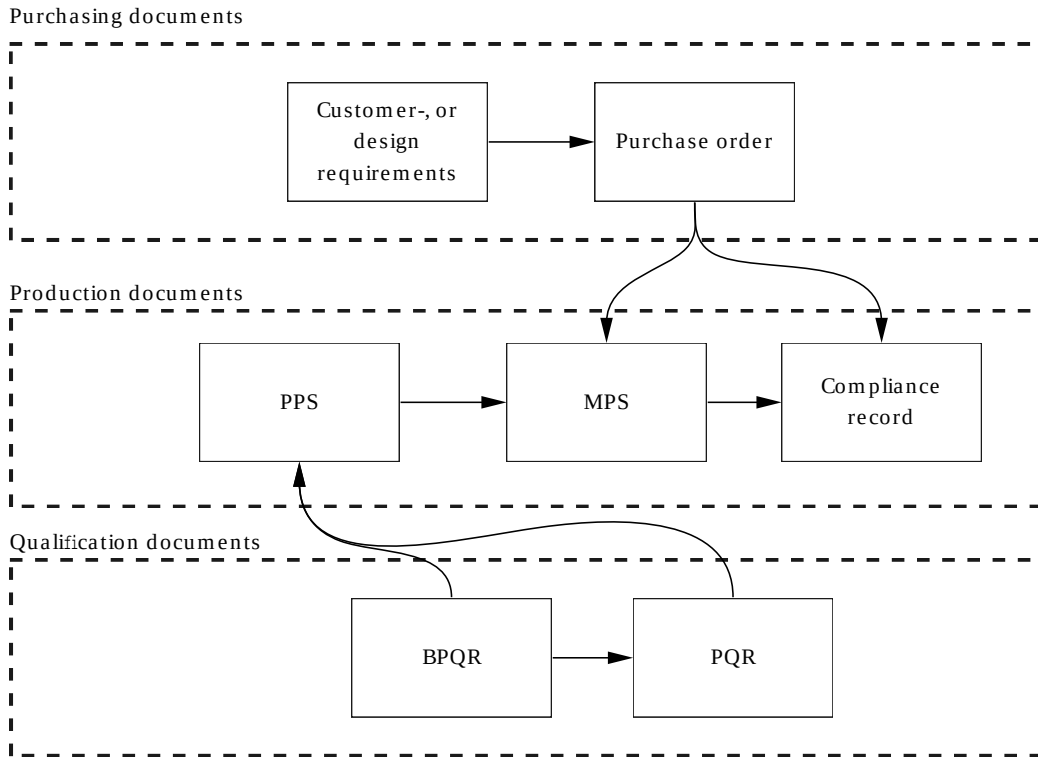


Figure 3-1 Overview of the defined documents and how they give input to each other

The manufacturer shall have a system in place for ensuring the traceability of orders and manufacturing documents, and the link between them. The traceability of manufacturing data, test results, feedstock materials, test coupons and specimen locations and orientations shall be ensured through the documentation and manufacturing logs kept by the manufacturer.

The duration of the documentation kept shall be agreed between the purchaser and manufacturer.

3.3 Work process

The work process steps described in this standard are illustrated in [Figure 3-2](#). The process and documentation are shown as a series of steps to indicate the link between the process steps and the documentation that is generated. In reality, the build process qualification, part qualification and production may be carried out in parallel, i.e. at the same time. However, the deliveries shall not be made before all the required steps have been carried out and documented.

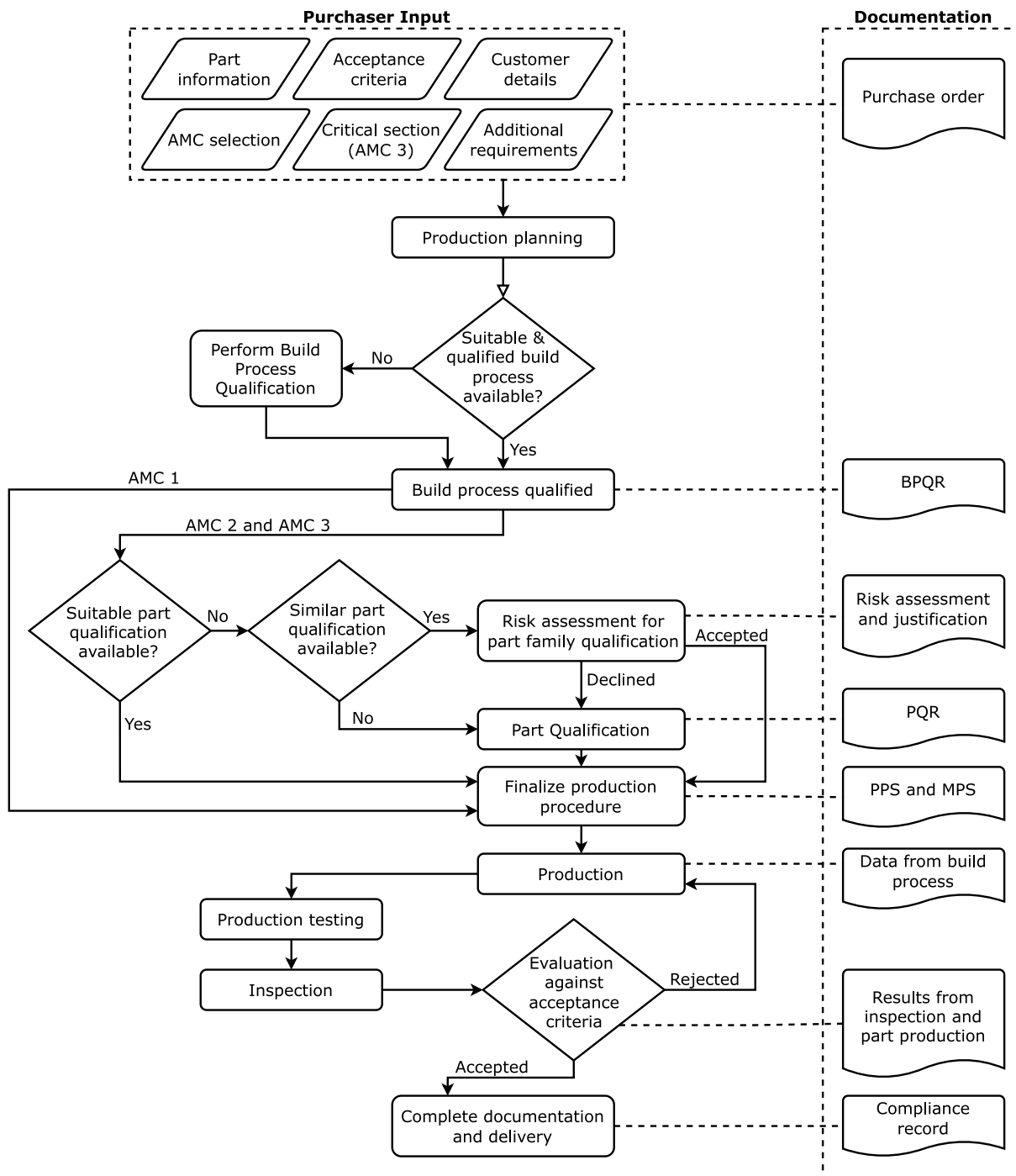


Figure 3-2 Flowchart showing process and documentation steps

3.4 Disclosure of information

The information generated through the ordering and manufacturing of a part may be of a business-critical nature and shall be kept confidential to different degrees. In general, the sharing party is responsible for protecting its confidential data, and the purchaser and manufacturer shall agree on the information to be shared and how.

As a recommendation, the documents should be prepared in such a way that the information in them may be shared between the parties, as described in [Table 3-1](#).

Table 3-1 Recommended level of confidentiality in project documentation

<i>Confidentiality level</i>	<i>Description</i>	<i>Documentation</i>
1	To be shared freely between the contracting parties	— Purchasing documents — MPS — Compliance record
2	To be shared with agreed IP protection	— PPS — BPQR — PQR
3	To be made available for review on the sharing party's terms	— Production logs from the production of the ordered part
4	Confidential information	— Underlying studies and trial data

SECTION 4 QUALIFICATION

4.1 General

This section provides requirements and recommendations for the qualification of processes, parts, manufacturers, and technologies.

AM parts made in accordance with this standard shall be produced by a qualified manufacturer using a qualified build process and part qualification, if applicable, and be built and manufactured within the limits defined for these qualifications.

The requirements for being a qualified manufacturer are stated in [4.4]. The requirements for build process qualification and its limits of qualification are given in [4.2], and those for part qualification are given in [4.3]. The required documentation from these qualification steps is described in [4.5]. Additional qualification requirements specific to each AM technology are given in the sections referenced in Table 4-1.

Table 4-1 References to the sections describing BPQ, part qualification and essential parameters for each AM technology

<i>AM technology</i>	<i>References to technology specific requirements for BPQ</i>	<i>References to technology specific part qualification</i>	<i>References to essential parameters</i>
DED-arc	[7.2.2]	[7.2.3]	[7.2.4]
DED-LB	[7.3.2]	[7.3.3]	[7.3.4]
PBF-LB	[7.4.2]	[7.4.3]	[7.4.4]
PBF-EB	[7.5.2]	[7.5.3]	[7.5.4]
BJT	[7.6.2]	[7.6.3]	[7.6.4]

The relevance of the different qualification steps depends on the AMC of the part to be produced. An overview is provided in Table 4-2. Parts without build process qualification or part qualification can be designated AMC 0 if they are in compliance with the requirements in [2.2].

Table 4-2 Overview of qualification steps for AMC 1, AMC 2 and AMC 3

<i>Qualification steps</i>		<i>AMC 1</i>	<i>AMC 2</i>	<i>AMC 3</i>
Build process qualification [4.2]		x	x	x
Part qualification [4.3]	Non-destructive testing		x	x
	Mechanical and metallurgical testing			x

4.2 Build process qualification

4.2.1 General

4.2.1.1 Build process qualification (BPQ) is the building of a pre-defined geometry or set of geometries, referred to as a standard qualification build, from which test specimens are extracted and tested. The standard qualification build shall be built with a defined set of essential parameters.

4.2.1.2 The purpose of the BPQ is to ensure that the build process and thermal post processing steps used for manufacturing parts are suitable for obtaining specified material properties. The results gained from the BPQ will provide a basis of comparison for similar types of tests performed during part qualification and production testing. The standard qualification build and relevant tests are unique for each AM technology and are detailed in the relevant section as given in [Table 4-1](#).

4.2.1.3 Additional application specific tests defined in the purchase order shall be included in the build process qualification or part qualification as relevant. Depending on the application, such testing may include, but is not limited to, fatigue testing, fracture toughness testing, corrosion testing, bend testing, etc.

4.2.1.4 The details of the BPQ shall be recorded and stored in a build process qualification record (BPQR), see [\[4.5.2\]](#).

4.2.2 Validity of qualification - essential parameters

4.2.2.1 A BPQ is a one-time qualification and is valid as long as essential parameters of the production are kept within the allowable limits of the parameters used during the qualification, and as long as production is done using machines and consumables stored and maintained as described in [\[5.5\]](#) and [\[5.6\]](#). Any deviation outside the allowable limit of qualification requires a new qualification. The essential parameters are defined for each AM technology, and can be found in the relevant section, see [Table 4-1](#).

4.2.2.2 A build process is limited to the manufacturing site at which the qualification was performed. A transfer of the build process qualification to another manufacturing site shall be documented and mutually agreed, and all procedures and essential parameters shall be maintained. The manufacturing site shall have the same quality control system as the site at which the qualification was performed. The manufacturing site shall verify that expected properties are achieved when transferring build processes to another site. The test scope of this verification may be a limited version of the standard BPQ scope, and shall be accepted by the purchaser.

4.3 Part qualification

4.3.1 General

4.3.1.1 Part qualification refers to the building and testing of a specific part. Relevant requirements given in applicable standards or purchasing documents shall be implemented.

4.3.1.2 The part qualification testing consists of two (2) tasks - a scope of mechanical and metallurgical testing and a scope of non-destructive testing and inspection. Mechanical and metallurgical testing only applies to AMC 3 parts, while non-destructive testing and inspection applies to both AMC 2 and AMC 3 parts.

4.3.1.3 The purpose of the part qualification testing is to ensure that the part(s) in question have the required and expected properties, as the properties of the AM technology may be highly design and geometry dependent. The results of the performed BPQ provide a basis for what should be expected during the part qualification. The scope of part qualification shall include the building and thermal post processing of the part, unless otherwise specified.

Guidance note:

The manufacturer responsible for the material properties may not perform all the subsequent post processing steps needed for the intended application. Thus, the part qualification scope is limited to the building and thermal post processing, unless otherwise specified.

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4.3.1.4 The details of the part qualification shall be recorded and stored in a part qualification record (PQR), as described in [4.5.3].

4.3.2 Validity of qualification and part families

4.3.2.1 A part qualification is by default valid for the specific part, and as long as the essential variables are kept within the allowable limits of the qualification.

4.3.2.2 A part qualification may be made valid for similar parts through defining and assessing the group of parts as a part family.

4.3.2.3 The use of a qualification for another part in the same part family shall be supported by a documented risk assessment. The risk assessment shall include the rationale for the part family designation, and the changes in design, manufacturing strategy or application with regard to the qualified part. The risks involved, and justification for acceptance of these risks, shall be included. If a risk assessment is carried out by the manufacturer, it shall be reviewed and accepted by the purchaser.

4.3.2.4 The risk assessment may conclude that a reduced testing scope may be relevant to assess the new part when a pre-existing part qualification is used. This shall be agreed on a case-by-case basis. For more guidance on defining part families, see App.C.

4.3.2.5 A new part qualified by a pre-existing part family qualification shall be considered the first production article in the context of inspection, see [6.6.2].

Guidance note:

The purpose of part families is to avoid repetition of control steps for parts with minor design changes compared to the qualified part which are unlikely to affect the integrity of the part.

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4.3.2.6 A part qualification is limited to the manufacturing site at which the qualification was performed. A transfer of the part qualification to another manufacturing site shall be documented, mutually agreed, and all procedures and essential parameters maintained. The manufacturing site shall have the same quality control system as the site at which the qualification was performed. If a part qualification is transferred to another manufacturing site, the first produced part shall be considered a first production article in the context of inspection, see [6.6.2].

4.4 Qualification of manufacturers and facilities

Manufacturing of AM parts according to this standard shall be done by qualified facilities or qualified manufacturers. A manufacturer may be deemed qualified if the facility and build process have been qualified, as further defined below:

- Facility qualification: the facility has equipment, procedures and competence in place to comply with the requirements given in Sec.5.
- Build process qualification: qualification of a build process as described in [4.2].

A qualified manufacturer shall ensure that production occurs in a safe and repeatable manner, and in accordance with the relevant requirements.

Qualification of manufacturers according to this standard, or other similar standards such as API 20S, may be done according to the procedure defined in DNV-SE-0568.

4.5 Documentation

4.5.1 General

4.5.1.1 This section describes the documentation requirements related to the build process qualification and part qualification.

4.5.2 Build process qualification record

4.5.2.1 All manufacturing shall be performed using a qualified build process described in a build process qualification record (BPQR). As a minimum, the BPQR shall contain the following information:

- a documentation summary, stating the identification number and date of qualification, manufacturer/sub-supplier and site, and relevant manufacturing standards
- description of AM method
- description of AM equipment
- description of the built qualification coupons
- essential parameters for the build process, given in the relevant section for the specific AM technology, including post processing details
- feedstock information, including material certificates or the certificate of conformity of feedstock used during build process qualification testing, including the powder state (i.e. virgin, reused or blended) if applicable
- post processing details and sequence
- the results of the build process qualification testing, including laboratory test reports and the orientation and location of test specimens
- NDT reports with a statement of results/compliance, if applicable.

Guidance note:

- In many cases, the creation of a preliminary PPS may be beneficial to set up the relevant build process before the qualification has been finalized.
- To ensure that the information needed for supporting a PPS is captured in the BPQR, it is recommended to include a summary of this information in the BPQR.

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4.5.2.2 A template for the BPQR is provided in [App.A](#). Another format may be used, as long as all the required information is provided.

4.5.3 Part qualification record

4.5.3.1 If the AMC of a part requires part qualification, the details of that qualification shall be given in a part qualification record (PQR), stating as a minimum:

- a documentation summary, with the identification number and date of qualification, manufacturer/sub-supplier and site, and relevant manufacturing standards
- the unique ID of the relevant part design, and the representative section or coupon built for qualification if different to the actual part
- all essential parameters used in building, and a description of the support structures, orientation, and nesting if relevant
- reference to a supporting BPQR
- material certificates or the certificate of conformity of feedstock used during part qualification, including the powder state (i.e. virgin, reused or blended) if applicable

- results from the part qualification testing, including laboratory test reports and the location and orientation of test specimens
- NDT reports including a statement of results/compliance.

4.5.3.2 A template for the PQR is provided in [App.A](#). Another format may be used, as long as all the required information is provided.

4.6 Qualification of new methods and technologies

Additive manufacturing is a rapidly developing technology. New and innovative methods, techniques and procedures may be used provided it is documented that the quality, safety, and reliability levels are equivalent to the requirements in this standard. [DNV-RP-A203](#) may be applied for the qualification of new technologies.

SECTION 5 REQUIREMENTS TO MANUFACTURERS

5.1 General

This section provides requirements and recommendations to manufacturers of additively manufactured parts. The objective of this section is to ensure that the manufacturer has systems and procedures in place to produce parts in accordance with purchaser requirements, with the desired quality, and in a repeatable and safe manner.

5.2 Quality management

The manufacturer shall have a certified quality management system (QMS) conforming to ISO 9001, or the equivalent.

The manufacturer should have a certified information security management system conforming to ISO/IEC 27001.

The manufacturer is responsible for ensuring that its sub-suppliers have equivalent quality systems.

Work instructions and procedures shall be in place to ensure the consistency and repeatability of a given process step that may influence the quality of the product. Such procedures may describe the handling of consumables, setting up of machines, etc. In addition to the requirements in this standard, requirements and relevant recommendations from machine/equipment manufacturers and consumable suppliers shall be integrated into the manufacturer's procedures.

Recommendations or limitations specified by the machine/equipment manufacturer and local regulations should be followed.

Numerical rounding related to test data shall be in accordance with ASTM E29.

5.3 Health, safety, security and environment

Local health, safety and environment (HSE) regulations shall be followed. HSE recommendations provided by the feedstock supplier and AM machine/equipment manufacturer shall be followed.

The manufacturer shall have HSE procedures in place that ensure the following as a minimum:

- Personal protective equipment (PPE) is made available to relevant personnel.
- Gloves and eye protection are used when handling feedstock. When handling powders, the necessary protective equipment is used to avoid inhalation and exposure to skin and eyes.
- Systems for identifying a gas leak or oxygen shortage are in place where gas is handled and stored.
- Measures are in place to avoid human contact with hot items such as produced parts or hot ends.
- Lifting equipment is available to avoid heavy manual lifting, e.g. for build platforms or build volumes.
- Heavy parts or equipment shall be properly secured to avoid rolling or falling.
- Consumables are disposed of according to requirements stipulated by the local authority or recommendations given by the consumable supplier.
- Powder outside of delivery compartments is stored and handled in a spark-free atmosphere to avoid ignition, combustion, or explosion, if applicable.
- Only machine operators have access to the area within the reach of moving machines during building, and precautions are put in place to avoid collisions with moving parts and robotic arms.
- When building is performed without controlled habitats, screens are used to minimize exposure to visible, and UV radiation and spatter. Proper ventilation is used to protect personnel from fumes.

5.4 Personnel

5.4.1 General

5.4.1.1 The manufacturer is responsible for supervising, training and maintaining the competence of its personnel for the assigned responsibilities at all its locations.

5.4.1.2 All personnel involved in AM production and quality control should be familiar with all process steps.

5.4.1.3 The operator of the AM machine/equipment shall have received training recommended by the machine/equipment manufacturer. The operator of the AM machine/equipment shall receive continuous training as new hardware/software is introduced. Documentation of the training received shall be available and linked to the relevant personnel.

5.4.1.4 The operator shall comply with vision requirements that meet the applicable requirements of ISO 9712.

5.4.2 Qualification of personnel

5.4.2.1 Operators shall be qualified according to the manufacturer's procedure for the qualification of personnel.

5.4.2.2 The manufacturer's procedures for the qualification of operators shall consist of a practical and a theoretical part. Recommended training given by AM machine/equipment manufacturers shall be included in the manufacturer's procedures.

- The practical training shall include a minimum of 40 hours of on-the-job training and shadowing of qualified personnel. After the 40 hours, the operator shall demonstrate the ability to carry out essential tasks, such as changing feedstock, the setup and operation of the equipment, relevant maintenance, and calibrations.
- Following the theoretical training, the operator shall demonstrate knowledge of the AM equipment, process and likely failure modes related to the manufacturing, and relevant procedures in case of emergency situations. This may be demonstrated e.g. by a multiple-choice test as defined by the manufacturer.

5.4.2.3 The manufacturer's operator qualification procedures shall specify the limitations on the machine setup and machine brand and model.

5.4.2.4 The operator qualification is valid as long as the operator regularly performs the tasks for which the operator is qualified. If the operator has not performed such tasks for a period of time defined by the manufacturer, the operator shall be re-qualified by a review of all work procedures and be supervised by a qualified operator during the first assignment. Through the re-qualification, the operator shall demonstrate the ability to carry out essential tasks and his/her knowledge of the AM equipment, process, failure modes and relevant procedures.

5.4.2.5 The operator qualification is not transferrable to other manufacturers.

5.4.2.6 Any national or internationally recognized certification or qualification schemes for AM operators may be used in lieu of the described requirements for internal operator qualification.

5.5 Consumables

5.5.1 General

5.5.1.1 Consumable suppliers shall have a certified quality management system (QMS) conforming to ISO 9001 or the equivalent.

5.5.1.2 The consumable supplier shall have systems in place to ensure the consistent quality and traceability of the feedstock products. See the requirements in [5.5.2] for wires and [5.5.3] for powders.

5.5.1.3 Instructions provided by the consumable supplier for preservation, storage and handling shall be available and adhered to. Feedstock shall not be used after the shelf-life period defined by the consumable supplier.

5.5.2 Wire feedstock

5.5.2.1 Wire feedstock shall be supplied in packaging that clearly states the supplier, product and trade name, batch number, wire diameter and date of production. A specification shall be supplied along with the wire stating information on the coating and chemical composition.

5.5.2.2 Wire shall be stored and handled according to the supplier's recommendations to ensure that any oil, grease, and other contaminants are avoided. Wire used for AMC 2 or AMC 3 parts shall be supplied with a type 3.1 inspection certificate according to EN 10204 or the equivalent as a minimum. Type 3.2 certificates may be requested by the purchaser. The testing of feedstock shall, as a minimum, be as specified in AWS A5.01 or ISO 14344, testing schedule 3 or H.

5.5.2.3 If available, wire developed for AM purposes should be used.

5.5.2.4 The oxidized end of a partly used wire should be cut off prior to reuse.

5.5.3 Powder feedstock

5.5.3.1 Powder shall be supplied in packaging protected from sunlight and moisture. The packaging shall clearly state the manufacturer, product and trade name, particle size distribution and lot number. The packaging shall be supplied with proof that the package has not been breached. A specification shall be supplied along with the powder, stating as a minimum the lot number, atomization atmosphere, chemical composition, density characteristics and flow characteristics.

5.5.3.2 Reactive powder shall be stored in a safe environment, e.g. in a fire cabinet, in an inert atmosphere.

Guidance note:

Reactive powders include, but are not limited to, magnesium, aluminium, and titanium alloys.

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5.5.3.3 Local regulations and supplier's recommendations for storing and handling shall be adhered to.

5.5.3.4 The powder shall be delivered with an inspection certificate of type 3.1 or 3.2 according to EN 10204, or the equivalent, or checked against defined requirements to ensure the correct and consistent quality of all lots. The powder shall be tested in accordance with a recognized standard, see ISO/ASTM 52907. As a minimum, the following properties shall be stated on the certificate or checked:

- powder particle size and distribution

- apparent density
- flowability
- chemical composition
- hydrogen, carbon, nitrogen, and oxygen content.

Guidance note:

Analysis of carbon, nitrogen, hydrogen and oxygen may need to be performed separately, as they are not always included in general chemical analyses.

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5.5.3.5 Unfused/unbound powder from a previous build job may be reused. Unfused/unbound powder shall be sieved, according to the machine manufacturer's recommendations or manufacturer's procedure, before reuse. Reuse shall be documented, and traceability maintained to the extent practically possible.

5.5.3.6 The manufacturer shall be able to document that the powder is still within the ranges defined in the relevant powder specification for the amount of powder reuse it operates with. The documentation shall include reference to the procedures for powder handling, storage, sieving and the performed testing.

5.5.3.7 The flowability and apparent density should be monitored regularly to discover potential changes in the powder as reuse increases. If a decrease in flowability is recorded, the reason for the decrease shall be assessed.

Guidance note:

An increase of powder flowability is expected upon reuse, as the particle size distribution is shifted upwards due to smaller particles being processed. A decrease in flowability may thus be an indication of e.g. moisture.

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5.5.3.8 All sieves shall have a certificate of conformity stating that they were manufactured according to ISO 9044 or ASTM E11 or the machine manufacturer's recommendations. The sieves shall be inspected, cleaned, and replaced according to the powder supplier's or machine manufacturer's recommendations.

5.5.3.9 The blending of batches or lots shall be documented, and traceability of the relevant batches/lots used for the blend shall be maintained. Blended batches or lots shall be of equivalent powder properties (chemical composition, particle size distribution, apparent density and flowability) and within the limits defined in the relevant specification. Powder batches or lots that do not conform to specification shall not be used for blending.

5.5.3.10 The powders shall be blended according to a qualified procedure to ensure a homogenous distribution of all powder lots or batches blended.

5.5.3.11 Any powder batch containing used powder shall be considered as used powder.

5.5.3.12 If powder is sourced from another supplier than the one used during the build process qualification or the part qualification, it shall be ensured that the properties of the powder are within the essential parameters.

5.5.4 Gases

5.5.4.1 Gases used in the manufacturing process shall meet the requirements specified in ISO 14175, AWS A5.32M/A5.32, or an equivalent standard.

5.5.4.2 Gas used for different process steps shall be delivered with an individual certificate for each such step.

5.5.5 Build plates and build substrate

5.5.5.1 Substrates and build plates shall be cleaned of any oil, grease, and other contaminants before use.

5.5.5.2 Build plates for PBF technologies shall be traceable to the manufacturer and have a unique identification for traceability.

5.5.5.3 Substrate material for DED technologies shall be of known origin and composition. If the substrate will be an integral part of the final part, then the chemical composition, supply condition and origin shall be documented.

5.5.6 Binder and cleaner

5.5.6.1 Binder and cleaner for BJT shall be used according to the manufacturer's recommendations and delivered with a certificate stating the following:

- supplier
- trade name
- material safety data
- expiry date.

5.6 Manufacturing equipment

5.6.1 General

5.6.1.1 All machines, equipment, measuring devices and testing equipment relevant for the manufacturing process, testing or safety shall be calibrated at regular intervals according to defined procedures, and inspected by accredited third parties, authorized internal departments or authorized external companies. The calibration shall be documented.

5.6.1.2 Instructions provided by the AM machine/equipment manufacturer for use and maintenance shall be available and adhered to. The manufacturing AM machine/equipment shall undergo maintenance and calibration according to the recommended schedule and regime supplied by the AM machine/equipment manufacturer.

5.6.1.3 Calibration shall be validated using calibrated equipment, in accordance with the machine/equipment manufacturer's recommendations, and be carried out after any hardware or software changes, maintenance or repair likely to influence the build process.

5.6.1.4 If machines and equipment from different machine manufacturers is combined in a system, the manufacturer shall ensure that the system works as intended and document the checks performed.

5.6.1.5 When a software or firmware update is performed, the effect of such updates on the quality and repeatability of the build process shall be evaluated by the manufacturer to determine the qualification validity. If the quality or repeatability is affected, a limited validation scope may be performed in stead of a new qualification. Such validation testing may be similar in extent to a production test. Similarly, when replacing or upgrading hardware elements, a similar evaluation shall be performed and validated as necessary.

SECTION 6 PRODUCTION

6.1 General

This section contains requirements and recommendations for how production shall be carried out during the different production steps, including production quality control. It is assumed that relevant qualification steps have been fulfilled. For qualification requirements, see [Sec.4](#).

The following requirements and recommendations are universal for all AM technologies. In addition to the provisions given in this section, AM technology-specific production requirements are given in [Sec.7](#).

If requirements defined by the purchaser are outside the manufacturer's qualification or capabilities, the purchaser shall be notified. If suggestions for design changes or adjustments are put forth to improve the production, the purchaser shall review and accept the design changes before production.

Guidance note:

Parts that are designed for conventional manufacturing may be requested, with potential challenges associated with production by AM. In such cases it should be agreed how to handle this (new design, additional supports, etc.). When relevant, the designer should be contacted when design changes are discussed. This is both to avoid complications regarding IP rights, and to ensure that changes in form resulting from the redesign have no negative impact on the function or integrity of the part.

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6.2 Developing procedures

6.2.1 Manufacturing procedure specification

6.2.1.1 A defined manufacturing procedure specification (MPS) shall be established by the manufacturer, outlining the plan to produce the part. The MPS shall address all factors that affect the quality, repeatability, and reliability of manufacturing, including subcontracted processes. Every principal production step within the scope of the manufacturer shall be addressed. The MPS shall as a minimum include:

- A descriptive title and a unique identification number for the MPS, with revision control.
- Reference to this standard.
- Manufacturer's organization name and site.
- AM technology.
- Feedstock.
- Part(s) and quality assurance test specimens to be produced, including acceptance criteria.
- Reference to applicable standards or specifications.
- Reference to the part production specification (PPS), including references to qualification records, see [\[6.2.2\]](#).
- Reference to manufacturing procedures.
- Safeguard to ensure traceability of the digital model and manufactured part, e.g. unique ID numbers.
- Design history, if applicable.
- The number of parts and quality assurance test specimens per build job and their orientation and location on the build platform. An image of the build platform shall be included. The location of support structures shall be included.
- Post processing steps, including procedures and the sequence.
- Testing or inspection requirements in addition to requirements given in this document, if required by purchaser.

6.2.1.2 When available, applicable standards and specifications for the relevant AM technology shall be referred to. Care should be shown when referring to non-AM standards and specifications, as assumptions may be made that are not relevant for the given AM technology.

Guidance note:

- The manufacturing procedure specification (MPS) is called the 'Manufacturing Plan' in ISO and ASTM standards on additive manufacturing. The MPS may be in the form of a physical document, or digitally stored. The MPS is intended for the manufacturer. It is up to the manufacturer to decide which details to share with the purchaser.
- The design history may be applicable if e.g. the manufacturer has made design changes to enable or improve the additive manufacturing process, thus establishing traceability between the requested, conventional part and the revised AM part.

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6.2.2 Part production specification (PPS)

6.2.2.1 The manufacturer shall establish a part production specification (PPS) that describes how the specific part shall be built and processed.

6.2.2.2 The PPS shall include:

- reference to the part design to be built
- reference to supporting qualification documents:
 - BPQR
 - PQR, for parts in AMC 2 or AMC 3
- feedstock information
- description of AM technology
- description of AM equipment, including relevant settings or adjustments necessary for the build process
- specification of all essential parameters
- minimum post processing details based on the qualifications and sequence.

6.3 Building the part

6.3.1 Build file preparation

6.3.1.1 The build file shall be prepared in accordance with the part production specification (PPS).

6.3.1.2 Additional test specimens for quality control may be requested by the purchaser for each build job. Additional test specimens may be built directly before or after the actual build job or included in the same build job.

6.3.1.3 Additional material shall be added to the part to allow for the dimensional and tolerance criteria to be met after post processing, e.g. heat treatment and machining.

6.3.1.4 Support structures should be added only for enabling or supporting the manufacturing of the part and shall not have any role or function for the finished part.

Guidance note:

Support structures may be used for heat dissipation or structural support during the build process. Such support structures may be manufactured with different parameters than the part itself, and the mechanical properties cannot therefore be regarded as equal to the part.

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6.3.1.5 Support structures shall be placed so that they can be removed during post processing. If they are intended not to be removed, they shall be modelled as an integral detail of the part.

6.3.1.6 The build job should be simulated, if practically feasible, to aid manufacturing and to avoid foreseeable problems during the build job.

Guidance note:

The part design, orientation and location of support structures may require iterations based on the results of the simulation.

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6.3.1.7 Simulation may be used to aid quality assurance. Simulation shall not be used as a replacement for other requirements specified in this document.

Guidance note:

For example, simulation may provide valuable information for determining the place from which a test specimen should be extracted, but not for determining whether or not testing shall be performed.

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6.3.1.8 The input used for simulation should, to the extent practically possible, be based on the parameters used in the relevant build process, and the relevant material properties. When possible, the manufacturer's own measurements of the material properties should be used.

6.3.1.9 Before finalization of the build file preparation, errors identified by the software shall be resolved and the sliced part shall be inspected. Care should be taken to avoid support structures that cannot be removed after building, e.g. in internal channels or features.

6.3.2 Build preparations

6.3.2.1 Build preparations shall be carried out according to the manufacturer's procedures, and include calibrations or verification of machine functionality, handling of feedstock and other consumables, and cleaning of equipment and tools.

6.3.2.2 Feedstock shall be handled according to the manufacturer's procedures to protect against impurities, humidity, etc., see [5.5].

6.3.2.3 The build platform or substrate material shall be properly cleaned and prepared before building, with no presence of oils or impurities at the surface, and with relevant surface properties, e.g. flatness and roughness. Chlorinated solvents or methyl alcohol shall not be used on stainless steels, nickel alloys, titanium, or titanium alloys.

6.3.2.4 The build chamber, if used, shall be free from impurities, e.g. in the case of previous build jobs with different feedstock.

6.3.2.5 If powder is used as feedstock, all tools and equipment used when handling powder shall not cause contamination or cross contamination. Dedicated tools and equipment for the material handled should be used.

6.3.3 Build requirements

6.3.3.1 The manufacturing shall be performed according to a qualified build process. The information in the relevant PPS shall be available to the operator(s). The PPS shall be presented in a language easily understood by the relevant personnel.

6.3.3.2 All essential parameters recorded during a build job shall be checked to ensure they are within the qualified range of the build process.

6.3.3.3 Procedures shall be in place describing actions during planned or unplanned process interruptions.

6.3.3.4 Any external intervention, or a process interruption not stated in the MPS, shall be recorded and communicated to the purchaser, along with the corresponding action performed.

Guidance note:

Planned process stops may be interpass cooling, loading of feedstock, etc. Unplanned process stops may be power outages, process errors, etc.

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6.3.3.5 Given parameters, defined for each AM technology in the relevant section for the specific AM technology, shall be logged and stored. Logged parameters shall be stored in a log-file and available for review after the build job has finished, and any deviation from the qualified values shall be stated.

6.3.3.6 The building shall be performed in an area protected from environmental impact, with systems for measuring the ambient temperature and humidity.

6.3.3.7 Any change, maintenance or repair of hardware or software that may affect the build process should be logged. The change, maintenance or repair of hardware or software mid-build shall lead to an assessment of the part and the purchaser shall be notified of the incident and checks performed to ensure the quality of the part. Corrective actions and continued production are subject to purchaser acceptance.

Guidance note:

Logging is also relevant when a piece of hardware is replaced with a new one of the same type, e.g. a recoater for PBF, or nozzle for DED.

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6.3.3.8 If errors and/or warnings are shown in the log file, the manufacturer shall evaluate if the error/warning will impact the quality of the part. The reason for the error/warning message shall be stated to the purchaser prior to delivery, along with the assessment made.

6.3.3.9 Data collected from monitoring systems may be stored in raw format or processed to reduce the data to a more useful form.

Guidance note:

Storing raw data relating to e.g. melt pool monitoring may require substantial storage capacity. Processing this data to create values such as the melt pool temperature, area or depth may be more efficient.

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6.4 Post processing

6.4.1 General

6.4.1.1 All post processing steps, including the removal of supports, shall be performed as specified in the PPS, in accordance with the manufacturer's procedures. Such steps include machining and thermal post processing, e.g. stress relief, annealing and hot isostatic pressing (HIP).

6.4.1.2 To evaluate whether stress relief is required, the following should be considered:

- alloy
- desired properties of the final part
- residual stresses predicted by analyses or prior experience
- design/geometry
- process parameters.

6.4.1.3 Stress relief may be performed separately or integrated with other thermal post processing procedures.

Guidance note:

It is recommended to carefully assess the need for stress relief for alloys with a high yield strength/ultimate tensile strength ratio (YS/UTS) and low elongation. Depending on the constraints during building, these alloys may have residual stresses closer to the UTS and a higher risk of premature failure.

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6.4.1.4 When applicable, stress relief shall be carried out within a defined time frame after the build job has finished to reduce the effect of post-build residual stress on the part. The time frame shall be defined by the manufacturer.

6.4.1.5 The part shall cool down to a temperature at which it can be handled safely before removal of powder, removal of clamping, cutting from substrate or similar handling.

6.4.1.6 All support structures should be removed from the part prior to delivery. If not, the support structure locations shall be clearly communicated in the delivery documents. Inert grit shall be used when applicable.

Guidance note:

When removing support structures, caution should be taken not to introduce defects through e.g. material removal, cold deformation or embedded particles (grit).

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6.4.2 Hot isostatic pressing

6.4.2.1 When required or requested, hot isostatic pressing (HIP) shall be performed according to a qualified procedure for the specific AM technology and material.

6.4.2.2 For parts that are designed for fatigue loading, post processing that includes HIP may be advantageous and should be considered.

6.4.2.3 HIP may be used as an argument for a reduction in the frequency of inspection, see [6.2.2]. Such a reduction shall be made in agreement with the purchaser.

Guidance note:

Hot isostatic pressing (HIP) is a process in which high pressure is applied to a part at elevated temperatures. The process may assist with the removal or reduction of internal porosity and internal cracks, and may therefore provide benefits for e.g. the fatigue properties and reduce the spread of mechanical results. HIP is applicable for all metallic materials.

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6.4.2.4 When HIP is used, it shall be performed prior to final machining.

Guidance note:

The HIP process may close or reduce pores and cracks that are not open to the surface. Final machining should therefore not be performed prior to HIP.

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6.5 Repair

Repair of a built part shall be agreed with the purchaser on a case-by-case basis. In such cases, repair shall be performed in accordance with qualified procedures and by qualified personnel.

A repair is considered an alteration of the part through re-heating or adding material using the same or another process. Grinding within the dimensional tolerances of the part and additional surface finishing is not considered a repair. Possible repair methods and corrective actions should be defined in the MPS.

6.6 Quality control and delivery

6.6.1 Production testing

6.6.1.1 Production testing refers to the testing of a specimen or part that is being produced at the same time, or during the same build job or production run, as the part to be delivered. The production test specimen shall undergo the same process steps as the part it represents. The purpose of production testing is to ensure the continued performance of the build process. The results obtained in the qualification activities described in [Sec.4](#) provide a basis for what should be expected from the production testing.

6.6.1.2 Additional requirements for production testing given in other applicable standards or in the purchasing documents shall be adhered to.

6.6.1.3 Rejected parts shall be quarantined and reported to the purchaser. If continuous rejection occurs, the cause of rejection shall be investigated and corrected. If rejection is caused by unsuitable build parameters or geometrical features, the build process and geometry shall be assessed, and relevant qualification steps shall be repeated if necessary.

6.6.1.4 If the parts in a build job are to undergo different heat treatments, or be heat treated at different times, additional production test specimens shall be added to ensure that the required production testing is performed for all heat treatment lots. Non-conformity of the production tests may be assessed on a case-by-case basis in cooperation with the purchaser.

6.6.1.5 Production testing is required for all AM parts. The extent of the production testing and type of tests carried out depend on the AM technology and AMC. References to relevant sections for each AM technology can be found in [Table 6-1](#).

Table 6-1 References to the relevant section for production testing depending on the AM technology

<i>AM technology</i>	<i>Reference to relevant section in this standard</i>
Directed energy deposition using arc, DED-arc	[7.2.9]
Directed energy deposition using laser beam, DED-LB	[7.3.9]
Powder bed fusion using laser beam, PBF-LB	[7.4.9]
Powder bed fusion using electron beam, PBF-EB	[7.5.9]
Binder jet technology, BJT	[7.6.9]

6.6.2 Inspection

6.6.2.1 Inspection refers to the quality control activities performed on parts to be delivered to ensure compliance with the applicable design and product standards, or the purchaser's specifications.

6.6.2.2 General requirements for inspection and non-destructive testing (NDT) and acceptance criteria are given in [Sec.8](#).

6.6.2.3 The scope of the inspection of the first production article and of subsequently built parts is given in [Table 6-2](#). When multiple parts are made at the same time, one of the parts may be named as the first production article.

6.6.2.4 The inspection shall be performed on 100% of the part as permitted by geometry. Limitations to the inspection shall be reported to the purchaser.

6.6.2.5 The results shall be reported and added to the compliance record for the delivery.

Table 6-2 Test scope for the inspection of built parts

AMC	Test method	Reference	Testing extent
AMC 1	Dimensional	[8.3.10]	First production article
	Visual inspection	[8.3.4]	All subsequently built parts
AMC 2	Dimensional	[8.3.10]	First production article
	Visual inspection	[8.3.4]	All subsequently built parts
	Surface NDT	[8.3.5]	First production article
	Volumetric NDT	[8.3.6]	Subsequent parts to be tested according to an agreed test frequency ¹
AMC 3	Dimensional	[8.3.10]	First production article
	Visual inspection	[8.3.4]	All subsequently built parts ²
	Surface NDT	[8.3.5]	
	Volumetric NDT	[8.3.6]	

1) The test frequency of the surface and volumetric inspection of AMC 2 parts is subject to agreement between the purchaser and manufacturer. A recommended frequency is minimum one article per build job.

2) Unless otherwise agreed with the purchaser.

6.6.3 Documentation, marking and delivery

6.6.3.1 The following documentation, as a minimum, shall be included in the compliance record for the delivery of a part:

- reference to the relevant purchase order
- supplier reference number
- the manufacturing technology used
- confirmation that a qualified build process, including post processing, has been used
- reference to applicable documentation, e.g. MPS
- release document, including documentation of final testing/inspection with referenced testing/inspection details
- any unplanned process interruptions or activities, if relevant
- feedstock material certificate or certificate of conformity, including state of powder if applicable, i.e. virgin, reused or blended
- results from tests carried out, including mechanical testing, dimension control and NDT. The location and orientation of the specimens shall be stated
- the acceptance criteria for the tests carried out, and a clear statement of whether the acceptance criteria have been met
- any other part- or application-specific tests performed, including results and acceptance criteria
- risk assessment from part family definition if a pre-existing part qualification was used
- design specification if the part was designed by the manufacturer on behalf of the purchaser.

6.6.3.2 When the manufacturer and purchaser are within the same company, internal documentation is acceptable as long as the required information is available.

6.6.3.3 The part(s) to be delivered shall be packaged in accordance with the manufacturer's procedure, or according to ASTM D3951, unless a specific requirement was requested by the purchaser. The marking of the part(s) and the delivery shall be as agreed.

6.6.3.4 When powder is used as feedstock, the packaging of the part(s) should contain a warning that loose powder may be present in the package.

SECTION 7 TECHNOLOGY-SPECIFIC QUALIFICATION AND PRODUCTION REQUIREMENTS

7.1 General

In this section specific requirements for qualification and production are presented for each AM technology. These requirements shall apply in addition to the general qualification and production requirements presented in [Sec.4](#) and [Sec.6](#).

The various AM technologies are differentiated from each other by the type of feedstock and the heat source or bonding mechanism used. The two (2) feedstock types considered in this standard are metallic powder and metal wires. The different heat sources are electric arc, laser beam and electron beam. Binder Jetting technology utilizes a binder for building the part and a sintering step to ensure metallic bonding. Further information for each AM technology is given in the relevant subsection.

The AM technologies covered in this section are listed below:

- [\[7.2\]](#) (DED-arc)
- [\[7.3\]](#) (DED-LB)
- [\[7.4\]](#) (PBF-LB)
- [\[7.5\]](#) (PBF-EB)
- [\[7.6\]](#) (BJT).

In general, orientations and directions shall be defined as described in ISO/ASTM 52921. For some systems, especially DED technologies, where the orientation of both the tool head and substrate base can be adjusted during the build process, the substrate base is a variable reference. The terminology described below and shown in [Figure 7-1](#) is therefore used for describing sampling for DED technologies:

- P-BD - parallel to build direction: the specimen shall be oriented normal to the build plane of the relevant layer.
- P-DP - parallel to deposition path: the specimen shall be oriented parallel to the deposition path at the relevant layer.
- N-DP - normal to deposition path: the specimen shall be oriented normal to the deposition path at the relevant layer and within the horizontal plane compared to the build direction.

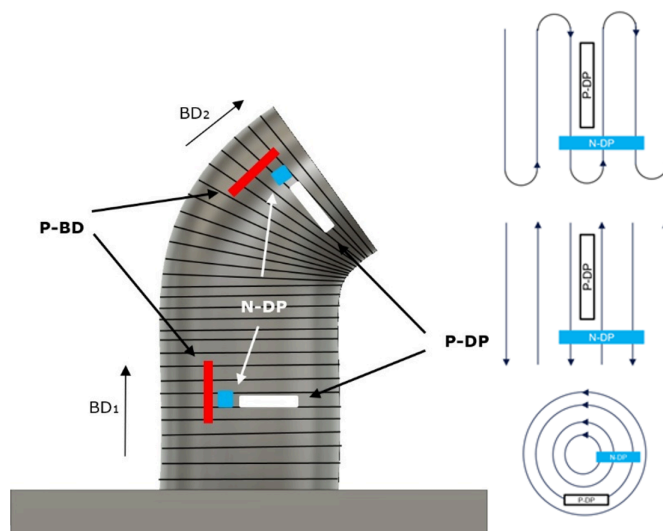


Figure 7-1 Example of how orientations of test specimens change as the build direction changes from BD1 to BD2. The blue specimens on the left figure are oriented inwards in the figure.

7.2 Directed energy deposition - arc

7.2.1 General

7.2.1.1 DED-arc technology falls under the category of directed energy deposition (DED). For a more in-depth description of the DED processes, see ASTM F3187.

7.2.1.2 DED-arc distinguishes itself from other DED systems by using an electric arc as a heat source and a metal wire as feedstock. The electric arc melts the wire, and the positioner deposits the metal according to a strategy defined by the computer control system.

7.2.1.3 The wire feedstock used for DED may be conventional welding wires, or wires specifically designed for AM purposes. Feedstock other than metal wire, e.g. metal powder, has not been considered in this section. If other types of feedstock are used, then requirements or essential parameters for this shall be agreed.

7.2.2 Build process qualification

7.2.2.1 All parts shall be built using a qualified build process. The build process qualification is performed through the building and thermal post processing of a standardized qualification build. If the build process is intended to be used for adding material or features to a substrate, the transition is included in the qualification testing scope. If additional tests are specified by the purchaser, then these shall be included, see [2.3] and [4.2.1.4].

7.2.2.2 All test specimens shall be produced with a defined set of essential parameters, defined in [7.2.4]. The material used for the extraction of samples shall be built as a thin coupon or a thick coupon. Qualification should be performed on both a thin coupon and a thick coupon. Depending on the deposition strategy, different geometries may be relevant, e.g. cylindrical or rectangular coupons.

7.2.2.3 The thick coupon is intended to represent a volume part, where a change in the build thickness has little or no effect on the material properties. The build thickness of the thick coupon shall be minimum 15 mm. The recommended width of the coupon is minimum 6 times the bead-width. The length of the coupon shall be at least equal to the width. Qualification using this coupon will also be valid for larger build thicknesses.

7.2.2.4 The thin coupon is indented to represent thin sections where a change in the build thickness may influence the material properties of the part. The build thickness used in the qualification determines the qualified range. The qualification will be valid for build thicknesses greater than the qualified build thickness.

7.2.2.5 For qualification of the deposited material, i.e. not considering any transitions to the substrate, the test specimens shall not be extracted from the first three (3) layers, due to chemical dilution and higher heat transfer. The test scope for testing thick and thin coupons is described in Table 7-1, and an example qualification build is shown in Figure 7-2.

Table 7-1 Test scope for build process qualification, given for each coupon

<i>Test method</i>	<i>Location (quantity)</i>	<i>Orientation, see [7.1] (quantity)</i>	<i>Testing according to</i>
Tensile testing	Anywhere (6) ¹⁾	P-BD (3), P-DP (3) ¹⁾	[8.2.2]
Impact testing ²⁾	Anywhere (1 set of 3 samples) ¹⁾	Notch axis oriented in P-BD (1 set) ¹⁾	[8.2.3]
Hardness measurements	Anywhere	The same orientation in the macrostructural assessment.	[8.2.4]
Macrostructural assessment	Anywhere	The surface normal for the investigated section in P-DP.	[8.2.5]
Microstructural assessment	T/4 and T/2 of both P-DP and N-DP, and a distance H/4 from top of build (4)	See [8.2.5]	[8.2.6]
Chemical analysis	Anywhere	Any	[8.2.8]
Volumetric NDT	100%	See [8.3.6]	[8.3.6]
1) If the build strategy is to align all layers in the same direction, i.e. parallel hatching, then an additional three (3) tensile specimens and one set of impact specimens shall be added to the thick coupon, if allowed by the geometry. The orientation shall be N-DP. 2) When the BPQ is done for both a thin coupon and thick coupon, impact testing may be waived for the thin coupon.			

7.2.2.6 If the build process is intended to be used where a start/stop point may be located inside a part, then a start/stop point should be included in the macrostructural assessment.

Guidance note:

Although avoiding start/stop points inside a part is good practice, unintentional process stops may cause a start/stop point to be located inside a part.

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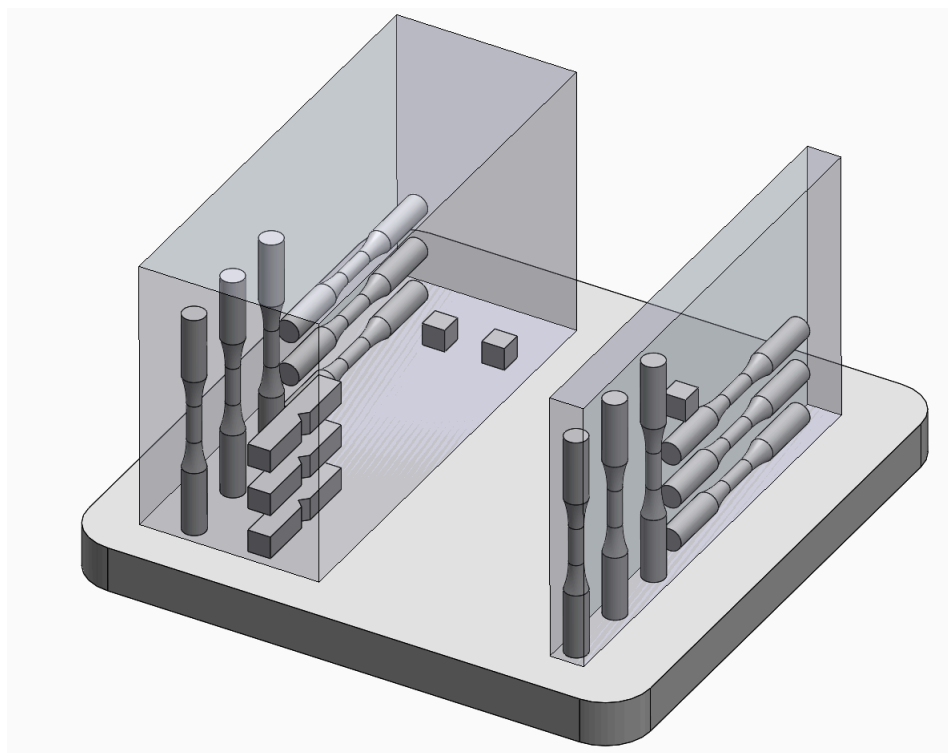


Figure 7-2 A schematic representation of a thick and thin coupon for build process qualification

7.2.2.7 When the substrate is included in the final part as shown in [Figure 7-3](#), or when the part is made using multiple AM processes, then the first layers and fusion boundary between the substrate and AM feature shall be qualified with an additional set of testing included in the BPQT. The qualification of the transition zone shall consider the substrate material, the fusion boundary and the first three (3) layers or minimum 5 mm. The test scope for qualification of a fusion boundary is described in [Table 7-2](#). If additional requirements for testing and qualification are specified by the purchaser or the design standard, then these shall apply. A qualification including the fusion boundary is illustrated in [Figure 7-4](#).

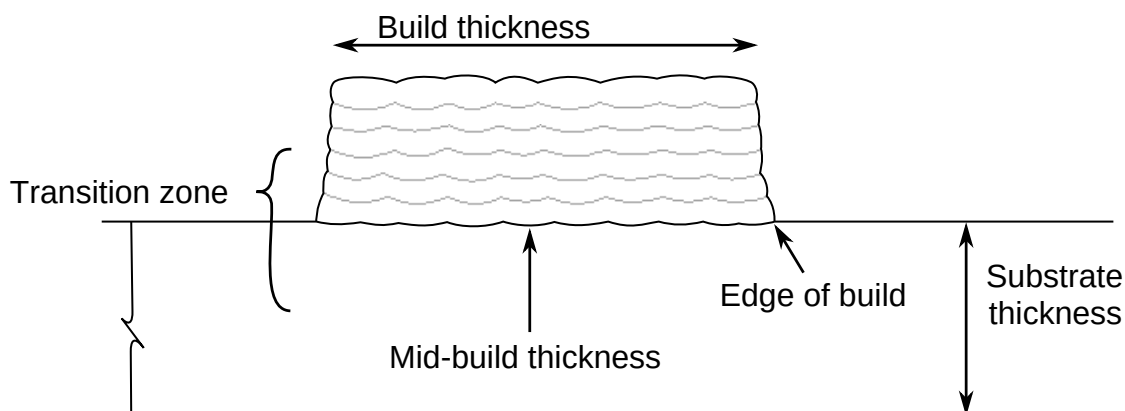


Figure 7-3 Schematic representation of the transition zone, deposited material, and substrate thickness

7.2.2.8 In cases where the transition between the substrate and feature is made using multiple steps with different procedures, each procedure shall be qualified with the deposit from the previous step used as the substrate. Previously deposited material, when used as a substrate, shall be considered unlisted base material. Qualification testing of multiple build steps may be performed separately or on the same part or coupon.

7.2.2.9 For weld overlay or hard facing where the feature is entirely enclosed in the first three (3) layers or within 5 mm thickness, whichever is greater, procedure qualification shall be in accordance with relevant internationally recognized standards, such as ISO 15614-7.

Table 7-2 Build process qualification for the fusion boundary between the AM feature and the substrate, given for each coupon

<i>Test method</i>	<i>Position and orientation</i>	<i>Reference</i>
Tensile testing ¹⁾	Two (2) transverse tensile tests that include the first three (3) layers of the deposited material and the heat affected zone.	[8.2.2]
Impact testing ¹⁾	Three (3) sets of three (3) specimens in the fusion zone. Notch sampling in the deposited material, fusion line and HAZ-2.	[8.2.3]
Hardness measurements	One (1) survey	[8.2.4]
Macrostructural assessment	Assessment of the macrostructure, typically in the same location as the hardness testing. Magnification 3x-5x. Includes start/stop location if relevant.	[8.2.5]

Test method	Position and orientation	Reference
Microstructural assessment	Assessment of built material and HAZ in two (2) locations: mid build-thickness and below the edge of the build, see Figure 7-3 . 100X-500X magnification.	[8.2.6]
Surface NDT	100% of the deposit and 25 mm on all sides of the boundary.	[8.3.5]
Volumetric NDT	100% of the volume and transition.	[8.3.6]
1) If the BPQ covers both the thin and thick coupon, the test specimens shall be taken from the thick coupon only. Impact testing shall be performed on the thin and thick coupons for C-Mn and low alloy steel.		

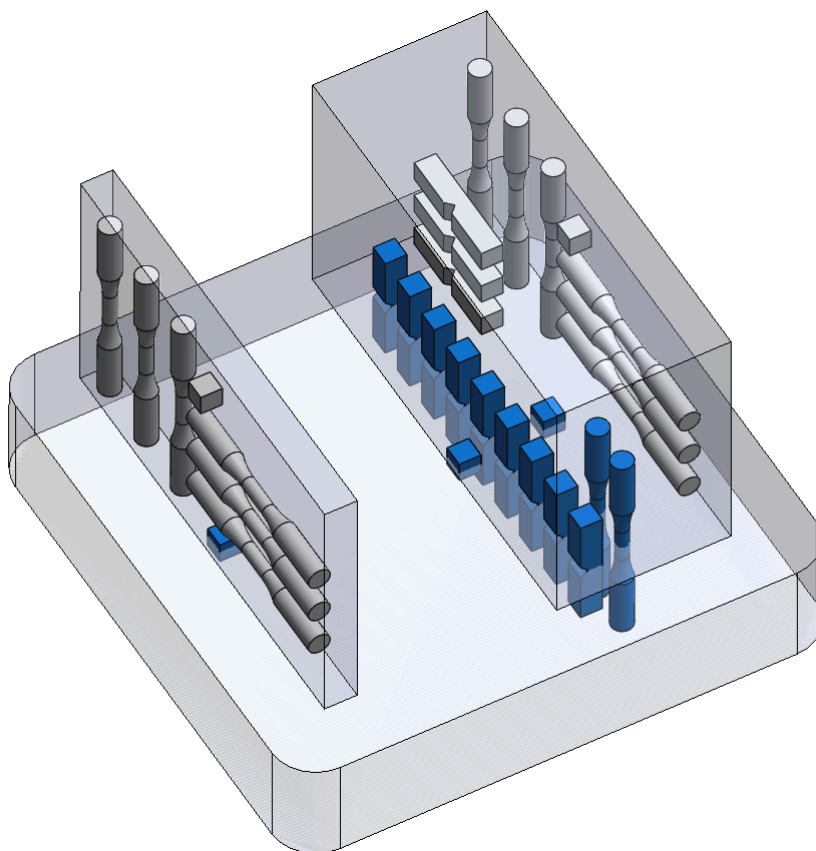


Figure 7-4 Schematic representation of the test specimens required in a build process qualification including the transition. Blue specimens indicate additional specimens required when including the transition zone. Tensile and impact tests have been waived for the thin coupon in the example as described in note 1 of [Table 7-2](#).

7.2.3 Part qualification

7.2.3.1 AMC 2 and AMC 3 parts shall undergo part qualification testing. For AMC 3 parts, the required test specimens for part qualification testing shall be extracted from the critical section(s).

Guidance note:

For large AMC 3 parts, it may not be economically viable to sacrifice an entire part. In such cases, representative testing may be performed by building the most critical sections using a representative thermal history of the full part.

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7.2.3.2 The part qualification is by default part-specific and linked to a specific job file, so that changes to the job file or other essential parameters may require a new qualification. In addition to the essential parameters given in [7.2.4], the part qualification is limited by the geometrical bounds of the part or part family. Guidance on the definition of part families is provided in App.C. The use of a prior part qualification for a new part shall be supported with documentation in accordance with [4.3.2].

7.2.3.3 The required tests to be performed are given in Table 7-3 and Table 7-4. If the geometry does not permit the extraction of the required test specimens, then geometrically representative prolongations or test blocks may, upon agreement with purchaser, be added to the build for testing.

7.2.3.4 Production testing shall be carried out on the part produced for part qualification. When part qualification is carried out on representative coupons or a prolongation to avoid sacrificing the first part, the actual part will be defined as the first production article, and be subject to inspection and production testing as described in [7.2.9]. Production testing shall also be included when the part is sacrificed in the part qualification. In such case, the additional prolongation or coupon shall be included as it will be in production and tested according to [7.2.9] to establish a reference for the future production testing of production parts.

Guidance note:

The inclusion of production testing during part qualification is intended to ensure that the properties achieved during production testing are representative of the part being manufactured.

- Example: if a smaller test coupon is intended to be built to check the production quality of a bigger part, it shall be ensured that the test coupon will represent the bigger part. The test coupon shall therefore be built and tested during part qualification. The test coupon may be used for production testing only if its mechanical properties are comparable to those of the part it represents.

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7.2.3.5 If the transition zone between a built DED feature and a substrate is considered a critical section, then the part qualification test scope shall be as specified for the build qualification shown in Table 7-2.

7.2.3.6 Additional testing may be required based on the part's intended application, see [2.3].

Table 7-3 Test scope for non-destructive part qualification - relevant for AMC 2 and AMC 3 parts

<i>Test method</i>	<i>Scope</i>	<i>Reference</i>
Visual inspection	100% as permitted by geometry	[8.3.4]
Dimensional check	As per purchaser requirement	[8.3.10]
Surface NDT	100% as permitted by geometry	[8.3.5]
Volumetric NDT	100% as permitted by geometry	[8.3.6]

Table 7-4 Test scope for mechanical and metallurgical part qualification - relevant for AMC 3 parts

Test method	Orientation, see [7.1] (quantity)	Specimen extraction	Reference
Tensile testing	P-BD (1), P-DP (1)	Mid thickness of the critical section(s) ¹⁾	[8.2.2]
Impact testing	Notch axis oriented in P-BD (1 set of three (3) specimens) ¹⁾	Mid thickness of the critical section(s) ¹⁾	[8.2.3]
Hardness measurements	As for macrostructural assessment	Through thickness of the critical section(s) ¹⁾	[8.2.4]
Macrostructural assessment	See [8.2.5]	Through thickness of the critical section(s) ¹⁾	[8.2.5]
Microstructural assessment	See [8.2.5]	T/4 and mid thickness of critical section(s) ¹⁾	[8.2.6]
1) See [2.3.2] for a definition of the critical section(s).			

7.2.4 Validity of qualifications

7.2.4.1 The PPS shall be supported by qualification testing as described in [7.2.2] and [7.2.3] and violation of the essential parameters defined in Table 7-5 and Table 7-6 shall be assessed in the following ways:

- Conventional: the essential parameters, given in Table 7-5 and Table 7-6, shall be specified. Upon successful part qualification testing, the qualification is valid within the limits of the essential parameters.
- Advanced monitoring system: the essential parameters given in Table 7-5, along with the unique job file, shall be defined. The job file shall contain definite information on the parameter values used during the different phases of the build job.

Guidance note:

The essential parameters in this section are established on a general basis for building with solid metal wire, and metal core wire, with gas shielding. When assessing whether other parameters or ranges should be used, then relevant standards describing similar welding processes may be evaluated, e.g. ASME BPVC.IX or relevant parts of ISO 15614.

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7.2.4.2 The essential parameters for DED-arc are divided into two (2) groups and given in respective tables:

- Process independent - Table 7-5, parameters related to machine/equipment, feedstock and post processing.
- Process dependent - Table 7-6, parameters related to the actual building of the part, e.g. voltage and speed.

7.2.4.3 The essential parameters shall be defined and included in the PPS. All parameters considered for a given setup shall be defined and included in the PPS.

7.2.4.4 For part qualifications, relevant for AMC 2 parts and AMC 3 parts, a change of part, or violation of the bounds defined for the specific part family, shall lead to a new part qualification. Part families are described in [4.3.2].

Table 7-5 Process independent essential parameters for DED-arc

Category	Parameter id	Essential parameters	Limits of qualification
1. Hardware and software description	1-1	Welding process	Any change of welding process.
	1-2	Wire system	Any change between single wire and multi wire.
	1-3	Power source	Any change in make, type and model ¹⁾ .
	1-4	Positioner	Any change in make, type and model ¹⁾ .
	1-5	Feedstock feed mechanism	Any change in make, type and model ¹⁾ .
	1-6	Computer control system	Any change in make, type and model ¹⁾ .
	1-7	Atmosphere control	Any change between local shielding, atmosphere-controlled chamber, vacuum chamber.
	1-8	Software and firmware	Change of software or firmware maker or product name, or major updates, see [5.6.1.5].
2. Materials and consumables	2-1	Substrate, when not included in the part	Change in the substrate material's thermal properties.
	2-2	Substrate, when included in the part	- For a substrate made by AM, change to the PPS used for making the substrate outside the ranges of the relevant BPQR and PQR. - For carbon and low-alloy steel, increase of grade. - Delivery condition of the base material, i.e change between forged or rolled (QT, normalized) and TMCP and casting. - Change in UNS for materials other than carbon and low alloy steel ²⁾.
	2-3	Wire diameter	Change of wire diameter outside the tolerance used in the qualification.
	2-4	Wire classification or trade name	Any change of type, classification and brand (brand not applicable for solid wire).
	2-5	Process gases (e.g. shielding gas, plasma gas)	Change of shielding gas group or reduction of purity level.
3. Post processing	3-1	Thermal post processing	Any changes in sequence and procedures.
1) Equivalent or improved equipment or software may be acceptable if validation tests are performed, see [5.6.1.5] 2) Similar UNS numbers may be validated by agreement and additional assessment.			

Table 7-6 Process dependent essential parameters for DED-arc

Category	Parameter id	Essential parameters	Limits of qualification
4. Process parameters	4-1	Preheating	Change of preheating method, and any reduction from the preheat temperature used during qualification.
	4-2	Interpass temperature	Any increase from the maximum interpass temperature used during qualification.

Category	Parameter id	Essential parameters	Limits of qualification
	4-3	Method for interpass cleaning	Any change resulting in poorer cleaning between layers.
	4-4	Forced cooling	Any change from forced cooling to natural cooling, or vice versa.
	4-5	Build thickness ¹⁾	Reduction of build thickness below the minimum build thickness used in the qualification.
	4-6	Deposition strategy ²⁾	Any change in deposition strategy.
	4-7	Degree of overlap, if multiple adjoining beads	Any change in percentage of overlapping beads.
	4-8	Build position, according to ISO 6947 or similar	Any change in build position.
	4-9	Wire feed rate	Change exceeding the programmed range of feeding rate used in the qualification.
	4-10	Nozzle to substrate distance, relevant for first layer	Any change in the distance between the gas shielding nozzle and the substrate, for the first layer.
	4-11	Nozzle to deposited material distance	Any change in the distance between the gas shielding nozzle and the deposited material.
	4-12	Hot or cold wire, if applicable	A change from hot to cold wire, or vice versa.
	4-13	Heat input/arc energy per ISO/TR 18491 or ISO 15614 level 1	Change exceeding the programmed range of heat input used in the qualification.
	4-14	Power supply	Any change in type of current and a change from normal to pulsed current or vice versa.
	4-15	Polarity	Any change in polarity.
	4-16	String/weave	A change from stringer to weave or vice versa.
	4-17	Shielding gas flow rate	Any change exceeding $\pm 10\%$ in volume per minute.
	4-18	Shielding gas nozzle dimensions, if applicable	Any change in size.
	4-19	Atmospheric oxygen content, if applicable ³⁾	Any increase compared to the oxygen content used during qualification.
1) If the BPQ was carried out with thin coupon geometry only, the qualification is limited to a maximum of 15 mm. If only a thick coupon was used, the minimum build thickness may be considered 15 mm. 2) The deposition strategy should be described, preferably with a sketch showing the tool path for a representative layer. The sketch shall show if the tool path is open-looped or close-looped and highlight the location of the start and stop areas and any sudden changes of direction. Several such descriptions may be necessary as the geometry changes with each layer. 3) Relevant for oxygen corrosion resistant alloys (CRAs), such as duplex stainless steel or titanium alloys.			

7.2.5 Build file preparation

7.2.5.1 The relevant CAD file shall be rendered for DED before path planning. Unless otherwise specified, 3 mm additional material should be added to all surfaces that will be machined.

7.2.5.2 The following aspects shall be evaluated in the design and path planning:

- Start-stop areas shall, to the extent possible, not be located at a critical section of the part, as defined by the purchaser.
- The build shall be planned to avoid collision between the part and the equipment during production.
- The axis of movement available in the specified equipment.
- If a significant amount of residual stress is expected in the part, even after stress relief, remedial actions to reduce the residual stress level should be considered.

7.2.6 Pre-build preparation

7.2.6.1 Calibration of positioning systems and process monitoring systems shall be performed according to the manufacturer's documented procedures.

7.2.6.2 Current contact tubes in welding torches and consumable feeding nozzles shall be regularly inspected for wear damage and replaced at documented intervals. Multiple current contact tube replacements may be needed within the same build process, depending on the build volume.

7.2.6.3 It should be assessed how to use clamps or fixtures on the substrate during building to manage distortions and residual stress. The clamping of the substrate should be described in the MPS.

Guidance note:

- The forces generated during build-up may be significant, causing the substrate to move.
- Clamping may induce or inhibit residual stresses.

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7.2.6.4 A mock-up run, or a dry run of the positioner, should be performed before building the intended part. The mock-up will indicate whether the height, width, overlap and start-stop areas are as intended.

Guidance note:

Small differences in height in the first layers may result in bigger deviations as the number of layers increases.

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7.2.7 Build requirements

7.2.7.1 It should be ensured that each layer is free from impurities and imperfections. Corrective actions such as cleaning, grinding or machining may be performed according to a qualified procedure.

7.2.7.2 When a build process is paused due to interruptions, the top layer shall be visually examined before continuing the build. The layer number at which the build was paused shall be logged. This requirement is not relevant for planned interruptions as long as the part is not cooled down below the minimum preheat temperature. If the temperature falls below the preheat temperature during the interruption, the build shall be re-heated above the preheat temperature before continuing. The preheat temperature shall in this case be confirmed in the top 50 mm of the build and measured in a way that ensures even heating.

Guidance note:

Different materials will react differently to interruptions to the preheating. Some materials may be exposed to hydrogen cracking in cases of high residual stress if allowed to cool over longer periods of time. Therefore, it should be assessed whether cooling below the preheat temperature during interruptions could be damaging to the build.

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7.2.7.3 If, a new feedstock batch is used in the middle of a build job, the effect of the batch change shall be evaluated. A procedure for evaluating the effect of batch changes shall be documented. The material deposited from the new feedstock batch should show no indication of a feedstock change.

7.2.7.4 As a minimum, the following parameters shall be logged during the building:

- arc voltage, current and travel speed
- wire feedrate
- gas flowrate
- oxygen content in atmosphere, if built in a closed chamber
- temperature in atmosphere, if built in a closed chamber
- Dew point, if built in a closed chamber.

7.2.7.5 The preheat and interpass temperatures shall be monitored in accordance with ISO 13916.

7.2.7.6 For oxygen-sensitive materials, measures shall be in place to control and monitor the oxygen level.

7.2.7.7 If the DED machine is equipped with systems for monitoring the process temperature, temperature logs shall be recorded.

7.2.7.8 Information from the build shall be summarized in a build report, and shall as a minimum include:

- a reference to the unique build job ID
- the unique ID of the applied parameter set
- the idle time between layers
- error messages or warnings.

7.2.8 Post processing requirements

7.2.8.1 When removing a part from the substrate after building, the first three (3) layers shall be cut from the part. Any deviation from this requirement should be compensated by additional testing of the first layers.

7.2.8.2 In addition to the general description of post processing in [6.4], heat treatment practices shall comply with the following requirements:

- 1) Heat treatment furnaces and equipment shall be calibrated according to ISO 10423/API 6A, SAE AMS-H-6875, ASTM A991 or equivalent.
- 2) Furnaces shall be equipped with automatic temperature indicating, controlling, and recording devices, which shall possess an accuracy of $\pm 1\%$ of their full-scale range.
- 3) Pallets or fixtures shall be used for placing parts during heat treatment and quenching to ensure the free circulation of air/atmosphere and quenching medium between each part. Parts shall not be placed directly onto the furnace floor. The piling of parts in baskets is not allowed.
- 4) For alloys requiring water quenching, the furnace unloading equipment shall be capable of transferring materials from the furnace into the quench tank within a specified maximum time, e.g. 60 s for duplex stainless steels and 90 s for low alloy steels.
- 5) For alloys requiring water quenching, quench tanks shall have dimensions and cooling facilities adequate to keep the water temperature below 50°C at any time during quenching. Quench tanks shall be

equipped with pumps or propellers to ensure satisfactory circulation and agitation during quenching. Quench tanks shall be equipped with continuous temperature monitoring/recording devices.

- 6) Heat treatment shall be controlled and documented through formal procedures.

Guidance note:

The requirement that the first three (3) layers must be removed due to diffusion and heat sink effects between the substrate and the part.

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7.2.9 Production testing

7.2.9.1 Production testing is required for all parts, with the scope and frequency described in [Table 7-7](#) and [Table 7-8](#). For parts intended for serial production, the frequency shall be agreed between the purchaser and manufacturer.

7.2.9.2 Production test specimens shall be extracted from a test coupon, a prolongation of the part, or a sacrificial part, as relevant. To avoid sacrificing a part for production testing, production testing coupons may be included in the part qualification, see [\[7.2.3.3\]](#).

7.2.9.3 If a prolongation or test coupon is used, it shall represent the thickness and deposition strategy of the critical section(s) of the part and its thermal behaviour shall be representative of the part. A test coupon shall accompany the part it represents through all thermal post processing steps.

Guidance note:

In this context representative thermal behaviour means that the volume will have a similar thermal cycle, e.g. similar peak temperature and heating and cooling rates locally during the build and thermal post processing.

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7.2.9.4 Additional testing may be required based on the part's intended application, see [\[2.3\]](#).

Table 7-7 Production testing scope

Additive manufacturing category	Test method (quantity)	Orientation and location
AMC 1	Tensile testing (1)	Samples and sections shall be in one of the orientations defined for BPQ or PQT, as relevant. ¹⁾
	Macrostructural assessment	
AMC 2 and AMC 3 ²⁾	Tensile testing (1)	
	Impact testing (1 set)	
	Hardness measurements	
	Macrostructural assessment	
	Chemical analysis	

1) Production testing shall be performed and oriented in the same manner as the production test specimens were included in the part qualification.

2) Additional testing may be specified by the purchaser, see [2.3].

Table 7-8 Frequency of production testing

<i>Additive manufacturing category</i>	<i>Frequency of testing</i>
AMC1 and AMC 2	As agreed with the purchaser.
AMC 3	Every build job.

7.3 Directed energy deposition - laser beam

7.3.1 General

7.3.1.1 DED-LB technology falls under the category of directed energy deposition (DED). For a more in-depth description of the DED processes, see ASTM F3187.

7.3.1.2 DED-LB distinguishes itself from other DED systems by using a laser as a heat source and wire or powder as feedstock. The laser melts the feedstock and the positioner deposits the metal according to a strategy defined by the computer control system.

7.3.1.3 The feedstock used for DED-LB may be wires or powders intended for conventional welding consumables or specifically designed for AM purposes.

7.3.2 Build process qualification

7.3.2.1 All parts shall be built using a qualified build process. The build process qualification is performed through the building and thermal post processing of a standardized qualification build. If the build process is intended to be used for adding material or features to a substrate, the transition is included in the qualification testing scope. If additional tests are specified by the purchaser then these shall be included, see [2.3] and [4.2.1.4].

7.3.2.2 All test specimens shall be produced with the set of essential variables defined in [7.3.4]. The material used for the extraction of samples shall be built as a thin coupon or a thick coupon as defined below. Qualification should be performed on both a thin coupon and a thick coupon. Depending on the deposition strategy, different geometries may be relevant, e.g. cylindrical or rectangular coupons.

7.3.2.3 The thick coupon is intended to represent a volume part, where a change in the build thickness has little or no effect on the material properties. The build thickness of the thick coupon shall be minimum 10 mm. The recommended width of the coupon is minimum six (6) times the bead-width. The length of the coupon shall be at least equal to the width. Qualification using this coupon will also be valid for larger build thicknesses.

7.3.2.4 The thin coupon is indented to represent thin sections where a change in the build thickness may influence the material properties of the part. The build thickness used in the qualification determines the qualified range. The qualification will be valid for build thicknesses greater than the qualified build thickness.

7.3.2.5 For qualification of the deposited material, i.e. not considering any transitions to the substrate, the test specimens shall not be extracted from the first three (3) layers or 5 mm, whichever is greater, due to chemical dilution and higher heat transfer. The test scope for testing DED-LB parts is described in Table 7-9 and an example qualification build is shown in Figure 7-5.

Table 7-9 Test scope for build process qualification, given for each coupon

<i>Test method</i>	<i>Location (quantity)</i>	<i>Orientation, see [7.1] (quantity)</i>	<i>Testing according to</i>
Tensile testing	Anywhere (6) ¹⁾	P-BD (3), P-DP (3) ¹⁾	[8.2.2]
Impact testing ²⁾	Anywhere (1 set of 3 samples) ¹⁾	Notch axis oriented in P-BD (1 set) ¹⁾ .	[8.2.3]
Hardness measurements	Anywhere	In the same orientation as macrostructural assessment.	[8.2.4]
Macrostructural assessment	Anywhere	The surface normal for the investigated section in P-DP.	[8.2.5]
Microstructural assessment	T/4 and T/2 of both P-DP and N-DP, and a distance H/4 from top of build (4)	See [8.2.5]	[8.2.6]
Chemical analysis	Anywhere	Any	[8.2.8]
Volumetric NDT	100%	See [8.3.6]	[8.3.6]
1) If the build strategy is to align all layers in the same direction, i.e. parallel hatching, then an additional three (3) tensile specimens and one set of impact specimens shall be added to the thick coupon, if allowed by the geometry. The orientation shall be N-DP. 2) When the BPQ is done for both a thin coupon and thick coupon, impact testing may be waived for the thin coupon.			

7.3.2.6 If the build process is intended to be used where a start/stop point may be located inside a part, then a start/stop point should be included in the macrostructural assessment.

Guidance note:

Although avoiding start/stop points inside a part is good practice, unintentional process stops may cause a start/stop point to be located inside a part.

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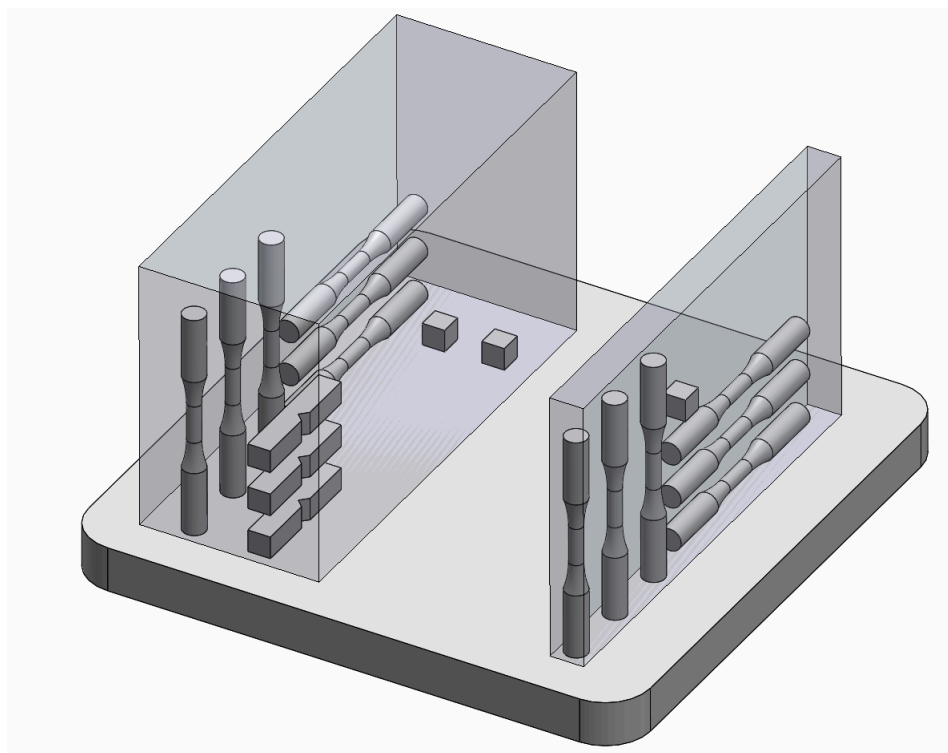


Figure 7-5 A schematic representation of a thick and thin coupon for build process qualification

7.3.2.7 When the substrate is included in the final part as shown in [Figure 7-6](#), or when the part is made using multiple AM processes, the first layers and fusion boundary between the substrate and AM feature shall be qualified using an additional set of testing included in the BPQT. The qualification of the transition zone shall consider the substrate material, the fusion boundary and the first three (3) layers or minimum 5 mm. The test scope for qualification of a fusion boundary is described in [Table 7-10](#). If additional requirements for testing and qualification are specified by the purchaser or the design standard, then these shall apply. A qualification including the fusion boundary is illustrated in [Figure 7-7](#).

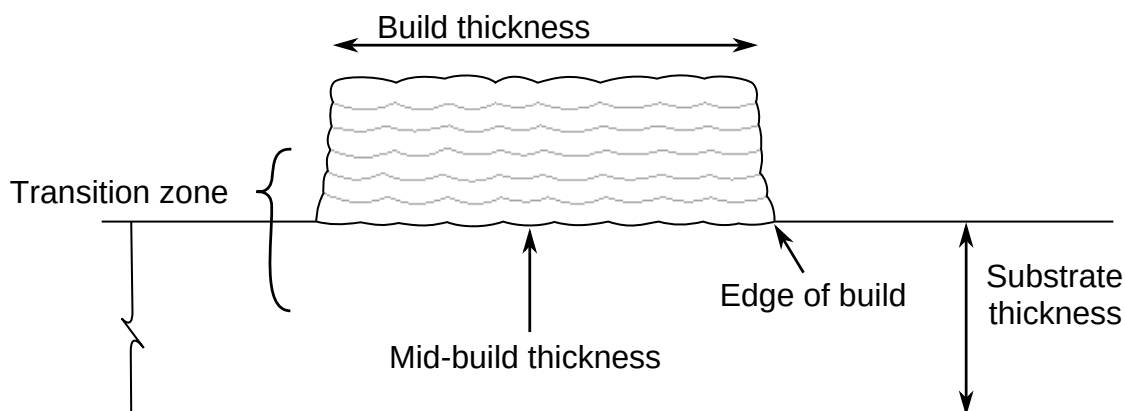


Figure 7-6 Schematic representation of the transition zone, deposited material, and substrate thickness

7.3.2.8 In cases where the transition between the substrate and feature is made using multiple steps with different procedures, each procedure shall be qualified with the deposit from the previous step used as the substrate. Previously deposited material, when used as a substrate, shall be considered unlisted base material. Qualification testing of multiple build steps may be performed separately or on the same part or coupon.

7.3.2.9 For weld overlay or hard facing, where the feature is entirely enclosed in the first three (3) layers or within 5 mm thickness, whichever is greater, procedure qualification shall be in accordance with relevant internationally recognized standards, such as ISO 15614-7.

Table 7-10 Build process qualification for the fusion boundary between the AM feature and the substrate, given for each coupon

<i>Test method</i>	<i>Position and orientation</i>	<i>Reference</i>
Tensile testing ¹⁾	Two (2) transverse tensile tests that include the first three (3) layers of the deposited material and the heat affected zone.	[8.2.2]
Impact testing ¹⁾	Three (3) sets of three (3) specimens in the fusion zone. Notch sampling in the deposited material, fusion line and HAZ-2.	[8.2.3]
Hardness measurements	One (1) survey	[8.2.4]
Macrostructural assessment	Assessment of the macrostructure, typically in the same location as the hardness testing. Magnification 3x-5x. Includes start/stop location if relevant.	[8.2.5]

Test method	Position and orientation	Reference
Microstructural assessment	Assessment of built material and HAZ in two (2) locations: mid build-thickness and below the edge of the build, see Figure 7-6 . 100X-500X magnification.	[8.2.6]
Surface NDT	100% of the deposit and 25 mm on all sides of the boundary.	[8.3.5]
Volumetric NDT	100% of the volume and transition.	[8.3.6]
1) If the BPQ covers both the thin and thick coupons, the test specimens shall be taken from the thick coupon only. Impact testing shall be performed on the thin and thick coupons, for C-Mn and low alloy steel if the minimum build thickness is equal to or greater than 6 mm.		

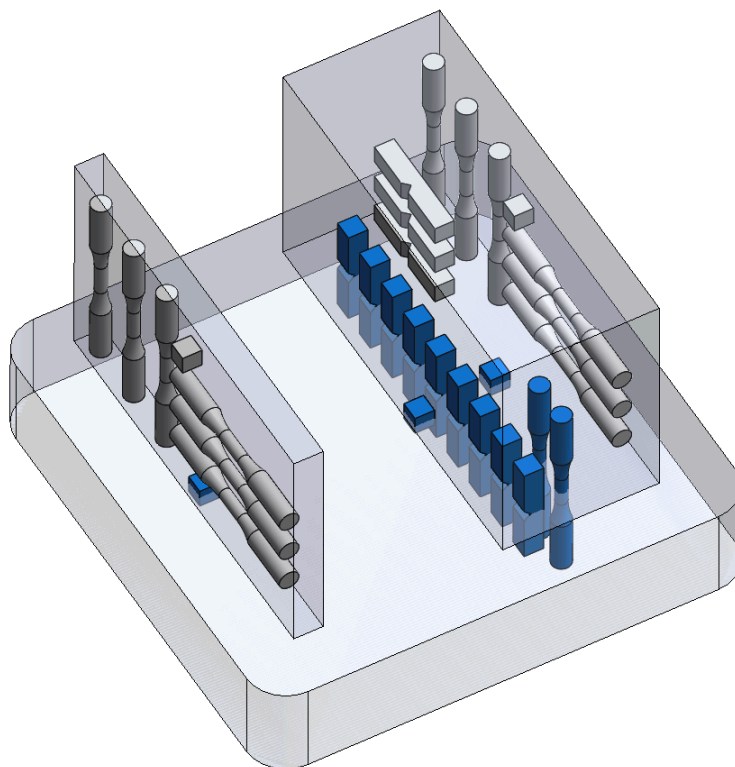


Figure 7-7 Schematic representation of the test specimens required in a build process qualification including the transition. Blue specimens indicate additional specimens required when including the transition zone. Tensile and impact tests have been waived for the thin coupon in the example as described in note 1 of [Table 7-10](#).

7.3.3 Part qualification

7.3.3.1 AMC 2 and AMC 3 parts shall undergo part qualification testing. For AMC 3, the required test specimens for part qualification testing shall be extracted from the critical section(s).

Guidance note:

For large AMC 3 parts, it may not be economically viable to sacrifice an entire part. In such cases, representative testing may be performed by building the most critical sections with a representative thermal history of the full part.

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7.3.3.2 The part qualification is by default part-specific and linked to a specific job file, so that changes to the job file or other essential parameters may require a new qualification. In addition to the essential parameters given in [7.3.4], the part qualification is limited by the geometrical bounds of the part or part family. Guidance on the definition of part families is provided in App.C. The use of a prior part qualification for a new part shall be supported with documentation in accordance with [4.3.2].

7.3.3.3 The required tests to be performed are given in Table 7-11 and Table 7-12. If the geometry does not permit the extraction of the required test specimens, then geometrically representative prolongations or test blocks may, upon agreement with purchaser, be added to the build for testing.

7.3.3.4 Production testing shall be carried out for the part produced for part qualification. When part qualification is carried out on representative coupons or a prolongation to avoid sacrificing the first part, the actual part will be defined as the first production article, and be subject to inspection and production testing as described in [7.3.9]. Production testing shall also be included when the part is sacrificed in the part qualification. In such case, the additional prolongation or coupon shall be included as it will be in production and tested according to [7.3.9] to establish a reference for the future production testing of production parts.

Guidance note:

The inclusion of production testing during part qualification is intended to ensure that the properties achieved during production testing are representative of the part being manufactured.

- Example: if a smaller test coupon is intended to be built to check the production quality of a bigger part, it shall be ensured that the test coupon will represent the bigger part. The test coupon shall therefore be built and tested during part qualification. The test coupon can be used for production testing only if its mechanical properties are comparable to those of the part it represents.

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7.3.3.5 If the transition zone between a built DED feature and a substrate is considered a critical section, then the part qualification test scope shall be as specified for the build qualification, shown in Table 7-10.

7.3.3.6 Additional testing may be required based on the part's intended application, see [2.3].

Table 7-11 Test scope for non-destructive part qualification - relevant for AMC 2 and AMC 3 parts

<i>Test method</i>	<i>Scope</i>	<i>Reference</i>
Visual inspection	100% as permitted by geometry	[8.3.4]
Dimensional check	As per purchaser requirement	[8.3.10]
Surface NDT	100% as permitted by geometry	[8.3.5]
Volumetric NDT	100% as permitted by geometry	[8.3.6]

Table 7-12 Test scope for mechanical and metallurgical part qualification - relevant for AMC 3 parts

<i>Test method</i>	<i>Orientation, see [7.1] (quantity)</i>	<i>Specimen extraction</i>	<i>Reference</i>
Tensile testing	P-BD (1), P-DP (1)	Mid thickness of the critical section(s) ¹⁾	[8.2.2]
Impact testing	Notch axis oriented in P-BD (1 set of three (3) specimens) ¹⁾	Mid thickness of the critical section(s) ¹⁾	[8.2.3]
Hardness measurements	As for macrostructural assessment	Through thickness of the critical section(s) ¹⁾	[8.2.4]
Macrostructural assessment	See [8.2.5]	Through thickness of the critical section(s) ¹⁾	[8.2.5]
Microstructural assessment	See [8.2.5]	T/4 and mid thickness of critical section(s) ¹⁾	[8.2.6]
1) See [2.3.2] for a definition of the critical section(s).			

7.3.4 Validity of qualifications

7.3.4.1 The PPS shall be supported by qualification testing as relevant for the given AMC, and violation of the essential parameters defined in Table 7-13 and Table 7-14 shall be assessed in the following ways:

- Conventional: the essential parameters, given in Table 7-13 and Table 7-14, shall be specified. Upon successful part qualification testing, the qualification is valid within the limits of the essential parameters.
- Advanced monitoring system: the essential parameters given in Table 7-13, along with the unique job file, shall be defined. The job file shall contain definite information on the parameter values used during the different phases of the build job.

7.3.4.2 The essential parameters for DED-LB are divided into two (2) groups and given in respective tables:

- Process independent - Table 7-13, parameters related to machine/equipment, feedstock and post processing.
- Process dependent - Table 7-14, parameters related to the actual building of the part, e.g. voltage and speed.

7.3.4.3 The essential parameters shall be defined and included in the PPS. All parameters considered for a given setup shall be defined and included in the PPS.

7.3.4.4 For part qualifications, relevant for AMC 2 and AMC 3 parts, a change of part, or violation of the bounds defined for the specific part family, shall lead to a new part qualification. Part families are described in [4.3.2].

Table 7-13 Process independent essential parameters for DED-LB

<i>Category</i>	<i>Parameter id</i>	<i>Essential parameters</i>	<i>Limits of qualification</i>
1. Hardware and software description	1-1	Power source	Any change in make, type and model ¹⁾ .
	1-2	Laser	Any change in make, type and model ¹⁾ .
	1-3	Positioner	Any change in make, type and model ¹⁾ .
	1-4	Feedstock feed mechanism	Any change in make, type and model ¹⁾ .

Category	Parameter id	Essential parameters	Limits of qualification
	1-5	Computer control system	Any change in make, type and model ¹⁾ .
	1-6	Atmosphere control	Any change between the local shielding, atmosphere-controlled chamber, and vacuum chamber.
	1-7	Software and firmware	Change of software or firmware maker or product name, and major updates, see [5.6.1.5].
2. Materials and consumables	2-1	Substrate, when not included in the part	Change of substrate material's thermal properties.
	2-2	Substrate, when included in the part	- For a substrate made by AM, change of the PPS used for making the substrate outside the ranges of the relevant BPQR and PQR. - For carbon and low-alloy steel, increase of grade. - Delivery condition of the base material, i.e change between forged or rolled (QT, normalized) and TMCP and casting. - Change in UNS for materials other than carbon and low alloy steel ²⁾.
	2-3	Feedstock	Any change of brand, manufacturing method, type or trade name.
	2-4	Wire diameter	Change of wire diameter.
	2-5	Powder characteristics	Change in composition, particle size distribution and rheological properties outside the tolerances given by the powder specification.
	2-6	Shielding gas and carrier gas	Change of nominal composition or reduction of purity level.
	2-7	Supplemental shielding gas	Addition or deletion of supplemental shielding and nominal composition of supplemental shielding gas.
3. Post processing	3-1	Thermal post processing	Any changes in sequence and procedures.
1) Equivalent or improved equipment or software may be acceptable if validation tests are performed, see [5.6.1.5]. 2) Similar UNS numbers may be validated by agreement and additional assessment.			

Table 7-14 Process dependent essential parameters for DED-LB

Category	Parameter id	Essential parameters	Limits of qualification
4. Process parameters	4-1	Preheating	Change of preheating method, and any reduction from the preheat temperature used during qualification.
	4-2	Interpass temperature	Any increase from the maximum interpass temperature used during qualification.
	4-3	Method for interpass cleaning	Any change resulting in poorer cleaning between layers.

Category	Parameter id	Essential parameters	Limits of qualification
	4-4	Forced cooling	Any change from forced cooling to natural cooling, or vice versa.
	4-5	Build thickness ¹⁾	Reduction of build thickness below the minimum build thickness used in the qualification.
	4-6	Deposition strategy ²⁾	Any change in deposition strategy.
	4-7	Degree of overlap, if multiple adjoining beads	Any change in percentage of overlapping beads.
	4-8	Build position, according to ISO 6947 or similar	Any change in build position.
	4-9	Powder-, or wire feed rate	Change exceeding the programmed range of feeding rate used in the qualification.
	4-10	Layer height	Change in programmed layer height.
	4-11	Nozzle to substrate distance, relevant for first layer	Any change in the nominal distance between the gas shielding nozzle and the substrate, for the first layer.
	4-12	Nozzle to deposited material distance	Any change in the nominal distance between the gas shielding nozzle and the deposited material.
	4-13	Energy density	Change exceeding the programmed range of energy density used in the qualification.
	4-14	Pulsing characteristics	Any change in pulsing characteristics.
	4-15	Laser spot size	Any change.
	4-16	String/weave	A change from stringer to weave or vice versa.
	4-17	Shielding gas flow rate	Any change exceeding $\pm 10\%$ in volume per minute.
	4-18	Shielding gas nozzle dimensions, if applicable	Any change in size.
	4-19	Atmospheric oxygen content, if applicable ³⁾	Any increase compared to oxygen content used during qualification.
1) If the BPQ was carried out with thin coupon geometry only, the qualification is limited to maximum 10 mm. If only a thick coupon was used, the minimum build thickness may be considered 10 mm. 2) The deposition strategy should be described, preferably with a sketch showing the tool path for a representative layer. The sketch shall show if the tool path is open-looped or close-looped and highlight the location of the start and stop areas and any sudden changes of direction. Several such descriptions may be necessary as the geometry changes with each layer. 3) Relevant for oxygen corrosion resistant alloys (CRAs), such as duplex stainless steel or titanium alloys.			

7.3.5 Build file preparation

7.3.5.1 The relevant CAD file shall be rendered for DED before path planning. Unless otherwise specified, 3 mm additional material should be added to all surfaces that will be machined.

7.3.5.2 The following aspects shall be evaluated in the design and path planning:

- Start-stop areas shall, to the extent possible, not be located at a critical section of the part, as defined by the purchaser.
- The build shall be planned to avoid collision between the part and the equipment during production.
- The axis of movement available in the specified equipment.
- If a significant amount of residual stress is expected in the part, even after stress relief, remedial actions to reduce the residual stress level should be considered.

7.3.6 Pre-build preparation

7.3.6.1 Calibration of positioning systems and process monitoring systems shall be performed according to the manufacturer's documented procedures.

7.3.6.2 Monitoring of the functionality of the heat source should be carried out at defined intervals, according to the machine manufacturer's recommendations. In the absence of such recommendations, a recommended monitoring scheme may be to monitor the heat loss of the laser for every build. If the heat loss is increasing over time, the cause of the heat loss should be investigated. To ensure that the laser spot size and shape are as expected, a burn test should be carried out when discrepancies are suspected or on a weekly basis.

7.3.6.3 Consumable feeding nozzles need to be regularly inspected for wear damage and replaced at documented intervals.

7.3.6.4 It should be assessed how to use clamps or fixtures on the substrate during building to manage distortions and residual stress. The clamping of the substrate should be described in the MPS.

Guidance note:

- The forces generated during build-up may be significant, causing the substrate to move.
- Clamping may induce or inhibit residual stresses.

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7.3.6.5 A mock-up run, or a dry run, of the positioner should be performed before building the intended part. The mock-up will indicate whether the height, width, overlap and start-stop areas are as intended.

Guidance note:

Small differences in height in the first layers may result in bigger deviations as the number of layers increases.

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7.3.7 Build requirements

7.3.7.1 It should be ensured that each layer is free from impurities and imperfections. Corrective actions such as cleaning, grinding or machining may be performed according to a qualified procedure.

7.3.7.2 When a build process is paused due to interruptions, the top layer shall be visually examined before continuing the build. The layer number at which the build was paused shall be logged. This requirement is not relevant for planned interruptions as long as the part is not cooled down below the minimum preheat temperature. If the temperature falls below the preheat temperature during the interruption, the build needs to be re-heated above the preheat temperature before continuing. The preheat temperature shall in such case be confirmed in the top 50 mm of the build and measured in a way that ensures even heating.

Guidance note:

Different materials will react differently to interruptions to the preheating. Some materials may be exposed to hydrogen cracking in cases of high residual stress if allowed to cool over longer periods of time. Therefore, it should be assessed whether cooling below the preheat temperature during interruptions could be damaging to the build.

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7.3.7.3 If a new feedstock batch is used in the middle of a build job, the effect of the batch change shall be evaluated. A procedure for evaluating the effect of batch changes shall be documented. The material deposited from the new feedstock batch should show no indication of a feedstock change.

7.3.7.4 As a minimum, the following parameters shall be logged during the building:

- power output, laser characteristics and travel speed
- wire or powder feed rate
- gas flow rate
- oxygen content in atmosphere, if built in a closed chamber
- temperature in atmosphere, if built in a closed chamber
- dew point, if built in a closed chamber.

7.3.7.5 The preheat and interpass temperatures shall be monitored in accordance with ISO 13916, or other relevant standards for a similar purpose.

7.3.7.6 For oxygen-sensitive materials, measures shall be in place to control and monitor the oxygen level.

7.3.7.7 If the DED machine is equipped with systems for monitoring the process temperature, temperature logs shall be recorded.

7.3.7.8 Information from the build shall be summarized in a build report, and shall as a minimum include:

- a reference to the unique build job ID
- the unique ID of the applied parameter set
- idle time between layers
- error messages or warnings.

7.3.8 Post processing requirements

7.3.8.1 When removing a part from the substrate after building, the first three (3) layers shall be cut from the part. Any deviation from this requirement should be compensated by additional testing of the first layers.

7.3.8.2 In addition to the general description of post processing in [6.4], heat treatment practices shall comply with the following requirements:

- 1) Heat treatment furnaces and equipment shall be calibrated according to ISO 10423/API 6A, SAE AMS-H-6875, ASTM A991 or the equivalent.
- 2) Furnaces shall be equipped with automatic temperature indicating, controlling, and recording devices, which shall possess an accuracy of $\pm 1\%$ of their full-scale range.
- 3) Pallets or fixtures shall be used for placing parts during heat treatment and quenching to ensure the free circulation of air/atmosphere and quenching medium between each part. Parts shall not be placed directly onto the furnace floor. The piling of parts in baskets is not allowed.
- 4) For alloys requiring water quenching, the furnace unloading equipment shall be capable of transferring materials from the furnace into the quench tank within a specified maximum time, e.g. 60 s for duplex stainless steels and 90 s for low alloy steels.
- 5) For alloys requiring water quenching, quench tanks shall have dimensions and cooling facilities adequate to keep the water temperature below 50°C at any time during quenching. Quench tanks shall be equipped with pumps or propellers to ensure satisfactory circulation and agitation during quenching. Quench tanks shall be equipped with continuous temperature monitoring/recording devices.
- 6) Heat treatment shall be controlled and documented through formal procedures.

Guidance note:

The requirement that the first three (3) layers must be removed is due to dilution and heat sink effects between the substrate and the part.

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7.3.9 Production testing

7.3.9.1 Production testing is required for all parts, with the scope and frequency described in [Table 7-15](#) and [Table 7-16](#). For parts intended for serial production, the frequency shall be agreed between the purchaser and manufacturer.

7.3.9.2 Production test specimens shall be extracted from a test coupon, a prolongation of the part, or from a sacrificial part, as relevant. To avoid sacrificing a part for production testing, production testing coupons may be included in the part qualification, see [\[7.3.3.3\]](#).

7.3.9.3 If a prolongation or test coupon is used, it shall represent the thickness and deposition strategy of the critical section(s) of the part and its thermal behaviour shall be representative of the part. A test coupon shall accompany the part it represents through all thermal post processing steps.

Guidance note:

In this context, representative thermal behaviour means that the volume will have a similar thermal cycle, e.g. similar peak temperature and heating and cooling rates locally during the build and thermal post processing.

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7.3.9.4 Additional testing may be required based on the part's intended application, see [\[2.3\]](#).

Table 7-15 Production testing scope

Additive manufacturing category	Test method (quantity)	Orientation and location
AMC 1	Tensile testing (1)	Samples and sections shall be in one of the orientations defined for BPQ or PQT, as relevant. ¹⁾
	Macrostructural assessment	
AMC 2 and AMC 3 ²⁾	Tensile testing (1)	
	Impact testing (1 set)	
	Hardness measurements	
	Macrostructural assessment	
	Chemical analysis	

1) Production testing shall be performed and oriented in the same manner as the production test specimens were included in the part qualification.

2) Additional testing may be specified by the purchaser, see [2.3].

Table 7-16 Frequency of production testing

<i>Additive manufacturing category</i>	<i>Frequency of testing</i>
AMC1 and AMC 2	As agreed with the purchaser.
AMC 3	Every build job.

7.4 Powder bed fusion - laser beam

7.4.1 General

7.4.1.1 This section describes requirements and recommendations related to production and qualification using the PBF-LB technology specifically. The requirements and recommendations specified in the section come in addition to the general requirements and recommendations provided in [Sec.4](#) and [Sec.6](#).

7.4.2 Build process qualification

7.4.2.1 The tests required to be performed during build process qualification (BPQ) are given in [Table 7-17](#) and together comprise the standard qualification build for PBF-LB.

7.4.2.2 Retesting is allowed in accordance with the relevant material standard. If retesting is carried out, this shall be stated in the resulting documentation of the build process qualification, and the results of the failed tests shall be included.

7.4.2.3 The standard qualification build shall be produced with all the essential parameters defined prior to starting the build job. The minimum essential parameters are given in [\[7.4.4\]](#).

7.4.2.4 The standard qualification build shall be produced during one build job, if possible.

Guidance note:

Machines have a limited substrate area available for building. In cases where all the test specimens do not fit on one substrate, several build jobs may be necessary.

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7.4.2.5 An AMC 1 or AMC 2 part may be produced at the same time as the standard qualification build.

7.4.2.6 If the machine uses several lasers, at least two (2) of the Z-oriented tensile test specimens, one positioned near the platform and one at elevated height, shall be built using each laser. Additionally, extra horizontal tensile tests shall be included when each adjacent laser builds one half of the specimen and overlaps in the middle.

Guidance note:

The configuration of the lasers' overlapping working areas could be different in different machines. However, horizontal tensile specimens should be built in the overlapping zones between two lasers, and each laser should be sampled at least once. Example: in a 2x2 laser configuration where lasers 1 and 2, 2 and 3, 3 and 4, and 4 and 1 overlap, horizontal tensile specimens with overlap should be built in these 4 regions. Samples from the region where all 4 lasers overlap is assumed to be covered by the other 4. The results of such tests may not be relevant if each laser operates alone on a part or specimen.

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Table 7-17 Test scope for build process qualification for PBF-LB

<i>Test method</i>	<i>Quantity and specimen orientation, according to ISO/ASTM 52921</i>	<i>Position in chamber</i>	<i>Testing requirement</i>
Tensile testing ¹⁾	1 x X	Anywhere	[8.2.2]
	1 x Y	Anywhere	

<i>Test method</i>	<i>Quantity and specimen orientation, according to ISO/ASTM 52921</i>	<i>Position in chamber</i>	<i>Testing requirement</i>
	10 x Z	Elevation: near build platform, and elevated. Elevation: near build platform, and elevated. Location: centre and every corner.	
Thin-walled tensile testing ²⁾	5 x Z	Elevation: near build platform. Location: centre and every corner.	
Impact testing	1 set, horizontal	Elevation: near build platform. Location: anywhere.	[8.2.3]
Hardness measurements	Any orientation	Elevation: near build platform. Location: anywhere. May be performed on another test specimen, or a separate cube.	[8.2.4]
Macrostructural assessment	Plane of examination: Surface normal in horizontal direction	Elevation: near build platform and at elevated part of chamber. Location: bulk ³⁾ . May be performed on another test specimen, or a separate cube.	[8.2.5]
Microstructural assessment			[8.2.6]
Porosity assessment			[8.2.7]
Chemical analysis	Any orientation	Elevation: any. Location: anywhere. May be performed on another test specimen, or a separate cube.	[8.2.8]
1) Additional horizontal tensile tests to be included if multiple lasers are used. 2) Thin-walled tensile specimens are recommended. 3) A porosity assessment at the surface of the specimen is required to check the interfaces between the contour scanning pattern and bulk scanning pattern. For this assessment, the sample shall not be machined.			

7.4.2.7 Figure 7-8 is added for clarification, to provide an overview of the test specimens given in Table 7-17. The recommended thin-walled tensile test specimens are added, as well as an additional horizontal tensile specimen, representing additional testing required if more than one laser is used by the machine.

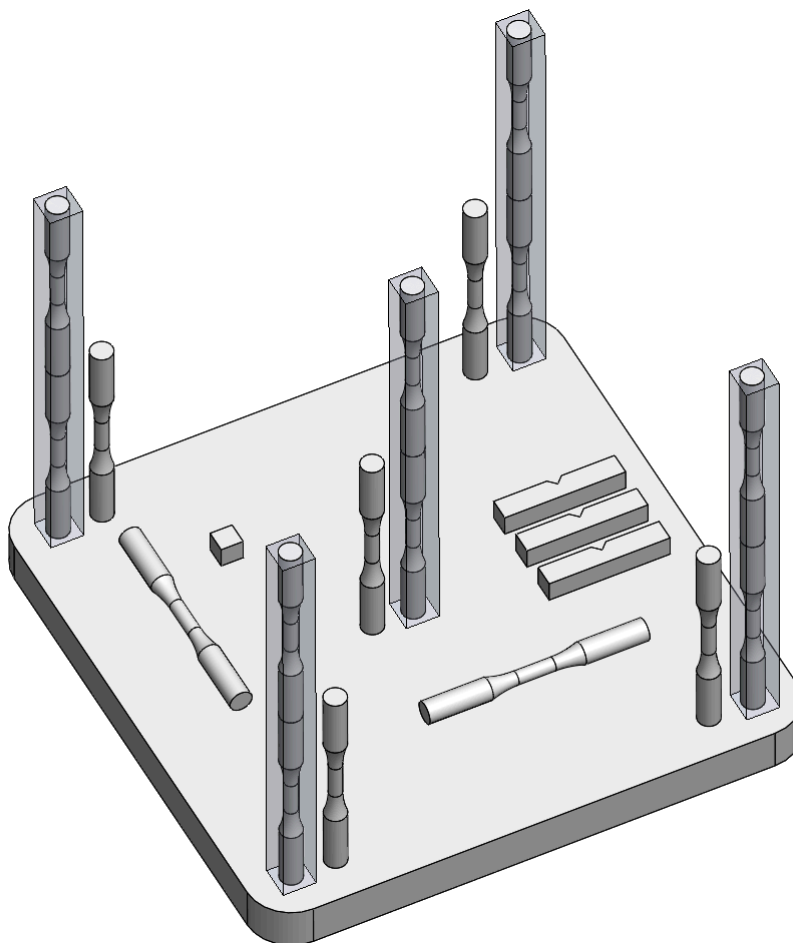


Figure 7-8 An illustration of how the standard qualification build may look during the build file preparation stage

Guidance note:

The elevated tensile specimens are added to control the effect of the distance to the substrate. If no significant difference is observed between the tensile specimens near the build platform and those at an elevated position, this behaviour is assumed for the rest of the chamber height as well.

To detect variation in porosity based on location on the build plate, density cubes may be included in the locations of the tensile bars.

It is recommended to include the thin-walled tensile specimens in the standard qualification build. These test specimens will represent parts with thin wall thicknesses that may have a different thermal history than those with thicker walls. If such tests are not performed, it may be relevant to include such tests in a part qualification.

There are some uncertainties with regard to thin-walled test specimens:

- Printability: thin and tall geometries may be difficult to build due to buckling. It may be relevant to add tubular supports around the tensile specimen or build a tube of relevant wall thickness with solid end-points to enable testing.
- Accuracy of result: the machining of thin specimens may be difficult, and testing as-printed may be necessary. Due to the rough surface of as-print specimens, wall thicknesses may vary. This variability may have a greater relative effect on thin-walled specimens than on normal sized specimens, and this may affect the interpretation of results.

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7.4.3 Part qualification

7.4.3.1 AMC 2 and AMC 3 parts shall undergo part qualification testing as described in Table 7-18 and Table 7-19.

- AMC 2 parts require non-destructive part qualification testing, in addition to mechanical testing according to the requirements for production testing, see [7.4.9].
- AMC 3 parts require non-destructive, mechanical and metallurgical part qualification testing.

7.4.3.2 If the relevant part has wall thicknesses below 1.5 mm, a separate thin-walled tensile test specimen, with relevant thickness, shall be built and tested during part qualification.

7.4.3.3 Additional testing may be required based on the part's intended application, see [2.3].

Table 7-18 Non-destructive test scope during part qualification - relevant for both AMC 2 and AMC 3 parts

<i>Test method</i>	<i>Scope</i>	<i>Testing requirement</i>
Visual inspection	100% as permitted by geometry	[8.3.4]
Dimensional check	In accordance with purchaser requirements	[8.3.10]
Surface NDT	100% as permitted by geometry	[8.3.5]
Volumetric NDT	100% as permitted by geometry	[8.3.6]

Table 7-19 Mechanical and metallurgical test scope for part qualification - relevant for AMC 3 parts

<i>Test method</i>	<i>Orientation and quantity</i>	<i>Specimen extraction</i>	<i>Testing requirement</i>
Tensile testing ¹⁾	1 x Z or parallel to greatest direction of force in the part's intended application.	Critical section(s) ²⁾	[8.2.2]
Hardness measurements	Plane of examination: Surface normal in horizontal direction	Critical section(s) ²⁾	[8.2.4]
Macrostructural assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.5]
Microstructural assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.6]
Porosity assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.7]
Production testing ³⁾	See [7.4.9]		

Test method	Orientation and quantity	Specimen extraction	Testing requirement
1) When extraction of tensile test specimens from a part is not possible due to size limitations, the part should be tested in a manner that represents its intended use. 2) See [2.3.2] for defining the critical section(s). 3) Intended specimens for subsequent production testing shall be included and tested in the part qualification. This is to ensure that the production test specimens are representative of the part's properties.			

7.4.4 Validity of qualifications

7.4.4.1 A build process qualification or part qualification is valid as long as the essential parameters used during manufacturing are kept within their limits of qualification. [Table 7-20](#) lists parameters that shall be stated during qualification testing and used during production.

7.4.4.2 A system or procedure shall be in place to ensure that the parameter set used in manufacturing is the same as the parameter set qualified in a BPQT, and that no changes have been made.

7.4.4.3 If the name or ID of a parameter set is changed, without any change to the parameters within, separate documentation shall be added to the relevant BPQR explaining the name change and confirming that the parameters are unchanged. The BPQR itself shall not be revised.

7.4.4.4 For part qualifications relevant for AMC 2 and AMC 3 parts, a change of part, or violation of the bounds defined for the specific part family, shall lead to a new part qualification. Part families are described in [\[4.3.2\]](#).

Table 7-20 Essential parameters for PBF-LB

Category	Parameter id	Essential parameters	Limits of qualification ¹⁾
Hardware and software description	1-1	Machine	AMC 1 and AMC 2: change of brand or model. AMC 3: change of serial number.
	1-2	Build plate	Thickness reduction of more than 10% from the thickness used in the qualification. Change of material specification, classification or form and shape.
	1-3	Recoater	Any change of type, configuration or material of recoater.
	1-4	Laser system and optics	AMC 1 and AMC 2: change of brand or model. AMC 3: change of serial number.
	1-5	Software and firmware	Change of software or firmware maker or product name, and major updates, see [5.6.1.5] .
	1-6	Sieve	Any change of type or mesh size of sieve.
Feedstock characteristics	2-1	Chemical composition	Any change outside the range given in the feedstock specification.
	2-2	Particle shape	Any change outside the range given in the feedstock specification.
	2-3	Particle size distribution	Any change outside the range given in the feedstock specification.

Category	Parameter id	Essential parameters	Limits of qualification ¹⁾
	2-4	Apparent density	Significant change compared to feedstock specification ²⁾ .
	2-5	Flowability	Significant decrease of flowability compared to feedstock specification ³⁾ .
	2-6	Inert process gas	Change of specified nominal composition.
Process parameters ⁴⁾	3-1	Parameter set	Change of parameters used for part.
Post processing	4-1	Thermal post processing	Any change of nominal heat treatment time, temperatures, or sequence.
1) If changes to the essential parameters are as described in this column, e.g. a sieve mesh size is changed, a new qualification of the build process is required. 2) When a significant change of apparent density is observed, the reason shall be assessed. 3) Upon reuse, the flowability is expected to increase, as the particle size distribution curve is shifted to a larger particle size. Any decrease in flowability may thus indicate other factors, such as an increase in humidity. 4) The production parameters not listed shall be considered locked. If parameters initially set by the machine manufacturer have been adjusted, these adjustments shall be communicated in the BPQR.			

7.4.5 Build file preparation

7.4.5.1 The total volume to be built in one build job shall not exceed the limitation stated by the machine manufacturer.

7.4.5.2 When adding supports and placing the parts on the build platform in the build preparation software, ISO/ASTM 52911 should be considered.

7.4.5.3 When placing a part on the build platform, it should be ensured that it can be removed after the build process without being damaged.

7.4.5.4 All build parameters shall be set in the build preparation phase. The parameters given by the parameter file shall not be adjusted at the machine.

7.4.6 Pre-build requirements

7.4.6.1 The recoater shall be inspected for any damage, deformations, or impurities before the start of a new build job.

7.4.6.2 Procedures for build preparation and maintenance shall be in place to ensure the cleanliness, traceability, and maintenance of build plates and heat shields.

7.4.6.3 If a machine is used for more than one powder alloy, a powder change procedure shall be in place. The powder change procedure shall be in accordance with the machine manufacturer's recommendations.

7.4.6.4 The powder shall be loaded into the machine in accordance with the machine manufacturer's recommendations.

7.4.6.5 Unless otherwise specified, the build chamber shall be purged of oxygen below a level of 1000 ppm before building starts.

7.4.6.6 The machine shall be calibrated in accordance with the requirements given in [5.6]. The procedure shall as a minimum include calibration of the:

- laser power
- laser focus
- laser alignment
- recoater alignment
- platform movement.

7.4.6.7 The calibration shall be validated as specified by the machine manufacturer.

7.4.7 Build requirements

7.4.7.1 The operator shall monitor the initial layers of the build job to ensure that the build is progressing as planned, and take appropriate action if necessary according to the manufacturer's procedure.

7.4.7.2 In the case of a planned or unplanned interruption to the build process, the part shall be inspected for signs of potential imperfections, such as shrinkage or discolouration, at the relevant layer. If signs are identified, they shall be recorded and rectified as relevant. These records, including the duration of the interruption or pause, shall be communicated to the purchaser for acceptance before delivery. For build jobs including AMC 3 parts, no unplanned interruption is allowed, and planned interruptions shall be included in the part qualification.

Guidance note:

A time limit for a build job pause is set to avoid any start-stop areas and uncertain thermal history in the finished part.

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7.4.7.3 For AMC 3 parts, the build job shall not be continued after individual part(s) or specimens are cancelled during a build job.

Guidance note:

Cancellation of one or more parts mid-build may be regarded as the removal of part(s) from the part qualification, and may lead to changes to the thermal history.

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7.4.7.4 If the recoater must be replaced mid-build, the build job shall be restarted from scratch. The recoater shall be inspected after every build job to see if it has been damaged during the build job.

7.4.7.5 As a minimum, the following parameters shall be logged at least once per layer throughout the entire build job:

- oxygen content in chamber atmosphere
- chamber temperature
- gas flow rate
- chamber pressure
- dew point
- time used per layer or idle time between layers.

7.4.7.6 The build report shall include:

- a reference to the unique build job ID
- the unique ID of the applied parameter set
- time used per layer or idle time between layers
- error messages or warnings.

7.4.7.7 The limits for the parameters given in [7.4.7.5] shall be defined and it shall be ensured that the build job is carried out within the defined ranges. If parameters are logged outside the defined ranges, these shall be logged along with a warning or error message.

7.4.8 Post processing requirements

7.4.8.1 All trapped powder shall be removed from the part(s) before thermal post processing. Stress relieving heat treatment may be performed before the removal of trapped powder. Any powder inside a part during stress relief shall not be reused.

Guidance note:

Powder trapped in small channels or suchlike may get stuck if stress relief is carried out prior to powder removal.

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7.4.8.2 Thermal post processing, such as stress relief, annealing and hot isostatic pressing, shall be performed in accordance with ASTM F3301 or a qualified procedure, see [6.4].

7.4.9 Production testing

7.4.9.1 Production testing with the scope described in Table 7-21 is required for all parts. For parts intended for serial production, the frequency shall be agreed between the purchaser and manufacturer.

7.4.9.2 The orientation of the production tensile test specimens should be according to the orientation of the part to be built, or according to the forces acting upon it in its intended application. If not specified, the orientation should be vertical.

Guidance note:

The building of hollow containers, in which powder will be trapped, should be considered. The trapped powder will represent the powder used for the build job and may be analysed for quality purposes.

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Table 7-21 Test scope for production testing for PBF-LB

<i>Additive manufacturing category</i>	<i>Test method</i>	<i>Quantity and orientation</i>	<i>Testing requirement</i>
AMC 1	Tensile testing	1 in Z direction	[8.2.2]
AMC 2	Tensile testing	1 in Z direction	[8.2.2]
	Impact testing	1 set, horizontal	[8.2.3]
AMC 3	Tensile testing	2 in Z direction, or in accordance with the orientation specified for part qualification.	[8.2.2]
	Impact testing	1 set, horizontal	[8.2.3]
	Hardness measurements	Same orientation as macrostructural assessment.	[8.2.4]
	Macrostructural assessment	Normal to build direction	[8.2.5]
	Porosity assessment	See [8.2.7]	[8.2.7]
	Chemical analysis	Any orientation.	[8.2.8]

7.5 Powder bed fusion - electron beam

7.5.1 General

7.5.1.1 This section describes requirements and recommendations related to production and qualification using the PBF-EB technology specifically. The requirements and recommendations specified in the section come in addition to the general requirements and recommendations provided in [Sec.4](#) and [Sec.6](#).

7.5.2 Build process qualification

7.5.2.1 The tests required to be performed during the build process qualification (BPQ) are given in [Table 7-22](#) and together comprise the standard qualification build for PBF-EB.

7.5.2.2 Additional material should be built if needed to maintain the build chamber temperature at the required level. Such material could be additional samples. Simulation should be used to assess whether additional material is needed.

7.5.2.3 Retesting is allowed in accordance with the relevant material standard. If retesting is carried out, this shall be stated in the resulting documentation of the build process qualification, and the results of the failed tests shall be included.

7.5.2.4 The standard qualification build shall be produced with all essential parameters defined prior to starting the build job. The minimum essential parameters are given in [\[7.5.4\]](#).

7.5.2.5 The standard qualification build shall be produced during one build job, if possible.

7.5.2.6 The standard qualification build may be produced within a limited volume within the build chamber. In such cases, it is the volume within which the test samples were placed that is being qualified.

Guidance note:

Some materials, e.g. nickel alloys, are usually built within a smaller build volume to better maintain the required process temperatures.

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7.5.2.7 An AMC 1 or AMC 2 part may be produced at the same time as the standard qualification build.

Table 7-22 Test scope for build process qualification for PBF-EB

<i>Test method</i>	<i>Quantity and specimen orientation, according to ISO/ASTM 52921</i>	<i>Position in chamber</i>	<i>Testing requirement</i>
Tensile testing	1 x X	Anywhere	[8.2.2]
	1 x Y	Anywhere	
	10 x Z	Elevation: near build platform, and elevated. Location: centre and at periphery.	
Thin-walled tensile testing	5 x Z	Elevation: near build platform. Location: centre and every corner.	
Impact testing	1 set, horizontal	Elevation: near build platform. Location: anywhere.	[8.2.3]
Hardness measurements	Any orientation	Elevation: near build platform. Location: anywhere. May be performed on another test specimen, or a separate cube.	[8.2.4]
Macrostructural assessment	Plane of examination: Surface normal in horizontal direction	Elevation: near build platform and at elevated part of chamber. Location: bulk. May be performed on another test specimen, or a separate cube.	[8.2.5]
Microstructural assessment			[8.2.6]
Porosity assessment			[8.2.7]
Chemical analysis	Any orientation	Elevation: any. Location: anywhere. May be performed on another test specimen, or a separate cube.	[8.2.8]

7.5.2.8 Figure 7-9 shows the positioning of the test specimens given in [Table 7-22](#).

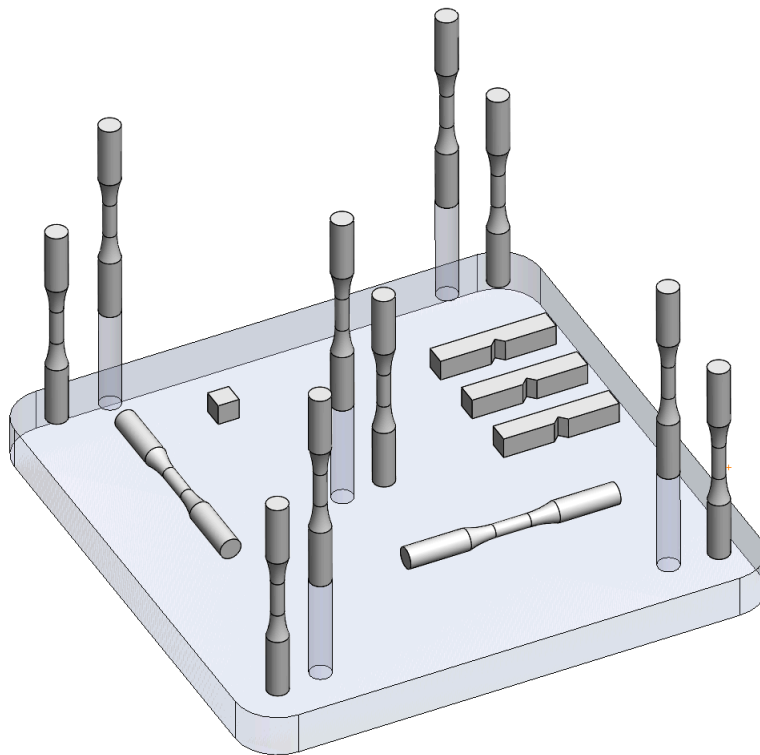


Figure 7-9 Standard build process qualification for PBF-EB

Guidance note:

To detect variation in porosity based on location on the build plate, density cubes may be included in the locations of the tensile bars.

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7.5.3 Part qualification

7.5.3.1 AMC 2 and AMC 3 parts shall undergo part qualification testing as described in [Table 7-23](#) and [Table 7-24](#).

- AMC 2 parts require non-destructive part qualification testing, in addition to mechanical testing according to the requirements for production testing, see [\[7.5.9\]](#).
- AMC 3 parts require non-destructive, mechanical and metallurgical part qualification testing.

7.5.3.2 If the relevant part has wall thicknesses below 1.5 mm, a separate thin-walled tensile test specimen, with relevant thickness, should be built and tested during part qualification.

7.5.3.3 Additional testing may be required based on the part's intended application, see [\[2.3\]](#).

Table 7-23 Non-destructive test scope during part qualification - relevant for both AMC 2 and AMC 3 parts

<i>Test method</i>	<i>Scope</i>	<i>Testing requirement</i>
Visual inspection	100% as permitted by geometry	[8.3.4]
Dimensional check	In accordance with purchaser requirements	[8.3.10]
Surface NDT	100% as permitted by geometry	[8.3.5]
Volumetric NDT	100% as permitted by geometry	[8.3.6]

Table 7-24 Mechanical and metallurgical test scope for part qualification - relevant for AMC 3 parts

<i>Test method</i>	<i>Orientation and quantity</i>	<i>Specimen extraction</i>	<i>Testing requirement</i>
Tensile testing ¹⁾	1 x Z or parallel to greatest direction of force in the part's intended application	Critical section(s) ²⁾	[8.2.2]
Hardness measurements	Plane of examination: Surface normal in horizontal direction	Critical section(s) ²⁾	[8.2.4]
Macrostructural assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.5]
Microstructural assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.6]
Porosity assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.7]
Production testing ³⁾	See [7.5.9]		
1) When the extraction of tensile test specimens from a part is not possible due to size limitations, the part should be tested in a manner that represents its intended use. 2) See [2.3.2] for defining the critical section(s). 3) Specimen intended for subsequent production testing shall be included and tested in the part qualification. This is to ensure that the production test specimens are representative of the part's properties.			

7.5.4 Validity of qualifications

7.5.4.1 A build process qualification or part qualification is valid as long as the essential parameters used during manufacturing are kept within their limits of qualification. Table 7-25 lists parameters that shall be stated during qualification testing and used during production.

7.5.4.2 A system or procedure shall be in place to ensure that the parameter set used in manufacturing is the same as the parameter set qualified in a BPQT, and that no changes have been made.

7.5.4.3 If the name or ID of a parameter set is changed, without any change of the parameters within, separate documentation shall be added to the relevant BPQR explaining the name change and confirming that the parameters are unchanged. The BPQR itself shall not be revised.

7.5.4.4 For part qualifications relevant for AMC 2 and AMC 3 parts, a change of part, or violation of the bounds defined for the specific part family, shall lead to a new part qualification. Part families are described in [4.3.2].

Table 7-25 Essential parameters for PBF-EB

Category	Parameter id	Essential parameters	Limits of qualification ¹⁾
Hardware and software description	1-1	Machine	Change of brand or model.
	1-2	Recoater	Any change of type, configuration or material of recoater.
	1-3	Electron gun column	Change of type of components in the column.
	1-4	Software and firmware	Change of software or firmware maker or product name, and major updates, see [5.6.1.5].
	1-5	Sieve	Any change of type or mesh size of sieve.
Feedstock information	2-1	Chemical composition	Any change outside the range given in the feedstock specification.
	2-2	Particle shape	Any change outside the range given in the feedstock specification.
	2-3	Particle size distribution	Any change outside the range given in the feedstock specification.
	2-4	Apparent density	Significant change compared to feedstock specification.
	2-5	Inert process gas	Change of specified nominal composition.
Process parameters ²⁾	3-1	Parameter set/process theme	Change of parameter set/theme used.
	3-2	Chamber vacuum pressure	Any change in programmed vacuum pressure.
	3-3	Column vacuum pressure	Any change in programmed vacuum pressure.
Post processing	4-1	Thermal post processing	Any change of nominal heat treatment time, temperatures, or sequence.
1) Adjustments to the essential parameters outside the limits of qualification, e.g. the vacuum pressure is changed or the process theme is updated, a re-qualification of the build process is required. 2) The production parameters not listed shall be considered locked. If parameters initially set by the machine manufacturer have been adjusted, these adjustments shall be communicated in the BPQR.			

7.5.5 Build file preparation

7.5.5.1 Parts shall be placed on the build platform with a defined minimum distance from each other.

7.5.5.2 There should be at least 2 mm of support structure between the build plate and a part. Test specimens can be placed directly on the build plate with an additional material allowance to allow for cutting.

Guidance note:

The build plate may be of a different material, and the 2 mm support structure may minimize diffusion. In addition, the powder spreading may be less stable close to the build platform.

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7.5.6 Pre-build requirements

7.5.6.1 Metallization shall be removed from all surfaces in the build chamber prior to the loading of powder.

7.5.6.2 The recoater shall be inspected for any damage, deformations, or impurities before the start of a new build job.

7.5.6.3 Procedures for build preparation and maintenance shall be in place to ensure the cleanliness, traceability, and maintenance of build plates and heat shields. It is recommended to monitor the thicknesses of build plates when they are re-used to avoid warping during the build.

7.5.6.4 If a machine is used for more than one powder alloy, a powder change procedure shall be in place. The powder change procedure shall be in accordance with the machine manufacturer's recommendations.

7.5.6.5 The powder shall be loaded into the machine in accordance with the machine manufacturer's recommendations.

7.5.6.6 The machine shall be calibrated in accordance with the requirements given in [5.6]. The procedure shall as a minimum include calibration of the:

- base plate height
- powder sensor positions
- recoater fetch and rest positions.

7.5.6.7 When the cathode is replaced, a beam calibration shall be performed, or the beam calibration shall be checked by running a verification plate. Cathode replacement and calibration shall be documented as recommended by the machine manufacturer.

7.5.7 Build requirements

7.5.7.1 The operator shall monitor the initial layers of the build job to ensure that the build is progressing as planned, and take appropriate action if necessary according to the manufacturer's procedure.

7.5.7.2 The parameters shall not be adjusted once the build job has started. The parameters used shall be documented.

7.5.7.3 In the case of a planned or unplanned interruption to the build process, the part shall be inspected for signs of potential imperfections, such as shrinkage or discolouration, at the relevant layer. If signs are identified, they shall be recorded and rectified as relevant. These records, including the duration of the interruption/pause shall be communicated to the purchaser for acceptance before delivery. For build jobs including AMC 3 parts, no unplanned interruption is allowed, and planned interruptions shall be included in the part qualification.

Guidance note:

A time limit for a build job pause is set to ensure enough heat in the chamber during the build process, and to avoid shrink lines and delamination.

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7.5.7.4 For AMC 3 parts, the build job shall not be continued after the cancellation of individual part(s) or specimens during a build job.

Guidance note:

Cancellation of one or more parts mid-build may be regarded as the removal of part(s) from the part qualification, and may lead to changes to the thermal history.

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7.5.7.5 If the recoater must be replaced mid-build, the build job shall be restarted from scratch. The recoater shall be inspected after every build job to see if it has been damaged during the build job.

7.5.7.6 As a minimum, the following parameters shall be logged at least once per layer throughout the entire build job:

- column temperature
- chamber vacuum pressure
- column vacuum pressure
- idle time between layers
- abnormal pulses per layer oxygen content in chamber atmosphere.

7.5.7.7 The build report shall include:

- a reference to the unique build job ID
- the unique ID of the applied parameter set
- the idle time between layers
- error messages or warnings.

7.5.8 Post processing requirements

7.5.8.1 All trapped powder shall be removed from the part(s) before thermal post processing.

Guidance note:

Powder trapped in small channels or similar may get stuck if stress relief is carried out prior to powder removal.

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7.5.8.2 Thermal post processing, such as stress relief, annealing and hot isostatic pressing, shall be performed in accordance with ASTM F3301, or a qualified procedure, see [6.4].

7.5.9 Production testing

7.5.9.1 Production testing with the scope described in Table 7-26 is required for all parts. For parts intended for serial production, the frequency shall be agreed between the purchaser and manufacturer.

7.5.9.2 The production test specimens shall be placed close to the part(s) to be built to represent the actual build environment.

7.5.9.3 The orientation of the production tensile test specimens should be according to the orientation of the part to be built, or according to the forces acting upon it in its intended application. If not specified, the orientation should be vertical.

7.5.9.4 The tensile specimens should be placed at locations in the build chamber representing the highest and lowest build temperature, see Figure 7-10 as an example.

7.5.9.5 It is recommended to build a cylinder in the full height of the build job, and from that machine a tensile witness sample. This enables the tensile witness sample to be extracted from a height which may need closer examination, e.g. due to a short process stop.

Table 7-26 Test scope for production testing for PBF-EB

<i>Additive manufacturing category</i>	<i>Test method</i>	<i>Quantity and orientation</i>	<i>Testing requirement</i>
AMC 1	Tensile testing	1 in Z direction	[8.2.2]
AMC 2	Tensile testing	1 in Z direction	[8.2.2]
	Impact testing	1 set, horizontal	[8.2.3]
AMC 3	Tensile testing	2 in Z direction, or in accordance with the orientation specified for part qualification.	[8.2.2]
	Impact testing	1 set, horizontal	[8.2.3]
	Hardness measurements	Same orientation as macrostructural assessment.	[8.2.4]
	Macrostructural assessment	Normal to build direction	[8.2.5]
	Porosity assessment	See [8.2.7]	[8.2.7]
	Chemical analysis	Any orientation.	[8.2.8]

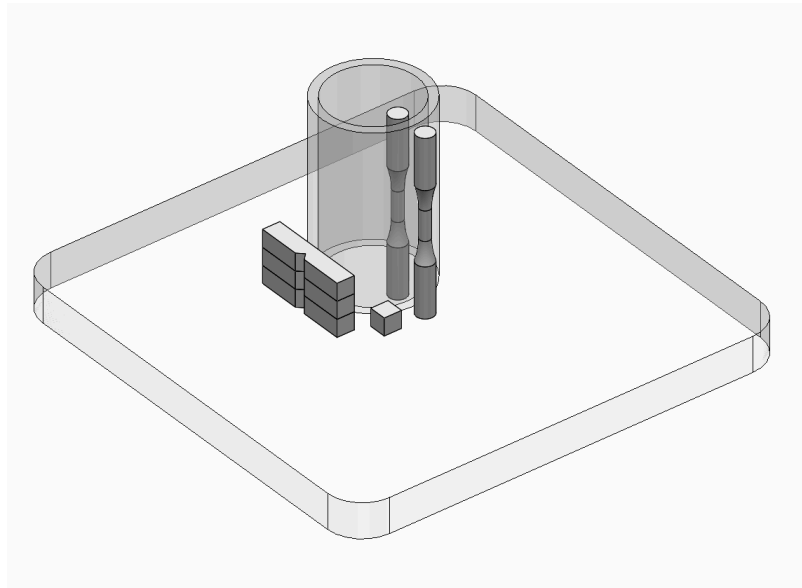


Figure 7-10 An example of the placement of production test specimens. In this example, one tensile specimen is placed in the hot zone of the build job, and one in the cold zone. The impact test specimens and cubes for porosity and macro are placed close to the part (cylinder).

7.6 Binder jetting technology

7.6.1 General

7.6.1.1 This section contains guidance on and requirements for additive manufacturing using binder jetting technology (BJT). The guidance and requirements given in this section come in addition to those given in [Sec.4](#) and [Sec.6](#) of this document.

7.6.1.2 The binder jetting build process consists of three (3) main steps: printing, curing and sintering. The process is similar to PBF-LB and PBF-EB, as all of these processes include a powder bed. In BJT, binder is used to connect to powder particles in a layer-by-layer manner. After the printing is complete, curing is performed. The curing is typically done without removing the parts from the build box. After curing, the parts are removed from the build box and loose powder is removed from the parts. The unsintered parts are called 'green parts'. The green parts are sintered at an elevated temperature, fusing the powder particles together. Shrinkage of about 20% is normal during the sintering process.

7.6.2 Build process qualification

7.6.2.1 The tests required to be performed during BPQT are given in [Table 7-27](#), and together comprise the standard qualification build. An example of a standard qualification build is shown in [Figure 7-11](#).

7.6.2.2 Since BJT is a multistage process, the positions in both the build chamber and sintering furnace are defined. The position in the sintering furnace is included to check for variations caused by the position within the furnace.

7.6.2.3 Retesting is allowed in accordance with the relevant material standard. If retesting is carried out, this shall be stated in the resulting documentation of the build process qualification, and the results of the failed tests shall be included.

7.6.2.4 The standard qualification build shall be produced with all the essential parameters defined prior to starting the build job. The minimum essential parameters are given in [\[7.6.4\]](#). The standard qualification build shall be produced during one manufacturing step, if possible.

Guidance note:

Machines have a limited substrate area available for building. In cases where all test specimens do not fit on one substrate, several build jobs may be necessary.

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7.6.2.5 The standard qualification build may be produced within a limited volume within the build chamber. In such cases, only the limited volume sampled will be qualified.

7.6.2.6 An AMC 1 or AMC 2 part may be produced at the same time as the standard qualification build.

Table 7-27 Test scope for build process qualification for binder jetting technology (BJT)

Test method	Quantity and specimen orientation, according to ISO/ASTM 52921	Position in chamber	Position in furnace	Testing requirement
Tensile testing	3 specimens (x, y and z)	Top corner of build box	Anywhere, together	[8.2.2]
	3 specimens (x, y and z)	Centre of build box		
	3 specimens (x, y and z)	Bottom corner for build box (opposing corner)		
	3 specimens in any direction (same direction)	Anywhere, together	Top, centre and bottom ¹⁾	[8.2.2]
Impact testing	1 set of three (3) specimens, notch orientation in Z	Anywhere	Anywhere	[8.2.3]
Hardness measurement	2 x any orientation	Anywhere	Anywhere	[8.2.4]
Macrostructural assessment	3 x any orientation	top bottom and centre	Anywhere	[8.2.5]
Microstructural assessment				[8.2.6]
Porosity assessment				[8.2.7]
Shrinkage from sintering	Shrinkage after sintering to be logged on all tensile specimens, by dimensional measurements.			
Chemical analysis	2 x any orientation	Anywhere	Anywhere	[8.2.8]
1) Or placed in the outer positions in the region of the furnace that will be used for sintering.				

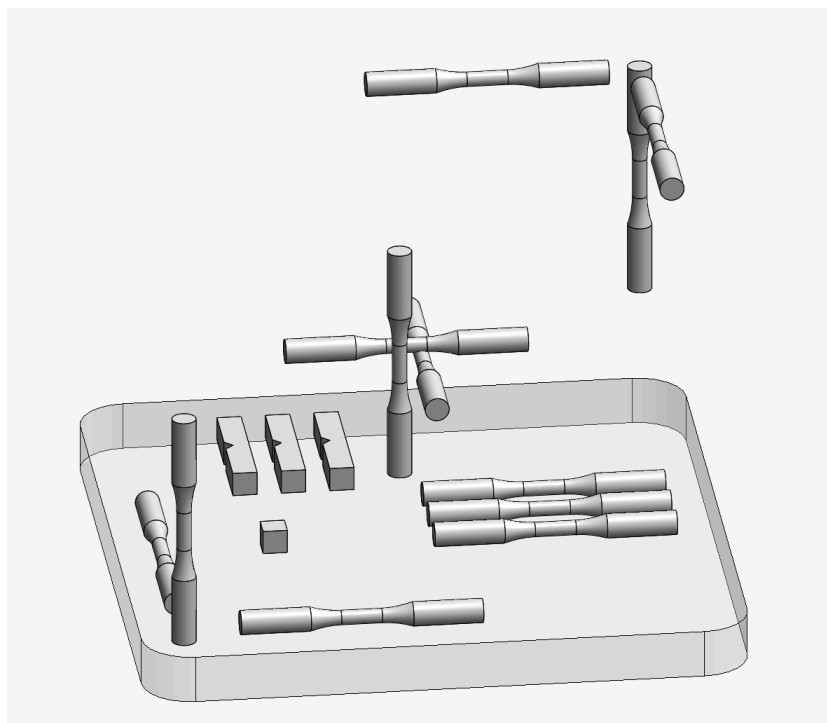


Figure 7-11 Example of specimens and orientation for BPQT

7.6.3 Part qualification

7.6.3.1 AMC 2 and AMC 3 parts shall undergo part qualification testing as described in [Table 7-28](#) and [Table 7-29](#).

- AMC 2 requires non-destructive part qualification testing, in addition to mechanical testing according to the requirements for production testing, see [\[7.6.9\]](#).
- AMC 3 requires non-destructive, mechanical and metallurgical part qualification testing.

7.6.3.2 Additional testing may be required based on the part's intended application, see [\[2.3\]](#).

Table 7-28 Non-destructive test scope during part qualification - relevant for both AMC 2 and AMC 3 parts

<i>Test method</i>	<i>Scope</i>	<i>Testing requirement</i>
Visual inspection	100% as permitted by geometry	[8.3.4]
Dimensional check	In accordance with purchaser requirements	[8.3.10]
Surface NDT	100% as permitted by geometry	[8.3.5]
Volumetric NDT	100% as permitted by geometry	[8.3.6]

Table 7-29 Mechanical and metallurgical test scope for part qualification - relevant for AMC 3 parts

<i>Test method</i>	<i>Orientation and quantity</i>	<i>Specimen extraction</i>	<i>Testing requirement</i>
Tensile testing ¹⁾	1 x Z or parallel to greatest direction of force in the part's intended application	Critical section(s) ²⁾	[8.2.2]
Hardness measurements	Plane of examination: Surface normal in horizontal direction	Critical section(s) ²⁾	[8.2.4]
Macrostructural assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.5]
Microstructural assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.6]
Porosity assessment	Plane of examination: Surface normal in horizontal direction	Thinnest, thickest, and critical section(s) ²⁾	[8.2.7]
Production testing ³⁾	See [7.6.9]		
1) When extraction of tensile test specimens from a part is not possible due to size limitations, the part should be tested in a manner that represents its intended use. 2) See [2.3.2] for defining the critical section(s). 3) Specimens intended for subsequent production testing shall be included and tested in the part qualification. This is to ensure that the production test specimens are representative of the part's properties.			

7.6.4 Validity of qualifications

7.6.4.1 A build process qualification or part qualification is valid as long as the essential parameters used during manufacturing are kept within their limits of qualification. Table 7-30 lists parameters that shall be stated during qualification testing and used during production.

7.6.4.2 A system or procedure shall be in place to ensure that the parameter set used in manufacturing is the same as the parameter set qualified in a BPQT, and that no changes have been made.

7.6.4.3 If the name or ID of a parameter set is changed, without any change to the parameters within, separate documentation shall be added to the relevant BPQR explaining the name change and confirming that the parameters are unchanged. The BPQR itself shall not be revised.

7.6.4.4 For part qualifications, violation of the bounds defined for the specific part family shall lead to a new part qualification. Part families are described in [4.3.2].

Table 7-30 Essential parameters for BJT

<i>Category</i>	<i>Parameter id</i>	<i>Essential parameters</i>	<i>Limits of qualification</i>
Print hardware and software	1-1	Build preparation software	Updates that are likely to affect the build quality or repeatability.
	1-2	BJT machine	Change of brand or model.
	1-3	Software and firmware	Change of software or firmware maker or product name, and major updates, see [5.6.1.5].
	1-4	Printhead	Change of brand or model.
	1-5	Powder reconditioning unit	Brand, model, serial no.: AMC1 or 2 change of brand or model, AMC3 change of serial no.
	1-6	Powder re-conditioning sieve	Type and mesh size.
	1-7	Powder delivery system	Component changes which are likely to change powder delivery to build area.
Feedstock and consumables	2-1	Chemical composition, powder	Any change outside the range given in the feedstock specification.
	2-2	Particle shape	Any change outside the range given in the feedstock specification.
	2-3	Particle size distribution	Any change outside the range given in the feedstock specification.
	2-4	Apparent density	Any change outside the range given in the feedstock specification.
	2-5	Tap density	Any change outside the range given in the feedstock specification.
	2-8	Flow rate	Change in flow rate.
	2-9	Binder	Change of nominal composition, brand or trade name.
Binder jetting process parameters	3-1	Layer thickness	Change of programmed value [μm].
	3-2	Saturation/print density parameter	Change of programmed value.
	3-3	Temperature	Change of $\pm 5^\circ\text{C}$.
	3-4	Binder drop volume/saturation	Change of parameter value.
	3-5	Gas atmosphere (if required)	Change of specified nominal composition.
	3-6	Build job parameter set	Change of parameters used for part.
	3-7	Roller speed	Change of programmed value, rpm.
	3-8	Roller pattern	Change in programmed value.
Curing	4-1	Curing oven	Change of brand or model.
	4-2	Atmosphere	Change of specified nominal composition.

Category	Parameter id	Essential parameters	Limits of qualification
	4-3	Oven temperature	Temperature, $\pm 7^{\circ}\text{C}$.
	4-4	Curing time	Any change in time cycle.
Sintering	5-1	Sintering furnace	Change of brand or model.
	5-2	Process temperature profile	Any change in time and temperature programme profile.
	5-3	Furnace temperature	Temperature, $\pm 7^{\circ}\text{C}$.
	5-4	Temperature control system	Any replacement of heating elements.
	5-5	Atmosphere	Any change in type, pressure, volume.
	5-6	Furnace setup	Change in furnace setup, e.g. heater elements, refractory, fans, control system.
Post processing	6-1	Thermal post processing	Changes in sequence or procedures.

7.6.5 Build file preparation

7.6.5.1 Parts should be placed or nested on the build platform with a defined minimum distance between parts of 1.5 mm - 2 mm to avoid the transfer and interaction of binder between parts.

7.6.5.2 There should be from 30 to 40 layers of powder below the first layer with binder to avoid distortion in the green part.

7.6.5.3 Parts should be positioned at an adequate distance from the side wall as specified by the machine manufacturer, in order to avoid the movement of green parts during the build process.

7.6.5.4 Depending on the AMC of the part, and specific customer requirements, production test specimens or additional test specimens may be required or requested.

7.6.6 Pre-build requirements

7.6.6.1 Preparation of the process shall be as per the machine manufacturer's specifications. Printhead operating parameters, binder grade selection, layer thickness and powder delivery settings shall be referenced against the defined parameter set for the part to be produced.

7.6.6.2 The recoaters shall be inspected for any damage, deformations, or impurities before the start of a new build job.

7.6.6.3 If a machine is used for more than one powder alloy, a powder change procedure shall be in place. The powder change procedure shall be in accordance with the machine manufacturer's recommendations.

7.6.6.4 The powder shall be loaded into the machine in accordance with the machine manufacturer's recommendations.

7.6.6.5 The machine printhead shall be inspected before each job to verify, the drop volume and positioner calibration.

7.6.6.6 A calibration certificate for the BJT machine shall be available. Calibration shall be checked and verified on a regular basis. The BJT machine shall be calibrated in accordance with the requirements given in [5.6]. The procedure shall as a minimum include calibration of the:

- binder drop volume
- print head position
- heater temperature
- build box volume and levelling
- X,Y,Z recoater position and levelling.

7.6.6.7 The calibration shall be validated as specified by the machine manufacturer.

7.6.6.8 Curing ovens shall be calibrated by qualified personnel in accordance with the curing oven manufacturer's specifications.

7.6.6.9 Heat treatment and sintering furnaces and equipment shall be calibrated according to ISO 17663, SAE AMS-H-6875, ASTM A991, SAE AMS2750 or the equivalent.

7.6.7 Build requirements

7.6.7.1 Unlike the other powder bed technologies described in this guideline, BJT requires curing and sintering steps after the addition of the binder. These steps are in this standard defined to be part of the build process, and thus requirements and recommendations regarding these steps are described in this section.

7.6.7.2 If a roller needs to be replaced during a build, the build job shall be cancelled and repeated. The roller shall be inspected after each build job for signs of wear or damage.

7.6.7.3 The following parameters shall be logged throughout the entire build job under an individual build ID:

- image of volume layout (nesting of parts)
- start and end time, and duration
- number of layers
- build packing efficiency
- software name and version
- machine type and serial number
- printhead type and serial number
- part file.

7.6.7.4 As a summary of the logged data, the build report shall include as a minimum:

- a reference to the unique build job ID
- the unique ID of the applied parameter set
- the idle time between layers
- error messages or warnings.

7.6.7.5 The parts or build volume shall be handled with care before and after the curing phase to ensure that the part is not damaged. Internal features may be damaged during this phase in a way that cannot be detected visually.

7.6.7.6 The curing cycle details shall be logged.

7.6.7.7 All loose powder shall be removed after curing.

7.6.7.8 Parts shall be carefully handled when being positioned in and extracted from the sintering furnace and placed on suitable carriers for sintering. Mishandling of green parts may lead to damage or cracking.

7.6.7.9 Parts should be placed in the sintering furnace in accordance with the intended orientation. For qualified parts, the parts and setters should be placed in accordance with the placement used during qualification.

7.6.7.10 Pre-defined test blocks may be built within the build volume to measure the variation in shrinkage and density during the sintering process across the entire build chamber.

7.6.7.11 The part(s) should not show discolouration after sintering.

Guidance note:

Discolouration after sintering may indicate air leakage.

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7.6.7.12 Any deviations from the qualified procedure in the sintering phase shall be documented.

7.6.7.13 If setters are used during sintering to avoid distortion, these should be coated with a barrier coating so that they do not stick to the part.

7.6.8 Post processing requirements

7.6.8.1 Due to the sintering stage of the metal binder jetting process, additional thermal post processing may not be applicable. If applicable, additional thermal post processing, such as annealing and hot isostatic pressing (HIP), shall be performed in accordance with ASTM F3301, or a qualified procedure, see [6.4].

7.6.9 Production testing

7.6.9.1 Production testing with the scope described in Table 7-31 is required for all parts. For parts intended for serial production, the frequency shall be agreed between the purchaser and manufacturer.

7.6.9.2 The orientation of the production tensile test specimens should be according to the orientation of the part to be built, or according to the forces acting upon it in its intended application. If not specified, the orientation should be vertical.

7.6.9.3 If the parts in a build job are to undergo different sintering cycles, or be sintered separately, additional production test specimens shall be added to ensure the required production testing is performed for all sintered lots. Nonconformity to the production tests may be assessed on a case-by-case basis in agreement with the purchaser.

Table 7-31 Test scope for production testing for BJT

<i>Additive manufacturing category</i>	<i>Test method</i>	<i>Quantity and orientation</i>	<i>Testing requirement</i>
AMC 1	Tensile testing	1 in Z direction	[8.2.2]
AMC 2	Tensile testing	1 in Z direction	[8.2.2]
	Impact testing	1 set, horizontal	[8.2.3]

<i>Additive manufacturing category</i>	<i>Test method</i>	<i>Quantity and orientation</i>	<i>Testing requirement</i>
AMC 3	Tensile testing	2 in Z direction, or in accordance with the orientation specified for part qualification.	[8.2.2]
	Impact testing	1 set, horizontal	[8.2.3]
	Hardness measurements	Same orientation as macrostructural assessment.	[8.2.4]
	Macrostructural assessment	Normal to build direction	[8.2.5]
	Porosity assessment	See [8.2.7]	[8.2.7]
	Chemical analysis	Any orientation.	[8.2.8]

SECTION 8 TESTING AND INSPECTION METHODS

8.1 General

This section describes requirements related to testing and inspection. The scope of testing is described for build process qualification testing, production testing, and part qualification testing for each AM technology and AMC part in [Sec.7](#).

Testing and inspection shall be performed after all the thermal post processing steps stated in the MPS have been performed.

8.2 Mechanical and metallurgical testing

8.2.1 General

8.2.1.1 Test laboratories performing the testing shall have a QMS conforming to ISO/IEC 17025 or the equivalent. If the laboratory is not certified according to ISO/IEC 17025, the QMS shall be approved by the purchaser, or the testing witnessed by an independent third party.

8.2.1.2 The test results shall be recorded and stored such that the test direction and build direction are clearly indicated, in addition to the test specimen types, geometries, and dimensions.

8.2.1.3 ASTM F3122 should be consulted before carrying out tests according to the standards referred to in this standard.

Guidance note:

Most testing standards referred to in this standard have been developed for conventionally manufactured materials. Many factors may influence the reported properties of additively manufactured materials, and this issue should be addressed to the extent reasonably possible.

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8.2.1.4 Unless otherwise specified by the purchaser, the specimens shall be in their final thermal heat-treated state and machined before testing.

8.2.1.5 For PBF and BJT: the coordinate system test direction and build direction shall be in accordance with ISO/ASTM 52921.

8.2.1.6 For PBF and BJT: the orientation of the test specimens may be adjusted to avoid recoater interference.

8.2.1.7 For DED: the test direction shall be given as a function of both the build direction and deposition path.

8.2.2 Tensile testing

8.2.2.1 Tensile testing shall be performed according to ASTM E8/E8M, ISO 6892-1, or the equivalent. The following parameters shall be reported:

- yield strength
- ultimate tensile strength
- elongation
- reduction of area.

8.2.2.2 Acceptance criteria shall be in accordance with the relevant design and material standards as stated in the MPS.

8.2.2.3 The orientation of the test specimens shall be reported along with the results.

8.2.2.4 When a part consists of a substrate and an AM feature, or multiple AM processes, tensile testing of the transition zone shall be performed according to ISO 4136 or ASTM E8/E8M or the equivalent. The scope of testing is described specifically for the qualification processes for each AM technology. Unless otherwise specified, the tensile strength shall not be less than the minimum value specified for the material with the lowest tensile strength. Specimen extraction for transverse tensile testing is shown in Figure 8-1. The specimen size for the transition zone should be the same as for the tensile testing of the deposited material if the geometry allows.

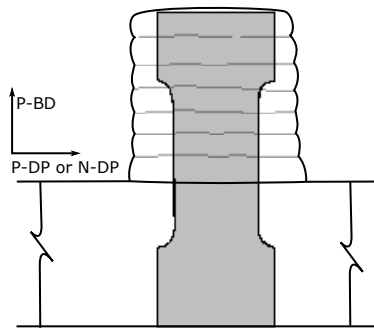


Figure 8-1 Sketch of transverse tensile testing specimen location

8.2.3 Impact testing

8.2.3.1 Impact testing shall be performed by means of Charpy V-Notch testing according to ISO 148-1, ASTM E23 or the equivalent.

8.2.3.2 For horizontally oriented impact test specimens, the notch shall be oriented parallel to the build direction.

Guidance note:

The notch of horizontally oriented test specimens is oriented parallel to the build direction to include as many build layers as possible in the notch.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

8.2.3.3 The acceptance criteria and test temperature shall be in accordance with the relevant design and material standard as stated in the MPS.

8.2.3.4 If no requirement of impact testing is stated for a part, impact testing shall nevertheless be performed as a quality control measure and carried out at room temperature. The production testing results shall be compared to the results obtained during the build process qualification. If the results are significantly below those which were obtained during the BPQ, appropriate measures shall be implemented to find the cause. When comparing results with the results achieved in the qualification, the size of the test specimens shall be the same.

8.2.3.5 The orientation of the test specimens and notch shall be reported along with the results.

8.2.3.6 When the part consists of a substrate and an AM feature, or when the part is made using multiple AM processes, the impact testing of the transition zone shall be performed as required by the qualification

processes for each AM technology. Three (3) sets of impact test specimens shall be taken from the transition zone and tested. Test locations are shown in [Figure 8-2](#). The specimen axis shall be placed normal to the fusion line with a tolerance within $\pm 20^\circ$. The notch axis shall be oriented parallel to the fusion line with the same tolerance, and the notches shall be placed in the following positions:

- in the deposited material, second layer or higher (deposit)
- on the fusion line
- in the HAZ in the substrate, 2 mm from the fusion line (HAZ-2).

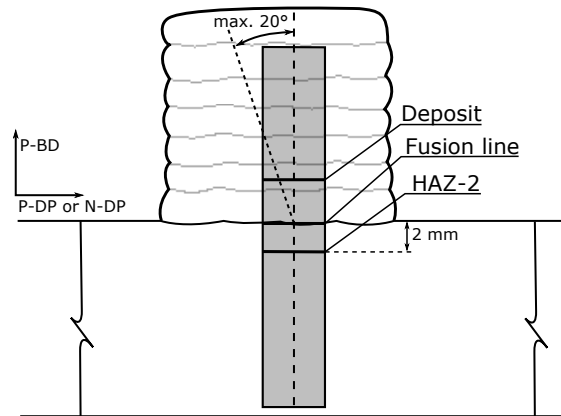


Figure 8-2 Charpy V-notch impact test locations for testing in a fusion zone between a substrate and a feature

8.2.4 Hardness testing

8.2.4.1 Hardness testing shall be performed according to a recognized international standard. Unless otherwise specified, the Vickers hardness test shall be used in accordance with ISO 6507-1 or ASTM E384.

8.2.4.2 Through-thickness measurements shall consist of at least five (5) measurements, equally spaced along the thickness.

8.2.4.3 Acceptance criteria shall be in accordance with the relevant design and material standard as stated in the MPS.

8.2.4.4 When hardness testing is required on the transition between a substrate and an AM feature, or multiple AM processes, one hardness survey shall be performed as shown in [Figure 8-3](#).

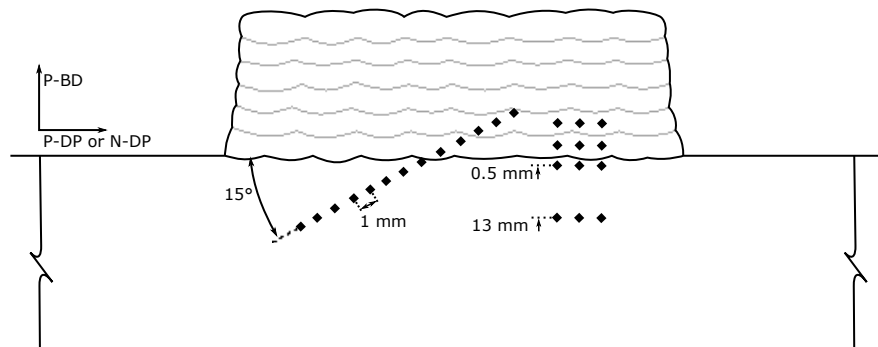


Figure 8-3 Hardness survey for testing in a fusion zone between a substrate and an AM feature

8.2.5 Macrostructural assessment

8.2.5.1 An assessment of the macrostructure shall be performed using 3-5x magnification, typically in the same location as the hardness testing. The macrostructure shall be free from defects characterized as unacceptable in the NDT acceptance criteria in [8.3.7] and [8.3.8].

8.2.5.2 The plane of examination shall be such that the height and width of individual beads may be established, as visualized in Figure 8-4. An assessment of the relative placement of the individual beads and overlap of beads shall be included in the examination.

Guidance note:

Fusion lines between layers may not be visible after thermal post processing.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

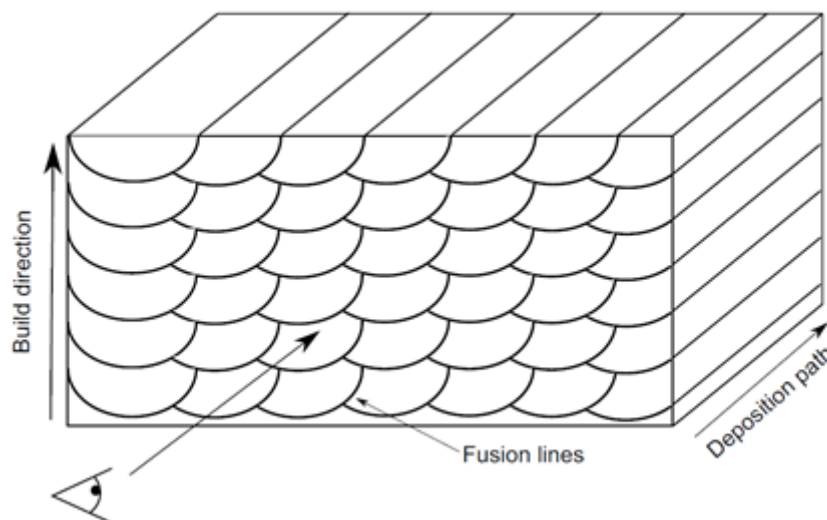


Figure 8-4 Schematic representation of test specimen orientation for macrostructural assessment

8.2.6 Microstructural assessment

8.2.6.1 Sample preparation for microstructural assessment shall be performed in accordance with ASTM E3 and ASTM E407, or the equivalent.

8.2.6.2 For PBF and BJT: an overview image shall be taken at approximately 50 x magnification for a similar assessment as described in [8.2.5].

8.2.6.3 Microstructural assessment shall be performed at minimum 100 x magnification.

8.2.6.4 The microstructure shall be checked against requirements in the relevant design or material standard. Design or material standards for the relevant AM technology shall be used if available. If no design or material standard is available, the acceptance criteria shall be defined. Care should be taken when using standards intended for conventional manufacturing methods.

Guidance note:

Additive manufacturing will create microstructures that are not necessarily directly comparable to available standards for conventional manufacturing methods. This should be taken into account when assessing the microstructure.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

8.2.6.5 For PBF: when available, the microstructure requirements given in PBF-specific material standards shall be adhered to. The following is a non-exhaustive list of available material standards for PBF at the time of writing:

- ASTM F2924 - Titanium-6 Aluminum-4 Vanadium
- ASTM F3001 - Titanium-6 Aluminum-4 Vanadium ELI (Extra Low Interstitial)
- ASTM F3055 - Nickel Alloy (UNS N07718)
- ASTM F3056 - Nickel Alloy (UNS N06625)
- ASTM F3184 - Stainless Steel Alloy (USN31603)
- ASTM F3302 - Titanium alloys.

8.2.7 Porosity

8.2.7.1 A porosity assessment shall be performed using appropriate methods, e.g. through digital imagery processing of micrographs.

8.2.7.2 Porosity may be determined using the parts to be delivered, test specimens, or the same specimens used for macro-/microstructural assessment, depending on the relevant control step, method used and part size. When porosity is measured on the same specimen used for macro-/microstructural assessment by means of digital imagery, the porosity shall be measured prior to etching.

8.2.7.3 For DED: the acceptance criteria for porosity shall be as described in the relevant NDT acceptance criteria.

8.2.7.4 For PBF: the sample on which porosity assessment at the surface is performed should not be machined. Both the contour and bulk should be visible in the plane of examination.

Guidance note:

An examination of the plane in which both the surface and bulk are visible allows any pores in the intersection between the contour and bulk laser scanning pattern to become visible.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

8.2.7.5 For PBF and BJT: the determined porosity shall not exceed the porosity stated in the relevant material specification. Unless otherwise specified, the maximum pore content shall be 0.5% for PBF, and 3%

for BJT. In addition, the diameter of a single pore shall be maximum three (3) times the layer thickness used. The pores shall be homogeneously spread throughout the material.

8.2.8 Chemical analysis

8.2.8.1 Chemical analysis shall be performed according to a recognized standard. For example, ASTM A751 for steel alloys. Appropriate methods and standards shall be used based on what is being analysed, and the amount and type of the relevant element.

8.2.8.2 The acceptance criteria shall be in accordance with the relevant design or material standard.

Guidance note:

When assessing the chemistry in the as-built condition, care should be taken not to compare it directly with the feedstock specification. The feedstock may be intentionally alloyed to compensate for a change in chemistry during building.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

8.2.8.3 For PBF: when available, chemical analysis shall be performed in accordance with PBF-specific material standards. See [8.2.6] for guidance on such standards.

8.3 Inspection and non-destructive testing

8.3.1 General

8.3.1.1 This section provides requirements and recommendations for the visual inspection and non-destructive testing (NDT) of additively manufactured parts.

8.3.1.2 Visual inspection and non-destructive testing shall be carried out as per documented and qualified procedures based on relevant standards, with the additional requirements given in this standard.

8.3.1.3 The build direction and any other factors that can influence the inspection shall be clearly indicated or communicated to the personnel performing the NDT.

8.3.1.4 Carbon steels and low alloyed steels with a tensile strength of less than 490 MPa, made without final heat treatment or stress relief, shall be inspected at least 24 hours after the build process. For steel with a tensile strength of more than 490 MPa, the NDT shall be performed after at least 48 hours. For materials and processes known and proven to be non-sensitive to delayed cracking, this requirement may be omitted.

8.3.1.5 Non-conforming parts shall be quarantined and reported to the purchaser. Non-conformity may be justified based on a special fitness for service analysis. The use of such analyses and acceptance of non-conformity shall be agreed with the purchaser and performed on a case-by-case basis.

8.3.1.6 The requirements given in this section are based on existing and established practices for non-destructive testing. The use of advanced NDT techniques or in-process monitoring techniques may replace the conventional inspection and testing of parts. However, such techniques shall have documented and qualified capabilities. See App.D for recommendations regarding the qualification of conventional and advanced techniques for simple and complex AM parts.

8.3.2 Personnel requirements

8.3.2.1 Companies performing NDT activities shall comply with the impartiality requirements of type A or type B inspection bodies in ISO/IEC 17020, as applicable.

8.3.2.2 Personnel responsible for NDT activities shall be certified to level 3 in accordance with a recognized certification scheme based on ISO 9712, ASNT Central Certification Program (ACCP), or an equivalent third-party certification scheme for the method(s) concerned. Such activities include administering the competence and qualification of the operators, state of the equipment and work procedures. The level 3 shall ensure that the application of NDT is considered at the appropriate step in the manufacturing process.

8.3.2.3 Personnel performing interpretations of results of NDT, shall be qualified and certified to minimum level 2 in accordance with a recognized certification scheme based on ISO 9712, ASNT Central Certification Program (ACCP), or an equivalent third-party certification system for the method(s) concerned. Employer-based certification schemes, such as EN 4179 or ANST SNT-TC-1A, may be accepted upon agreement.

8.3.2.4 Personnel performing visual inspection shall be qualified in accordance with the manufacturer's documented procedures or a third-party scheme that meets the requirements of ISO 9712 level 2 for Visual Testing (VT) or the equivalent.

8.3.2.5 For methods and techniques for which no recognized certification scheme exists or is being developed, it shall be the NDT service provider's responsibility to provide evidence of appropriate experience and training per its written practice. See [App.D](#) for additional guidance.

8.3.2.6 The NDT operators shall have adequate training in the particular AM deposition methods and associated discontinuities and indications.

8.3.3 Scope of inspection

8.3.3.1 The scope of inspection is defined for each AM technology based on the AMC. When NDT is specified in the manufacturing process, inspection shall apply to 100% of all components. Where NDT cannot be performed due to geometry or material characteristics, the scope of inspection will be agreed by the parties.

8.3.3.2 Reducing the test scope may be relevant for AM processes that produce many parts in one build or in a continuous production run without interruption, or a change in parameters or consumables. At agreed frequencies, a part shall be sampled from the build or build series and inspected 100%. The first production article shall always be inspected. If a part is found unacceptable, then the entire build or build series shall be tested.

8.3.3.3 Reduction of the test scope, or waiving of NDT processes, may be implemented based on continuous achievement of mutually agreed performance targets, i.e. the rejection rate.

8.3.3.4 Further information about the scope of inspection is given in the AM technology-specific sections.

8.3.4 Visual inspection

8.3.4.1 All parts shall be visually inspected as per EN 13018 or the equivalent on all accessible surfaces in the final condition to be supplied to the purchaser. The light intensity at the examined surface shall be minimum 500 lx, recommended 1000 lx. Visual aids (mirrors, borescope, cameras, etc.) shall be applied where direct visual inspection is not possible.

8.3.4.2 Parts shall be free from cracks, surface pores, distortions, significant discolouration, surface contaminants or any other imperfections which, due to their nature, degree, or extent, will interfere with the use or subsequent manufacturing operations of the part(s).

8.3.5 Surface testing

8.3.5.1 When surface NDT is required, it shall be performed, using a suitable technique on all the accessible surfaces of the part in its final delivery condition, after thermal post processing and machining. Examples of surface testing techniques are given in [Table 8-1](#). Other techniques may be used, see [App.D](#) for recommendations.

Table 8-1 Overview of applicable surface NDT methods and standards

<i>Method</i>	<i>Standards</i>
Magnetic particle testing (MT)	ISO 9934
Penetrant testing (PT), colour contrast or fluorescent technique	ISO 3452
Eddy current testing (ET)	ISO 15549

8.3.6 Volumetric testing

8.3.6.1 When volumetric NDT is required, it shall be performed after final thermal post processing.

8.3.6.2 The use of standards intended for conventional manufacturing methods shall be justified by the manufacturer. Examples of volumetric testing techniques are given in [Table 8-2](#). Other techniques may be used, see [App.D](#) for recommendations.

Table 8-2 Overview of some applicable volumetric NDT methods and standards

<i>Method</i>	<i>Standards</i>
Conventional ultrasonic testing (UT)	ISO 16810
Radiographic testing (RT)	ISO 5579, Class B
Computed tomography (CT)	ISO 15708

8.3.6.3 The volumetric method shall be optimized to detect defects in all potential directions and orientations.

8.3.6.4 Ultrasonic calibration blocks and reference blocks shall have acoustic properties that are equivalent to those of the object to be tested, or be made so that differences in acoustic properties may be compensated for. In order to achieve this, the blocks and the parts being tested should be built under similar conditions, e.g. build parameters, build direction and feedstock, as the part being tested.

8.3.6.5 For UT of parts made by powder bed AM technologies, i.e. PBF and BJT, unless otherwise specified in the design or product standard, a reference reflector of 1.2 mm flat-bottom hole or equivalent notch may be used for sensitivity setting.

8.3.7 Acceptance criteria for directed energy deposition

8.3.7.1 Unless acceptance criteria are defined by the design or product standard, the criteria in [\[8.3.7.2\]](#), shall be applied.

8.3.7.2 For DED methods, the acceptance criteria shall follow the acceptance criteria defined for irregularities in welds, corresponding to ISO 5817 level C for AMC 1 and AMC 2 parts, and level B for AMC 3 parts. Correlation between ISO 5817 and method and acceptance standards for specific NDT method is given in ISO 17635.

8.3.8 Acceptance criteria for powder bed fusion and binder jetting technology

8.3.8.1 Unless acceptance criteria are defined by the design or product standard, the following criteria shall be applied.

8.3.8.2 For powder bed technologies, e.g. PBF and BJT, the acceptance criteria for surface irregularities are given in [Table 8-3](#).

8.3.8.3 Indications shall be evaluated to determine if they are of a cosmetic nature, or if they are process related. Cosmetic indications are not associated with the AM process steps, but with handling, transportation, or storage. Cosmetic indications shall be evaluated according to a visual inspection criterion that ensures the agreed surface conditions. Material defects or abnormalities arising from any steps in the manufacturing process, i.e. building, grinding, machining, heat treatment, shall be evaluated per the acceptance criteria.

8.3.8.4 Indications shall be categorized as linear indications if the length to width ratio is 3 or higher. If the length to width ratio is less than three (3), the indication shall be considered a round indication. The size is measured by the indication size and not the actual flaw causing the indication.

Table 8-3 Acceptance criteria for surface inspection for PBF and BJT

<i>indication type</i>	<i>AMC 2</i>	<i>AMC 3</i>
<i>Machined surfaces</i>		
Linear indications	$\leq 2 \text{ mm}$	$\leq 1 \text{ mm}$
Round indications	$\leq 3 \text{ mm}$	$\leq 2 \text{ mm}$
Grouped indications ¹⁾	$\leq 5 \text{ mm}$	$\leq 3 \text{ mm}$
<i>As printed surfaces</i>		
Linear indications	$\leq 3 \text{ mm}$	$\leq 2 \text{ mm}$
Round indications	$\leq 4 \text{ mm}$	$\leq 3 \text{ mm}$
Grouped indications ¹⁾	$\leq 6 \text{ mm}$	$\leq 5 \text{ mm}$
1) Indications with a distance between them of less than two (2) times their largest dimension are considered grouped indications. The largest dimension enclosing the group is evaluated against the acceptance criteria.		

8.3.8.5 For powder bed methods, e.g. PBF and BJT, the acceptance criteria for volumetric irregularities are given in [Table 8-4](#).

8.3.8.6 Indications measuring 1 mm or more in its longest dimension are considered relevant indications. If inspected by UT with a reference curve corresponding to a 1.2 mm FBH, indications giving a response equal to or higher than 6 dB below the reference curve are considered relevant. For thicknesses of less than 10 mm, the size to be considered relevant should be considered and agreed.

Table 8-4 Acceptance criteria for volumetric inspection for PBF and BJT

<i>Defect type</i>	<i>AMC 2</i>	<i>AMC 3</i>
Relevant linear or planar indications (such as cracks, lack of fusion etc.)	Not accepted	Not accepted
Relevant single irregularity	< 10% of thickness but max. 3 mm ¹⁾	< 10% of thickness but max. 2 mm ¹⁾
Grouped indications ²⁾	Diameter of cluster envelope < 50% of thickness	Diameter of cluster envelope < 25% of thickness
1) For thickness less than 10 mm, the acceptable size will be determined by what is considered relevant, thus this definition should be considered and agreed. 2) Grouped indications are indications with a distance between them of less than two (2) times their largest individual dimension.		

8.3.9 Inspection of repairs

8.3.9.1 When a component is repaired and inspected, the inspection should to the extent possible be carried out in the same way as on the original part. For conventionally made parts, e.g. pipes, forgings and castings, that have been in service and repaired by using AM, the requirements for the inspection shall be agreed with the original part manufacturer. When repairing AM parts due to damage or flaws from the manufacturing process, the repair shall be carried out using a qualified process and inspected as the original part.

8.3.9.2 The addition of features to already finished components by the use of AM shall not be considered a repair, but a design change. In these cases, special consideration shall be given to the boundary between the part and AM feature.

8.3.9.3 Machining within the design tolerances shall not be considered a repair.

8.3.10 Dimensional control

8.3.10.1 A dimensional check shall be performed using an appropriate method based on relevant sizes and tolerances.

8.3.10.2 The acceptance criteria shall be in accordance with the information given by the purchaser.

APPENDIX A DOCUMENTATION SUMMARIES AND EXAMPLES

A.1 General

This appendix contains summarized lists of documentation requirements and examples of templates for purchase order information, criteria specifications, part production specifications, build process qualification records and part qualification records. Other formats may be used, as long as all the required information is detailed.

A.2 Documentation requirements

Lists summarizing the documentation requirements given in this standard are shown in [Table A-1](#) to [Table A-8](#).

Table A-1 Purchase order information based on [\[2.3\]](#)

<i>Documentation requirement</i>
Customer organization, contact person and shipping address.
Traceability or marking requirements.
Required delivery date for single orders, or required delivery frequency and quantity for multiple orders.
Disposal of rejected material.
Reference to this standard.
Preservation and shipment requirements
Clear definition of the part(s) through digital model(s) in a format accepted by the manufacturer. The part definition shall contain information on the unit of measurement used. The unit system used shall be either SI or the US customary system of units.
Requested AMC.
Quantity of part(s).
Material selection.
For AMC 3 only: definition of critical section(s) of the requested part(s), see [2.3.2] .
Post processing, e.g. thermal processing, surface treatment or machining.
Surfaces of the part(s) where supports are not to be added, if applicable.
Dimensional tolerances, preferably through product manufacturing information or drawings.
Mechanical, chemical and metallurgical acceptance criteria.
NDT acceptance criteria.
Inspection and test plan, indicating the frequency of production testing and NDT.
Surface condition, e.g. as-printed, machined, processed by blasting, tribo-finishing, electrochemical polishing, or passivation treatment.

Table A-2 MPS, Manufacturing procedure specification based on [6.2.1].

<i>Documentation requirement</i>
A descriptive title and a unique identification number for the MPS, with revision control.
Reference to this standard.
The manufacturer's organization name and site.
AM technology.
Feedstock.
Part(s) and quality assurance test specimens to be produced, including acceptance criteria.
Reference to applicable standards or specifications.
Reference to a part production specification (PPS), including references to qualification records, see [6.2.2].
Reference to manufacturing procedures.
Safeguard to ensure traceability of the digital model and manufactured part, e.g. unique ID numbers.
Design history, if applicable.
The number of parts and quality assurance test specimens per build job and their orientation and location on the build platform. An image of the build platform shall be included. For PBF parts, the location of support structures shall be included.
Additional post processing steps not defined in the PPS, including procedures and the sequence.
Testing or inspection requirements in addition to requirements given in this document, if required by the purchaser.

Table A-3 Part production specification based on [6.2.2]

<i>Documentation requirement</i>
The unique ID of the relevant part design.
Reference to a supporting build process qualification record (BPQR).
Reference to a part qualification record (PQR).
Feedstock information.
Description of AM technology used and relevant machine settings.
Essential parameters, see Sec.7.
Thermal post processing steps and sequence.

Table A-4 BPQR, Build process qualification record based on [4.5.2]

<i>Documentation requirement</i>
A documentation summary, stating the identification number, date of qualification, manufacturer/sub-supplier, site and relevant manufacturing standards.
Description of AM method.
Description of AM equipment.

Description of the built qualification coupons.
Essential parameters for the build process, given in the relevant subsection for the specific AM technology in Sec.7 , including post processing details.
Feedstock information, including material certificates or the certificate of conformity for the feedstock used during build process qualification testing, including the powder state (i.e. virgin, reused or blended) if applicable.
Post processing details and sequence.
The results of the build process qualification testing, including laboratory test reports and the orientation and location of test specimens.
NDT reports with a statement of results/compliance, if applicable.

Table A-5 PQR, Part qualification record based on [\[4.5.3\]](#)

<i>Documentation requirement</i>
A documentation summary, stating the identification number, date of qualification, manufacturer/sub-supplier, site and relevant manufacturing standards.
The unique ID of the relevant part design, and the representative section or coupon built for qualification if different than the actual part.
All essential parameters used in building, and a description of the support structures, orientation, and nesting if relevant.
Reference to a supporting qualified build process, BPQR.
Material certificates or the certificate of conformity for feedstock used during part qualification, including the powder state (i.e. virgin, reused or blended) if applicable.
Results from the part qualification testing, including laboratory test reports and the location and orientation of test specimens.
NDT reports with a statement of results/compliance.

Table A-6 Compliance record and marking based on [\[6.6.3\]](#)

<i>Documentation requirement</i>
Reference to relevant purchase order.
Supplier reference number.
The manufacturing technology used.
Confirmation that a qualified build process, including post processing, has been used.
Reference to applicable documentation, e.g. MPS.
Release document, including documentation of final testing/inspection with referenced testing/inspection details.
Any unplanned process interruptions or activities, if relevant.
Feedstock information, including the state of the powder if applicable, i.e. virgin, re-used or blended.
Results from tests carried out, including mechanical testing, dimension control and NDT. The location and orientation of the specimens shall be stated.
The acceptance criteria for the tests carried out, and a clear statement of whether or not the acceptance criteria have been met.

Any other part or application specific tests performed, including results and acceptance criteria.
Risk assessment from part family definition if a pre-existing part qualification was used.
Design specification if the part has been designed by the manufacturer on behalf of the purchaser.

Table A-7 Test results (for reporting test data in e.g. a BPQR)

<i>Documentation requirement</i>
Tensile testing: test procedure, orientation/location/specimen ID, orientation, dimensions [mm], yield strength [MPa], ultimate tensile strength [MPa], elongation [%], reduction of area [%], failure type and location.
Impact testing: test procedure, orientation/location/specimen ID, test result.
Microstructural assessment: test procedure, orientation/location/part ID, macrograph of sample, microstructural assessment at 100x, result compared with requirements in the relevant design or material standard.
Porosity measurement: test procedure, orientation/location/part ID, acceptance criteria, description/image of area(s) measured.
Hardness: test procedure, orientation/location/part ID, test result.
Visual inspection: test procedure, orientation/location/part ID, test result.
Surface testing: test procedure, orientation/location/part ID, test result.
Volumetric testing: test procedure, orientation/location/part ID, test result.

Table A-8 Data from the build process

<i>Documentation requirement</i>
Parameters defined for each AM technology in Sec.7 shall be logged and stored.

A.3 Examples of documentation formats

A.3.1 General

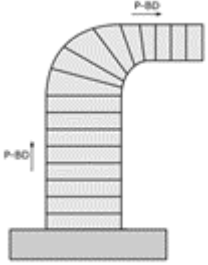
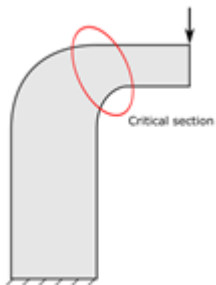
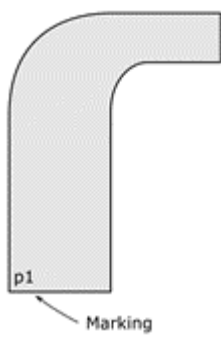
This appendix provides examples of how the documentation can be prepared for purchase orders, PPS, BPQS, and PQR. This is only meant for illustration purposes. Any formats may be used, as long as all the required information is included.

A.3.2 Purchase order information

Examples of a summary of purchase order information and of the specification of additional testing and criteria are shown in [Table A-9](#), [Table A-10](#), [Table A-11](#) and [Table A-12](#).

Table A-9 Example of purchase order information about the part

Purchaser organization:	Name	Purchaser contact person:	Name
Shipping address:	Address		
Additive manufacturing standard:	DNV-ST-B203	Additive manufacturing category (AMC):	AMC3

Additional relevant standards:	...		
Part(s): <i>Insert reference to part digital model.</i>	Part: p1 	Quantity of part(s):	10
Critical section(s): <i>Description, image or similar.</i>		Surfaces where supports are not to be added, if applicable. <i>Description, image or similar.</i>	No supports are added
Material:	<i>Insert material grade, consumable etc.</i>	Manufacturing method	DED-arc
Post processing steps	- Machining description - heat treatment description		
Traceability/marking: <i>Digital traceability, physical marking, etc.</i>	Part to be marked during final machining 		
Delivery date, if applicable:	01.01.2021	Delivery frequency, if applicable:	
Handling of rejected material:	Rejected material shall be recycled according to ...		
Additional requirements:	None		

A.3.3 Acceptance criteria

Table A-10 Example of acceptance criteria

Dimensional tolerances, e.g. minimum wall thickness	
Ultimate tensile strength	Min: 500 MPa
	Max: 700 MPa
Yield strength	Min: 400 MPa
	Max: 700 MPa
Impact toughness	Test temp: 0°C
	Individual [min]: 21 J
	Average [min]: 27 J
Hardness values	Min HV10: 200
Porosity percentage	Max: 4%

A.3.4 Non-destructive testing

Table A-11 Example of NDT requirements

<i>Criteria</i>	<i>Requirement</i>
Visual inspection acceptance criteria	DNV-ST-B203 [8.3.4], ...
Surface NDT acceptance criteria	DNV-ST-B203 [8.3.5], ...
Volumetric NDT acceptance criteria	DNV-ST-B203 [8.3.6], ...

A.3.5 Additional testing

Table A-12 Example of additional testing

<i>Test</i>	<i>Description</i>
Fatigue testing	Fatigue specimens according to ASTM E466 — Number of specimens: 30 — Position of specimens: Distributed around build platform — Orientation of specimens: 10 z-direction, 10 x-direction, 10 y-direction
Bend testing	Three-point bend test according to ISO 5173 — Specimen dimensions: width 50 mm x thickness 5 mm x length 80 mm — Support diameter/distance: 20 mm/32.5 mm — Orientation of specimens: length oriented in z

Test	Description
...	

A.3.6 Part production specification

The PPS shall contain the essential parameters and a reference to the BPQR, and to the PQR if relevant. An example of the essential parameters is provided in [Table A-13](#).

Table A-13 Example of a part production specification

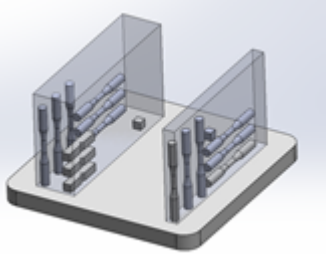
PPS no.:	PPS.01	Revision no:	Rev0
Supporting BPQR ID no.:	BPQR.01	Date:	01.01.2021
Manufacturer/company:	Name	Approved by:	Name
Site:	Location	Approver name:	Name
Part ID:	SN000001	PQR ID no.:	PQR.01
Build file ID:	Build file	Build file version:	v0
AM process:	DED-arc. Plasma welding combined with cold wire feeding (description of AM process and necessary settings if necessary).		
Welding process:	DED-arc, plasma	Feedstock feed mechanism:	Feeder type
Power source:	Equipment brand and model	Positioner:	Positioner brand and model
Computer control system:	Type of system	Build preparation software:	Software and version
Atmosphere control, temperature:	ambient	Atmosphere control, shielding	Inert gas chamber (< 50 ppm O ₂)
Consumable classification or trade name:	(consumable classification or trade name)	Wire diameter:	Ø 1.2 mm
...			

A.3.7 Build process qualification record

The BPQR shall contain a documentation summary, the essential parameters used, material certificates, and the results from the BPQT and NDT. An example of a documentation summary is given in [Table A-14](#). The BPQR should refer to a PPS or a preliminary PPS for essential variables, as shown in the example in [Table A-13](#). The material certificates are designed by the material supplier and are not included. Examples of reporting test data from e.g. the BPQT are shown in [\[A.4\]](#).

Table A-14 Example of a documentation summary for the BPQR

BQR ID no.:	BQR01	Date:	Date
Manufacturer/company:	Name	Approved by:	Name
Operator name:	Name	Approver name:	Name
AM process:	DED-arc		

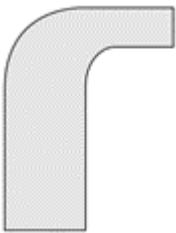
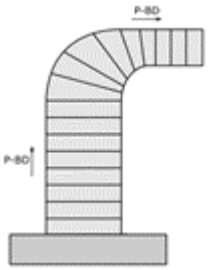
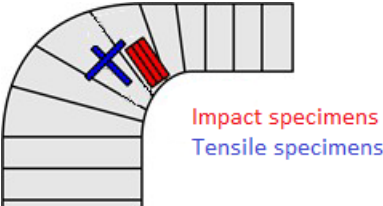
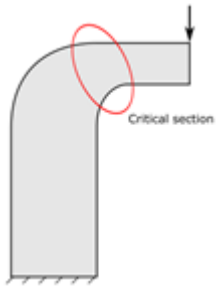
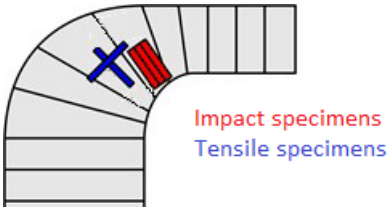
Part 3D-model no.:	01	Build file ID no.:	01
Build process qualification test setup according to DNV-ST-B203:			
			
Deposition strategy:		Cross hatching, 90 degrees per layer	
Tensile specimens, thick coupon: 3 x P-BD direction (T1, T2, T3) 3 x P-DP direction (T4, T5, T6). Tensile specimens, thin coupon: 3 x P-BD direction (T7, T8, T9) 3 x P-DP direction (T10, T11, T12)		ISO 6892-1 Round tensile specimens Ø 6 mm	
Impact specimens, thick coupon: 3 x notch in P-BD direction (C1, C2, C3)		ISO 148-1 Charpy V-notch 10 mm x 10 mm	
2 x cube for macrostructure, microstructure, porosity and hardness, from the thick and thin coupon (M1 and M2)		20 mm x 20 mm x 20 mm	
Chemical analysis sample from remaining material		ASTM A751	
Log of voltage during build:		Log of current during build:	
Log of wire feed rate during build:		Log of gas flow rate during build:	
Log of oxygen content in atmosphere 1:		Log of temperature in atmosphere 2:	

A.3.8 Part qualification record

The part qualification record shall contain a documentation summary, a reference to material certificates, results from part qualification testing and an NDT report. An example of a documentation summary is shown in [Table A-15](#).

Table A-15 Example of a documentation summary for a part qualification record

PQR ID no.:	PQR01	Date:	Date
Manufacturer/company:	Name	Approved by:	Name
Operator name:	Name	Approver name:	Name
AM process:	DED-arc	Reference to BPQR ID no.:	BPQR01
Part description:	Part for illustration		
Part 3D-model no.:	01	Build file ID no.:	01

Image of part: insert image of part taken from CAD viewer or build preparation software.			
Image of build platform: insert image of build platform taken from build preparation software. The image shall clearly show the orientation, placement and number of part(s) and test specimens.			
Specimens:	Tensile	1 x P-BD 1 X P-DP	
	Impact	1 set (3) P-BD	
	Material sample	Sample for hardness testing, and macrostructural and microstructural assessment	
Definition of critical area(s): insert description of critical area, preferably including image.			
Date of test specimen production:			
Image of test specimen extraction:			
Log of voltage during build:		Log of current during build:	

Log of wire feed rate during build:		Log of gas flow rate during build:	
Log of oxygen content in atmosphere during build ¹⁾ :		Log of temperature during build ²⁾ :	
1) Relevant for oxygen sensitive materials, such as duplex or titanium alloys. 2) If built in a closed chamber.			

A.4 Examples of reporting of test results

A.4.1 General

This subsection shows examples of how to report test results. The examples are given in [Table A-16](#) to [Table A-27](#).

A.4.2 Tensile testing

Table A-16 Example of tensile test description

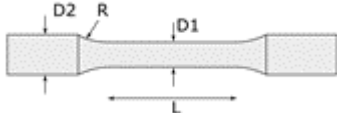
Test procedure:	Tensile test, extensometer
Reference to standard:	ISO 6892-1
Specimen type:	Round tensile bar 
Specimen dimensions:	D1 = 6 mm, D2 = 10 mm, R = 60 mm, L = 30 mm
Acceptance criteria:	Yield strength: min 300 MPa, max 1000 MPa Tensile strength: min 500, max 1000 MPa
Testing performed by:	Name
Date of testing:	Date

Table A-17 Example of tensile test results

	<i>Specimen ID</i>	<i>Orientation</i>	<i>Diameter</i>	<i>Yield strength</i>	<i>Tensile strength</i>	<i>Elongation</i>	<i>Reduction of area</i>	<i>Failure type and location</i>
Acceptance criteria:	N/A	N/A	N/A	Min: Max:	Min: Max:	Min: Max:		Failure in gauge section
Test result 1	T1	P-BD	6.06	700	900	18	60	Cup cone, gauge section

	<i>Specimen ID</i>	<i>Orientation</i>	<i>Diameter</i>	<i>Yield strength</i>	<i>Tensile strength</i>	<i>Elongation</i>	<i>Reduction of area</i>	<i>Failure type and location</i>
Test result 2	T2	P-BD	6.07	701	901	17	61	Cup cone, gauge section
Test result 4	...	...	...	...	...			

A.4.3 Impact testing

Table A-18 Example of impact test description

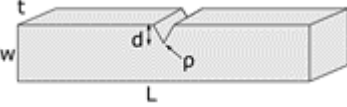
Test procedure:	Charpy V-notch
Reference to standard:	ISO 148-1
Specimen type:	V-notch 
Specimen dimensions:	W=10, t=10, L=50, d= , p=
Acceptance criteria:	Average (minimum acceptable value): 27 J Individual (minimum acceptable value): 21 J
Testing performed by:	Name
Date of testing:	Date

Table A-19 Example of impact test results

	<i>Specimen ID</i>	<i>Orientation</i>	<i>Width</i>	<i>Thickness</i>	<i>Area</i>	<i>Impact toughness [J]</i>	<i>Shear percentage [%]</i>
Acceptance criteria	N/A	N/A	N/A	N/A	N/A	Average (min): 27 Individual (min): 21	N/A
Test result 1	C1	Notch in P-BD direction	10	10		66	15
Test result 2	C2	Notch in P-BD direction	10	10		50	30
...	...	...	...	...	...	...	...
Average value						55	20

A.4.4 Bend testing

Table A-20 Example of bend test setup

Test procedure:	Three-point bending, support diameter: 20 mm, support distance: 32.5 mm
Reference to standard:	ISO 5173
Specimen type:	
Specimen dimensions:	50 mm x 5 mm x 80 mm
Acceptance criteria:	N/A
Testing performed by:	Name
Date of testing:	Date

Table A-21 Example of bend test results

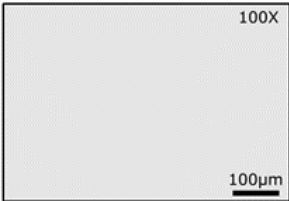
	<i>ID</i>	<i>Orientation</i>	<i>Width</i>	<i>Thickness</i>	<i>Support diameter</i>	<i>Support distance</i>	<i>Bending angle</i>	<i>Result</i>
Acceptance criteria	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Test result 1	B1	P-BD	49.6	5.05	20	32.5	180°	
Test result 2	B2	P-DP	49.9	5.04	20	32.5	180°	

A.4.5 Microstructural assessment

Table A-22 Example of microstructure test setup

Test procedure:	— Cross section polished. Final step: 1µm diamond paste — Cross section etched with HNO3 etchant — Image of microstructure obtained by optical microscope
Reference to standard:	
Specimen type:	Cube, M1
Specimen dimensions:	50 mm x 5 mm x 80 mm
Acceptance criteria:	N/A
Testing performed by:	Name
Date of testing:	Date

Table A-23 Example of microstructure test results

Acceptance criteria:	N/A		
Description/image of area(s) analysed:	XZ-cross section of cube	Magnification:	200X
Part thickness at sectioning plane:	100 mm	Orientation:	XZ-plane
Image of microstructure:			

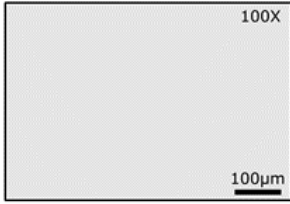
A.4.6 Porosity

Table A-24 Example of porosity test setup

Test procedure:	— Cross section polished. Final step: 1µm diamond paste — Image of microstructure obtained by optical microscope — Porosity percentage calculated by ...
Reference to standard:	
Specimen type:	Cube, M1
Specimen dimensions:	20 mm x 20 mm x 20 mm
Acceptance criteria:	N/A
Testing performed by:	Name
Date of testing:	Date

Table A-25 Example of porosity test results

Acceptance criteria:	Maximum 4% porosity		
Description/image of area(s) analysed:		Magnification:	100X
Part thickness at sectioning plane:	100 mm	Orientation:	XY-plane
Measured percentage porosity:	2%		

Image of microstructure	
-------------------------	---

A.4.7 Hardness testing

Table A-26 Example of hardness test setup

Test procedure:	— Cross section polished. Final step: 1µm diamond paste — Vickers hardness (HV10) measurements done on a ... system
Reference to standard:	
Specimen type:	Cube, M1
Specimen dimensions:	20 mm x 20 mm x 20 mm
Acceptance criteria:	N/A
Testing performed by:	Name
Date of testing:	Date

Table A-27 Example of porosity result

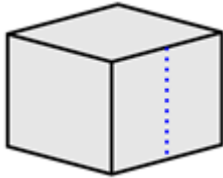
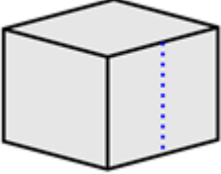
Acceptance criteria:	Minimum value of HV10 : 200		
Description/image of area(s) analysed:	Hardness measured along path parallel to z-direction in the xz-plane. Distance between measurement was 1 mm. 		
Part thickness at sectioning plane:	100 mm	Orientation:	XY-plane

Image of hardness measurements and location:											
Hardness measurement and location:	<table border="1" data-bbox="488 562 688 720"> <thead> <tr> <th>Location</th><th>Hardness (HV10)</th></tr> </thead> <tbody> <tr> <td>0</td><td>400</td></tr> <tr> <td>1</td><td>400</td></tr> <tr> <td>2</td><td>400</td></tr> <tr> <td>3</td><td>400</td></tr> </tbody> </table>	Location	Hardness (HV10)	0	400	1	400	2	400	3	400
Location	Hardness (HV10)										
0	400										
1	400										
2	400										
3	400										

APPENDIX B SELECTION OF AMC

B.1 General

The intention of this appendix is to give guidance on the selection of additive manufacturing category (AMC). The AMCs are defined in [2.2] in terms of the criticality of the part produced.

The selection of the AMC should be a risk-based decision. Risk is a function of the probability of failures and the consequence of failures. This standard gives directions on how to ensure consistently high integrity in the produced parts, which leads to risk mitigation by lowering the probability of failure. However, there are aspects of the probability that are not addressed in this standard, such as usage with higher utilization. The overall risk shall be addressed on a case-by-case basis in a risk assessment, and some recommended topics for assessment are listed in Table B-1.

Table B-1 A non-exclusive list of topics for risk assessment in the process of selecting the AMC

<i>Topic</i>	<i>Description</i>	<i>Effect on the AMC</i>
Consequence of failure	In a scenario where the part fails in operation, the consequence of that failure may result in — health and safety being compromised — long repair or lead time, causing operational downtime — damage to equipment or facilities	A higher consequence of failure may be an argument for using a higher AMC.
Safety factor	An increase in the safety factor will increase the confidence in a part. This could be done either by selecting a material that is superior compared to the requirements, increasing wall thicknesses, or similar.	A higher safety factor has a positive impact on the confidence in the final part, and may be an argument for the selection of a lower AMC.
Hot isostatic pressing (HIP) and machining	Hot isostatic pressing and subsequent machining may result in greater control of some of the defects that may be present in the final part.	The adoption of HIP may be used as an argument for the selection of a lower AMC.
Manufacturing experience of the type of part, or the specific build	The degree of proven experience of the manufacturer.	May be an argument for lowering or increasing the AMC.

When choosing the AMC based on the uncertainty in the production, the aim of increasing the AMC is to capture variations caused by the part geometry and individual production batches. Increasing the AMC will trigger more quality control measures in production, and is intended to ensure that the production is representative and repeatable compared to the build process qualification. If there is a low risk associated with the representativeness and repeatability of the build process, a lower AMC could be considered. Further, AMC 3 has additional quality control measures to ensure that a specific part geometry has acceptable properties. When the part's geometry may cause critical areas in the part where unacceptable properties are more likely to occur, or when there is a lot of uncertainty about the representativeness of the BPQ considering the specific part geometry, AMC 3 should be chosen.

Based on the aspects described in Table B-1, the designer may define the AMC for specific regions of the part. The definition of AMC shall in such case be accepted by the purchaser.

APPENDIX C PART FAMILIES

C.1 General

This appendix is intended to aid the definition of part families. By defining part families, the qualification of parts becomes a more cost effective and efficient process. Through leveraging the qualification process for parts that are likely to both behave and be used similarly, repetitive testing can be reduced significantly.

The appendix provides guidance on the use of part families and how such part families may be defined. Part families should be defined with regard to both their application and the AM technology, as each poses unique risks.

C.2 Using an existing qualification for a new part

The use of part families may impose some risk, depending on the changes made to a new part compared to the qualified one. When using part families, the purchaser and manufacturer should be aware of these potential risks and have a systematic approach for assessing their relevance and significance.

Therefore, the use of an existing qualification for a new part should be supported by an appropriate risk assessment, and the risk assessment should be documented. The risk assessment should include the rationale for putting the new part in the same part family as the qualified part, as well as what risks may be involved and the justification for accepting these risks. This risk assessment should include both application and manufacturing aspects. A non-exhaustive list for the application and different AM technologies is provided in [C.3].

For parts where there is lower risk involved in using an existing qualification, a reduced testing approach may be applicable. These should be handled on a case-by-case basis and the justification should be included in the risk assessment.

C.3 Defining part families

C.3.1 Application

The definition of part families with regard to application should be done by the purchaser or designer, based on knowledge about the application and potential failure modes. Table C-1 contains topics that should be assessed, and recommendations for when a part can be considered defined as a member of a part family.

Table C-1 Application topics and recommendations to be assessed when defining part families

<i>Topic</i>	<i>Recommendations</i>
The part's geometry	Changes to the general geometry and scale should be limited.
The part's application	Changes to the application should be assessed, i.e. function, service environment and relevant failure modes in design.
The critical section	Changes to the definition of critical sections should be assessed. The critical section should remain the same as the critical section for the qualified part.
Acceptance criteria	The part should have the same or less stringent acceptance criteria as the qualified part.
Design stresses/utilization	The utilization of the part expressed as design stresses compared to the acceptance criteria should be equal to or lower than that of the qualified part.

C.3.2 AM technology

The definition of part families with regard to design and AM technology should be used in addition to the part family definition for application given in Table C-1. Table C-2, Table C-3, and Table C-4 contain topics that should be assessed related to design and manufacturing methods and include recommendations for when a part can be considered allocated to a part family. These definitions are dependent on the AM technology used, as there are unique risks associated with each technology.

The experience of the relevant manufacturer in building similar parts should be considered in the risk assessment.

Table C-2 General topics and recommendations to be assessed when defining part families

<i>Topic</i>	<i>Recommendations</i>
Essential parameters	Changes of essential parameters should be kept within the window of tolerance used for the qualified part. ¹⁾
Special features	Changes to the design in the critical section should be assessed. Additional features added to or removed from a critical section may impact the material properties. Features include, but are not limited to, holes, notches, splines, and fillets.
Replanning/reslicing ²⁾	Changes to the path planning or slicing should be assessed. The planning should be similar to the planning used in the qualification.
1) Keeping the process within the qualified essential parameters is a requirement detailed in the main chapters of this standard. 2) Replanning or reslicing may be necessary for building a new part. Depending on how path planning or slicing is performed, there may be a degree of randomness involved.	

Table C-3 Powder bed specific topics and recommendations to be assessed when defining part families

<i>Topic</i>	<i>Recommendations</i>
Active boundary box	The active boundary box in the build chamber of the machine should be assessed. An increase in the size may for example result in a lower build temperature.
Wall thickness	A change of wall thickness to a lower or higher wall thickness than the qualified part should be considered. The wall thickness may impact the cooling rate, thus leading to deformation of the part, residual stress, microstructure technology etc.
Thick-thin transitions	Changes to thickness transitions should be assessed. Abrupt thickness transitions may cause local changes to the microstructure, defects, delaminations, etc.
Main orientation and placement	Changes to the main orientation and relative placement of the parts and specimens in the build chamber should be assessed. The same orientation and relative placements as in the qualification should be used.
Support structures	The type and extent of support structures should remain similar.
Adding or removing parts	The addition or removal of parts to/from a qualified build job should be assessed.

Guidance note:

For any changes of a qualified part in powder bed technologies, the part should be re-sliced and the build job may therefore not be identical to the qualified part. For crucial applications, the same build file as used for qualification provides the most confidence.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

Table C-4 Directed energy deposition specific topics and recommendations to be assessed when defining part families

<i>Topic</i>	<i>Recommendations</i>
Active build platform area	Changes to the active area on the build platform should be assessed. The active area on the build platform should be similar to that for the qualified part, i.e. the actual area of the substrate where deposition occurs.
Surface-volume ratio	Changes to the surface to volume ratio of the part during deposition should be assessed. Changes to this may impact the cooling rate during deposition.
Deposition strategy	The deposition strategy should not deviate significantly from that used in the qualification. This means that: — the relative locations of the start and stop positions should be similar — the effect on additional features in the deposition strategy should be assessed — the main orientation of the part to be built should be similar, keeping a similar deposition strategy as a function of the build direction.

Guidance note:

The deposited and heat-treated material is the basis for qualification of DED parts. This means that a bigger deposited piece may be machined down to various sizes, all of which can be regarded as qualified. Note that the relevant heat treatment should be performed prior to such machining. Any additional surface treatment after machining is outside the scope of this qualification and should be assessed separately.

---e-n-d---o-f---g-u-i-d-a-n-c-e---n-o-t-e---

APPENDIX D NON-DESTRUCTIVE TESTING AND IN-PROCESS MONITORING

D.1 General

The design possibilities provided by additive manufacturing are allowing features and geometries combining several manufacturing techniques. While this gives designers and manufacturers a great tool to increase their efficiency and optimize their product, the higher complexity in a part's structure will also make inspecting those parts more difficult. In response to the growing complexity of parts, purchasers need to be increasingly open to accepting new ways of ensuring the quality of their parts. This appendix gives a framework for assessing the certainty of an inspection technique, for both simple and complex scenarios. By providing increased confidence, the framework may allow the conventional NDT performed on the final part to be replaced by novel and innovative techniques, either in process or on the final shape.

D.2 Definitions

Categorization of part complexity and technique novelty is used in this appendix to identify the necessary qualification activities for a specific case. The categories are defined in [Table D-1](#) and [Table D-2](#).

Table D-1 Grouping of parts based on inspectability, defined in ASTM E3166

<i>Category</i>	<i>Short description ¹⁾</i>
Group 1: Simple parts	Simple well-established designs that do not capitalize on the advantages of AM.
Group 2: Optimized standard parts	Parts based on conventional designs, but with some of the advantages of AM.
Group 3: Parts with embedded features	Parts with internal, embedded features such as tubes or channels.
Group 4: Design-to-constraint AM parts	Parts appearing to be free formed without straight lines or parallel surfaces and with no analogue made by conventional subtractive techniques.
Group 5: Lattice structures	Parts with thin free form metallic lattices that have high strength-to-weight ratios, increased surface areas, and tailored stiffness and damage tolerance.
1) For complete definitions, see ASTM E3166.	

Table D-2 Levels of novelty for inspection techniques

<i>Level</i>	<i>Description</i>
Level A: Established technique	Conventional NDT methods and advanced NDT methods with proven abilities and established standards.
Level B: Advanced, but partly established technique	Advanced NDT methods without proven abilities for the given application, or previously proven methods without established standards.
Level C: Advanced and innovative technique	Novel and innovative techniques without prior experience on the given application or established practices.

D.3 Assessment of inspection techniques

To determine with confidence that an inspection technique is suitable, the following aspects shall be taken into consideration.

- the uncertainty related to the inspection technique

— the complexity of the part with respect to inspectability.

In some instances, an established technique may be used in an innovative manner, e.g. the use of automation in combination with a conventional method. In these cases, it is the aspects of the system that are innovative that should be considered at a higher level, thus needing increased confidence. This assessment shall be performed at the beginning of the assessment process.

The control steps to be taken in each case are categorized as described in [Table D-3](#).

Table D-3 Description of control steps for the assessment of techniques

<i>Control steps</i>	<i>Description</i>
Procedure review	Review and confirmation of compliance with established standards and/or specifications.
Procedure validation	Confirmation of a procedure's ability to detect given material defects with a certain degree of confidence.
Technique qualification	Defining, performing, and documenting the technique's ability to detect given material defects with a certain degree of confidence.

The matrix in [Table D-4](#) shows the extent of method validation that should be performed based on these criteria.

Table D-4 Control steps needed for confirmation of inspection techniques

	<i>Level A: Established technique</i>	<i>Level B: Advanced, but partly established, technique</i>	<i>Level C: Advanced and innovative technique</i>
Group 1: Simple parts	Procedure review	Procedure validation	Procedure validation
Group 2: Optimized standard parts	Procedure review	Procedure validation	Technique qualification
Group 3: Parts with embedded features	Procedure validation	Procedure validation	Technique qualification
Group 4: Design-to-constraint AM parts	Procedure validation	Technique qualification	Technique qualification
Group 5: Lattice structures	Technique qualification	Technique qualification	Technique qualification

D.4 Assessment of competence

In general, the degree of assessment of the technician's competence is described in [Table D-5](#).

In cases where a conventional technique is used in a complex way, and where personnel qualifications do not exist, the qualification scheme for the conventional technique may be used in addition to in-house training on the specific equipment and methods.

Table D-5 Assessment of personnel competence

<i>Technique</i>	<i>Personnel competence assessment</i>	<i>Description</i>
Level A: Established technique	Certificate review	Review and confirmation of compliance with established standards and/or specifications.

<i>Technique</i>	<i>Personnel competence assessment</i>	<i>Description</i>
Level B: Advanced, but partly established, technique	Personnel verification	Controlled spot check of competence by verification of records, confirmation of competence by interviews, examination etc. according to established standards or specifications.
Level C: Advanced and innovative technique	Personnel qualification	Definition of a training scheme with a relevant curriculum, requirements regarding experience, personnel examinations and the documentation of competence.

D.5 Methodology

The methodology for assessing the abilities of an inspection technique and the organizational system is presented in this section and is based on [DNV-RP-A203](#). The process is performed through defining critical claims or properties that state the abilities of the system and assessing and documenting the confidence in each of those claims or properties.

The properties listed in [Table D-6](#) may be proved correct using rationale and gathered facts, which can be based on already established practices, historical data, modelling and experimental data. Each point is then assessed in terms of level of certainty, see [Figure D-1](#).

Experimental validation of the procedure may be done by testing and confirming each relevant discontinuity type of relevant size and shape prescribed by the acceptance criteria.

An inspection technique may be experimentally qualified for a given set of discontinuity types and types of parts. The discontinuity types that will be validated or qualified are then detected by the technique in question and confirmed by another qualified technique, destructive or non-destructive. Detection and confirmation shall be performed on several discontinuities of relevant size and shape to achieve statistical confidence.

Table D-6 Properties to be proved

<i>Application</i>	<i>No.</i>	<i>Properties</i>	<i>ID</i>	<i>Description</i>
General	1.	Operators have the appropriate skill level, mandate, and time to carry out their inspection activities with the right quality.	a)	The operators are trained and work according to established work procedures.
			b)	The organization regularly follows up its operators to maintain and improve their competence.
			c)	The organization has a training scheme in place that includes a relevant curriculum and defined requirements regarding experience, personnel examinations and the documentation of competence.
General	2.	The organization has processes in place to ensure consistency over time and between projects.	a)	Processes are described and implemented in management and governance documents. The processes include: — inspection processes — maintenance and calibration of inspection equipment — reporting — handling of deviations and corrective actions.

<i>Application</i>	<i>No.</i>	<i>Properties</i>	<i>ID</i>	<i>Description</i>
Procedure review	3.	The inspection procedure ensures the correct quality and confidence	a)	The procedure has been developed according to recognized standards and industry practices.
			b)	The procedure's scope of application is in line with recognized standards and the parts it will be used for.
Procedure validation	4.	The inspection procedure is capable of detecting of relevant defects and suitable for the scope of application	a)	The procedure is in line with existing standards and industry practices.
			b)	The procedure's scope of application is suitable for the parts it will be used for.
			c)	The technique is capable of detecting and categorizing of relevant defects of relevant size within its scope of application.
Technique qualification	5.	The technique is capable of identifying process abnormalities	a)	Abnormalities in the build process can be identified with defined thresholds of criticality.
Technique qualification	6.	The technique is capable of detecting and categorizing the following indication types of relevant size	a)	Able to detect lack of fusion-type defects.
			b)	Able to detect pores and porosity.
			c)	Able to detect cracks.
			d)	Able to detect other defect types as listed in ASTM E3166, if relevant.
Technique qualification	7.	The technique can size indications with sufficient accuracy	a)	Indications can be confidently sized with a specified accuracy.
Technique qualification	8.	The technique's range of application is well defined	a)	A procedure for the specific application has been established.
			b)	The technique works on the relevant type of materials.
			c)	The technique covers the material thickness and mass.
			d)	The technique is suitable for the relevant level of complexity with regard to access and inspectability.

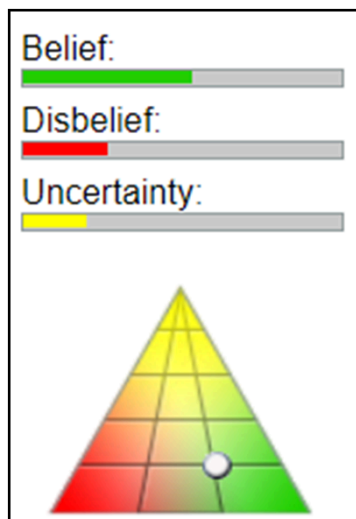


Figure D-1 The assessment of each property can be done continuously based on the degree of proof, and associated uncertainty, illustrated in the example through belief/disbelief and certainty.

The assessment of certainty may be based on experimental data, documented experience, mathematical modelling, theoretical assessments, or a combination of multiple elements. Each of the points in [Table D-6](#) shall be evaluated before a final qualification statement is prepared based on the complete assessment. Strengths and weaknesses shall be emphasized in the documentation.

D.6 Building an inspection programme

Based on a combination of different techniques, all aspects of a part's quality can be verified. As a general principle, the inspection programme shall as a minimum include enough inspection techniques to cover each relevant defect category once with a known degree of certainty. The purchaser may specify a larger scope for critical components.

D.7 Documentation

The capabilities of an inspection technique or an inspection programme shall be documented through a definition of its limits of application and of the proof supporting those limits. The qualification details and proof should be documented in a way that is practical to update and expand upon, e.g. in a database of information or a report. However, quality checked revisions of the conclusions should be dated and approved. Limitations on the inspection technique and inspection programme shall be clearly emphasized and communicated to the designer at the design stage.

CHANGES – HISTORIC

May 2020 edition

Amendments September 2021

<i>Topic</i>	<i>Reference</i>	<i>Description</i>
Rebranding to DNV	All	This document has been revised due to the rebranding of DNV GL to DNV. The following have been updated: the company name, material and certificate designations, and references to other documents in the DNV portfolio. Some of the documents referred to may not yet have been rebranded. If so, please see the relevant DNV GL document. No technical content has been changed.

Changes May 2020

This is a new document.

About DNV

DNV is the independent expert in risk management and assurance, operating in more than 100 countries. Through its broad experience and deep expertise DNV advances safety and sustainable performance, sets industry benchmarks, and inspires and invents solutions.

Whether assessing a new ship design, optimizing the performance of a wind farm, analyzing sensor data from a gas pipeline or certifying a food company's supply chain, DNV enables its customers and their stakeholders to make critical decisions with confidence.

Driven by its purpose, to safeguard life, property, and the environment, DNV helps tackle the challenges and global transformations facing its customers and the world today and is a trusted voice for many of the world's most successful and forward-thinking companies.

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